

A MERGED RESEARCH REPORT

Differential Algebra and Differential Equations over the Surreal and Surcomplex Numbers

The finite-primitive criterion, the phase obstruction,
and support-certified coordinate systems

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Abstract

Eight independently written manuscripts developed the same theory of the Berarducci–Mantova derivation ∂ on the surreal class field $\mathbf{No}$, normalized by $\partial\omega = 1$, and of its unique extension to the surcomplex field $\mathbf{No}[i]$. This report is their union. Seven of them share a twelve-item spine of scalar and workspace theory; the eighth carries the finite matrix stratum and joined later.

The organizing identity is exact: the image of the logarithmic derivative is

$$\{\partial y/y : y \in \mathbf{No}[i]^\times\} = \mathbf{No} + i\mathcal{A}, \quad \mathcal{A} := \partial\mathfrak{m} = \partial\mathcal{O},$$

where $\mathcal{O}$ is the class of finite surreals and $\mathfrak{m}$ its ideal of infinitesimals. Hence $\partial y = (a+ib)y$ has a nonzero surcomplex solution exactly when b has a *finite* surreal primitive. The two printed forms $\mathbf{No} + i\partial\mathfrak{m}$ and $\mathbf{No} + i\partial\mathcal{O}$ agree only through a proved lemma, and $\mathcal{A}$ is a *strict* subclass of $\mathfrak{m}$: the coefficient ω^{-1} is infinitesimal while its primitive $\log \omega$ is infinite. The purely infinite part of the accumulated phase is an $\mathbb{R}$ -linear surjection onto the class of purely infinite surreals, a complete gauge invariant, and the middle term of a four-term exact sequence. From it follow sharp support and iterated-logarithm thresholds, a decision procedure for rational coefficients, the pointwise maximality of the strip $\mathbf{No} + i\mathcal{O}$ as the domain of any derivation-compatible exponential, the nonexistence of a nonzero solution of $\partial y = iy$ or $\partial^2 y + y = 0$, a complete constant-coefficient and constant-matrix classification in which only real characteristic roots contribute, a triangular fundamental-matrix criterion, exact resolvents with factorially divergent Hahn solutions, differentially stable set-sized workspaces, and oscillatory Picard–Vessiot extensions with differential Galois tori — at the cost, in the real form, of the H -field ordering rule.

A later part classifies *finite matrix* systems. Every $\partial y = Ay$ with A an ordinary finite matrix over $\mathbf{No}[i]$ is gauge equivalent to $i \operatorname{diag}(\partial P_1, \dots, \partial P_n)$ with each P_j purely infinite, and the multiset $\{P_j\}$ is a complete invariant of differential equivalence; for a Hermitian H it is computed by integrating the ordinary algebraic eigenvalues of H and keeping the purely infinite parts of the primitives, with neither commutation nor a spectral gap assumed. From that follow a gap-free finite-primitive perturbation theorem, an exact kernel count, normalized non-Abelian integration, the complete cone of positive invariant metrics, a real rotation-block descent, a positive Ermakov–Pinney dichotomy, an exact Hahn expansion for the oscillatory Airy equation, and the phase-lattice Picard–Vessiot torus of an arbitrary finite system. A deliberate counterexample — a nonnormal coefficient with instantaneous eigenvalues $\pm i$ and a fundamental matrix already in the base field — marks the boundary: the non-Hermitian case is not classified. That part also invokes a separately imported linear-splitting and first-order surjectivity theorem, which is not a consequence of the scalar criterion and is flagged wherever it is used.

A deliberately separate half treats *coordinate* differential equations in fixed-common-domain Hahn rings $\mathcal{O}(U)((t^\Gamma))$: existence, uniqueness, support certificates, variation of constants, a Liouville determinant identity, monodromy valued in $\operatorname{GL}_d(K_\Gamma)$, and a polynomially nonlinear initial-value theorem, all built on strong summability and an ordered-word support lemma proved here in full. The two halves share notation and almost nothing else; several theorems are true for one operator and false for the other on the same printed equation, and the report keeps their type distinctions intact rather than flattening them.

The Berarducci–Mantova construction is imported, not reproved. Nothing here is claimed as new, refereed, or machine-checked.

Status. A merged exposition assembled from eight independent manuscripts: seven pinned to VladimirReshetnikov/Surreal at commit `e260237db9b71da8b74a0c13c8e6355119091100`, and the eighth, which supplies Part III, pinned to the later revision `aa846271b4dcae2c055b216126a87210292ec19b`. Deep structural inputs are attributed; the connecting deductions are proved in the text. No priority claim, no solution of a named published open problem, no independent refereeing, and no proof-assistant verification is asserted. Section 31 is the consolidated register of what is *not* claimed; it is part of the result, not a disclaimer appended to it.

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Part I

Foundations, and the distinctions that must survive

1 What this report is

1.1 Eight manuscripts, one theory

This report is the union of eight independently written manuscripts on the differential algebra of the surreal and surcomplex numbers. Seven of them form the original core: they were prepared separately, they are all pinned to the same documentation snapshot, and they prove — in seven notations — the same twelve-item spine:

- (i) the Berarducci–Mantova derivation ∂ extends uniquely to $\mathbf{No}[i] = \mathbf{No}[i]$, because differentiating $i^2 + 1 = 0$ forces $\partial i = 0$; its constant field is exactly $\mathbb{C}$;
- (ii) derivatives of finite elements are infinitesimal;
- (iii) an infinitesimal exponential and logarithm, defined as Hahn sums, with their chain rules;
- (iv) a branch-free polar decomposition with a finite angle;
- (v) a primitive normalized by a vanishing ω^0 coefficient;
- (vi) the rank-one criterion — $\partial y = (a + ib)y$ has a nonzero surcomplex solution if and only if b has a finite primitive — and the exact logarithmic-derivative image;
- (vii) the purely infinite obstruction, its exact sequence, and a gauge normal form;
- (viii) $\partial y = iy$ and $\partial^2 y + y = 0$ have only the zero solution;
- (ix) there is no globally derivation-compatible complex exponential;
- (x) the constant-coefficient classification, in which only real characteristic roots contribute, with its constant-matrix corollary;
- (xi) a set-sized differential localization;
- (xii) an oscillatory Picard–Vessiot extension with no new constants.

Every one of the seven proves items (i)–(vi) and (viii)–(ix). The remaining four are proved by most but not all: the gauge normal form of (vii) is absent from three members, its exact sequence from a fourth; the constant-coefficient classification (x) is absent from one member, which explicitly declines it, and its constant-matrix corollary from two others; and at least five prove each of (xi) and (xii). Section C gives the member-by-member record, and no claim below rests on a member that does not prove what it is cited for. Keeping them apart would mean printing one theory seven times, including seven proofs of $\partial i = 0$ and seven proofs that $\partial y = iy$ has no nonzero solution. Each is nevertheless the unique source of some genuinely distinct payload, and those payloads are all built on the shared spine and cannot be extracted without carrying it along. Section C records, member by member, what each contributed.

The eighth manuscript joined later and is different in kind. It does not reprove the spine; it cites it, names the gap this report’s own scope ledger recorded — no classification of arbitrary

nontriangular matrix systems — and closes the Hermitian half of it. Its mathematics is Part III, and it is pinned to a later revision, `aa846271b4dcae2c055b216126a87210292ec19b`, of the same repository. Several of the twelve spine items are reproved inside it by its own routes, and where that happens the routes are recorded as agreement (Theorems 15.1 and 15.2) rather than printed twice. It also brings one *new imported input*, the linear-splitting and first-order surjectivity theorem of Theorem 15.6, which the first seven do not use and which the register of non-claims accordingly records as an import rather than as a proved consequence.

Merging carries a specific risk, which Section 5 addresses before any theorem is stated: several different objects in these manuscripts are called “phase”, and two nearly identical classes are both called the exponential domain. A merged sentence that unifies either word, or either symbol, proves false statements. This report fixes one name and one symbol per object, keeps the two classes visibly distinct, and states the distinction wherever a reader might carry one sense into another.

1.2 The documented gap

The seven audits agree on the gap and on its character. All are pinned to the same snapshot of VladimirReshetnikov/Surreal:

`e260237db9b71da8b74a0c13c8e6355119091100`

That snapshot describes fifteen research packages covering surreal normal forms and constructions, surcomplex analysis and geometry, polynomial algebra, trigonometry, foundations and symbolic computation [15]. Differentiation is not absent from that collection. What is absent is the step from the collection’s careful *distinctions* between kinds of derivative to existence, obstruction and support-control *theorems* for differential equations.

Three boundaries in the existing documentation delimit the gap precisely. The trigonometry report constructs finite phases and states explicitly that it chooses no derivation on **No**, and that its classification of global phase extensions is not a classification of differential-equation solutions [17]. The computer-algebra report separates an external-variable derivative, a formal monomial derivative and a derivation of the surreal scalar field, and its subsection “Differential equations must name their category” issues a uniqueness warning rather than developing a solution theory [19]. The foundations report insists that a fine derivative, a formal-series germ, a coherent Hahn datum and a scalar-field derivation stay four different interfaces [20]. The analysis report goes further still: its derivation-obstruction theorem already shows that one chosen canonical global surcomplex exponential is incompatible with an internal scalar derivation normalized by $\partial\omega = 1$ [18]. That existing theorem is prior coverage, acknowledged as such; Theorem 11.4 below is strictly stronger in its phase-independence and is compatible with it.

Every one of the seven audits was targeted: the catalogue, the specialist READMEs, and selected source passages, at one pinned revision. None was a line-by-line certification of the fifteen reports or of their archived sources, and none used the absence of a code-search hit as evidence of absence. Section 31 records those audit limits in full.

1.3 What is imported and what is proved

The deep input is the existence of the derivation, together with its strong additivity, its constant field, its compatibility with the surreal exponential, its surjectivity and its *H*-field positivity. These are imported from Berarducci and Mantova [1] and are not reproved; their construction through log-atomic numbers, path derivatives and nested truncation ranks is not reproduced. The

positive-support summability lemma is imported in its standard form from Neumann [10], and its ordered-word core is additionally proved in full in Section A because the original paper was not retrieved. Restricted composition and the global composition-incompatibility theorem for ω -series are separately imported from [2] and are flagged wherever used. The universal H -field results of [3] are background for Parts I, II, IV and V, not premises; in Part III alone they become a premise, because Theorem 15.6 transfers along the elementary embedding of [3] two further published facts — splitting of monic scalar operators into first-order factors and surjectivity of nonzero scalar operators [4, Lemma 2.4.19]. That import is a genuine addition to the hypothesis base of this report and is flagged at every use (Theorem 15.7, item N53). Ordinary complex initial-value theory [12] supplies the analytic ingredient of the coordinate half, and Picard–Vessiot terminology is used in the standard sense of [13, 14].

Everything else is proved here. None of it is claimed to be new.

Remark 1.1 (Reading order). Part I fixes conventions, imports the derivation, derives smallness, and installs the vocabulary discipline. Readers who want only the headline should then read Section 8. Part II is the scalar theory over $\mathbf{No}[i]$; Part III raises it to finite matrix systems and contains the spectral–integral theorem Theorem 16.5; Part IV is its set-sized and Hahn workspaces; Part V is the separate coordinate theory, whose only dependence on Part II is negative — it explains what the scalar obstruction does *not* forbid. Part VI collects the computational contracts and the register of non-claims.

2 Conventions, classes, and normal forms

2.1 Class-theoretic ground rules

We reason classically: ZFC with definable classes, or NBG with choice, or any conservative class-theoretic presentation with enough replacement for the constructions displayed below. The symbols $\mathbf{No}$ and $\mathbf{No}[i]$ denote proper classes; a “class field” assertion means that the field operations are class functions satisfying the usual identities, and a class-level statement is a statement about elements.

Every individual number, every normal-form support, every family that is summed, every polynomial, and every matrix below is a *set*. No proper-class family of nonzero terms is summed and no sum is indexed by $\mathbf{No}$.

Warning 2.1 (No class of proper-class cosets). The suggestive notation $\mathbf{No}/\mathcal{O}$ must never be implemented as a collection whose members are cosets of $\mathcal{O}$: a coset of a proper class is not an object that can belong to another class. Throughout this report it denotes the *represented* quotient, namely the class $\mathcal{P}$ of purely infinite normal forms together with the projection pp of equation (2.3). Every obstruction value is an actual surreal number, hence a set. Likewise, automorphism groups over a proper-class base are encoded by constant multipliers of finitely many generators, not by forming a class whose members are proper-class maps; and rank-one equations are classified by their scalar coefficients together with an explicitly stated gauge relation, not by a class-sized collection of modules. Inside a fixed set-sized field, ordinary set quotients are of course legitimate; Section 20 supplies such fields.

2.2 Normal forms and the three-part splitting

$\mathbf{No}$ is real closed and $\mathbf{No}[i] = \mathbf{No}[i]$ is algebraically closed. We use the Conway normal form, the real surreal (Gonshor) exponential and its inverse log on positive elements as background [1, 7]. A real surreal has a unique expansion

$$x = \sum_{\alpha \in S} r_\alpha \omega^\alpha, \quad r_\alpha \in \mathbb{R} \setminus \{0\}, \quad (2.1)$$

with S a reverse well-ordered set of surreal exponents. For surcomplex numbers the coefficients lie in $\mathbb{C}$. It is often convenient to reverse the orientation and write

$$t^\gamma := \omega^{-\gamma}, \quad (2.2)$$

so that a support is well ordered in increasing γ and positive exponents describe infinitesimal monomials. We abbreviate $t = t^1 = \omega^{-1}$.

Warning 2.2 (Two different exponent constructions). In equation (2.1) and equation (2.2), $\gamma \mapsto \omega^{-\gamma}$ is the Conway *monomial map*. For an arbitrary surreal γ it must not be silently replaced by $\exp(-\gamma \log \omega)$: the omega-map and general surreal exponentiation are different constructions. The identity $\partial(\omega^r) = r\omega^{r-1}$, equivalently $\partial(t^r) = -r t^{r+1}$, is asserted for *ordinary real* r only, where the two interpretations agree. The safe universal identity is $\partial(\omega^\gamma) = \omega^\gamma \partial \log(\omega^\gamma)$ with the actual Gonshor logarithm of the monomial. A symbolic implementation needs different constructors for the two operations. For the record, the expression $\exp(\gamma \log \omega)$ has derivative $\exp(\gamma \log \omega)((\partial \gamma) \log \omega + \gamma/\omega)$, which is not a general rule for the monomial map.

Write

$$\mathcal{O} = \{x \in \mathbf{No} : |x| < n \text{ for some } n \in \mathbb{N}\}, \quad \mathfrak{m} = \{x \in \mathbf{No} : |x| < 1/n \text{ for every } n \geq 1\}.$$

Thus $\mathcal{O}$ is the convex valuation ring of finite elements and $\mathfrak{m}$ is its maximal ideal, with $\mathcal{O} = \mathbb{R} \oplus \mathfrak{m}$. Let $\mathcal{P}$ be the additive class of *purely infinite* normal forms: those all of whose exponents in equation (2.1) are strictly positive, together with 0. It is an additive real vector class. It is *not* the class of all infinite surreals. Splitting the support of equation (2.1) at exponent 0 gives the unique additive decomposition

$$x = \text{pp}(x) + \text{ct}(x) + \varepsilon, \quad \text{pp}(x) \in \mathcal{P}, \quad \text{ct}(x) \in \mathbb{R}, \quad \varepsilon \in \mathfrak{m}, \quad (2.3)$$

and we write $\text{fin}(x) = \text{ct}(x) + \varepsilon = x - \text{pp}(x)$. All of pp, ct and fin are additive and $\mathbb{R}$ -linear, and extend $\mathbb{C}$ -linearly to $\mathbf{No}[i]$ componentwise. So $\mathbf{No} = \mathcal{P} \oplus \mathbb{R} \oplus \mathfrak{m}$ and $\mathcal{O} = \mathbb{R} \oplus \mathfrak{m}$, and x is finite exactly when $\text{pp}(x) = 0$.

Warning 2.3 (ct is not a standard part and is not multiplicative). $\text{ct}(x)$ is the coefficient of the monomial $\omega^0 = 1$, defined for *every* surreal, including infinite ones. The standard part st is defined only on $\mathcal{O}$, where it is a ring homomorphism with kernel $\mathfrak{m}$; the two agree only there. ct is additive and real-linear but is not multiplicative: $\text{ct}(\omega) = \text{ct}(\omega^{-1}) = 0$ while $\text{ct}(\omega \cdot \omega^{-1}) = 1$. None of pp, ct, fin is a field homomorphism.

For complex values, conjugation fixes $\mathbf{No}$ and sends i to $-i$, and $|u + iv| = (u^2 + v^2)^{1/2}$. Put $\mathcal{O}_{\mathbb{C}} = \mathcal{O} + i\mathcal{O}$ and $\mathfrak{m}_{\mathbb{C}} = \mathfrak{m} + i\mathfrak{m}$; standard part is taken coordinatewise and lands in $\mathbb{C}$. An element of modulus one is necessarily finite, whatever the birthdays of its coordinates.

2.3 Summability is a support condition

Definition 2.4 (Strong summability). A set-indexed family $(x_j)_{j \in J}$ of Hahn normal forms is *summable* (equivalently *strongly summable*) when $\bigcup_j \text{supp } x_j$ is well ordered in the increasing-exponent convention of equation (2.2) and, at every exponent, only finitely many x_j have a nonzero coefficient. Its sum is formed coefficientwise, by those finite coefficient sums.

Warning 2.5 (What summability is not). Strong summability is a statement about supports, not about convergence. It is strictly stronger than “valuations increase”. It licenses coefficientwise rearrangement of a summable family, and nothing else: it does not authorize exchanging two infinite sums, and *writing* a formal geometric or exponential series is not by itself a summability proof. None of the Hahn identities below is asserted to be a limit of partial sums in the fine surreal topology, in which set-indexed convergence is degenerate. Conversely, a fixed-rank set-sized Hahn field does have nontrivial valuation-topological convergence, and full-class topological statements must not simply be transferred to it. The three notions — fine topology, fixed-rank valuation topology, strong summability — are never identified here.

A concrete separator: with $\Gamma = \mathbb{Z}^2$ lexicographically ordered, $v(t) = (0, 1)$ and $v(u) = (1, 0)$, one has $v(t^n) < v(u)$ for every ordinary n . The identity $\sum_{n \geq 0} t^n = (1 - t)^{-1}$ is strongly summable, yet its partial sums do not converge in that valuation topology. At higher rank $n\sigma$ need not become cofinal, and Section A establishes no such cofinality; no argument in this report uses one.

The combinatorial engine is the positive-support lemma.

Lemma 2.6 (Positive-support calculus). *Let Γ be an ordered abelian group and let $S \subseteq \Gamma_{>0}$ be well ordered. Then the additive monoid*

$$S^* = \{0\} \cup S \cup (S + S) \cup (S + S + S) \cup \dots$$

is well ordered, and for each $\gamma \in \Gamma$ only finitely many finite ordered words $(s_1, \dots, s_n)$ in S satisfy $s_1 + \dots + s_n = \gamma$. Consequently, if h has positive support, then for ordinary real or complex coefficients a_n the family $(a_n h^n)_{n \geq 0}$ is summable.

This is the lemma underlying the classical Hahn–Neumann construction [10], used in the summability preliminaries of [1, 2]. Section A proves it in full, by way of Higman’s finite-word lemma [11], and stresses that the count is over *ordered* words: that stronger form is what noncommuting matrix products require in Sections 24 and 27, and an unordered or multiset count will not carry those theorems.

Warning 2.7 (Splitting a differentiated sum incurs two obligations). If a family $(a_k x^k)_k$ is summable, strong additivity gives $\partial \sum_k a_k x^k = \sum_k \partial(a_k x^k)$. To split the right-hand side into a coefficient-derivative sum and a variable-derivative sum, both separated families must *themselves* be summable. Summability of a family of sums does not imply summability of its summand families, because cancellation can hide a violation. Every Taylor evaluation used below carries the required positive-support certificate; Theorem 4.5 is the one place where the summability of the two split families is *derived* rather than assumed, and it says exactly which extra structure makes that possible.

3 The imported derivation, and smallness as a theorem

3.1 The imported package

Theorem 3.1 (Berarducci–Mantova: imported input). *There is a distinguished derivation $\partial : \mathbf{No} \rightarrow \mathbf{No}$ — the simplest Berarducci–Mantova derivation — with the following properties.*

- (BM1) *Leibniz*: $\partial(xy) = x\partial y + y\partial x$.
- (BM2) *Constant field*: $\ker \partial = \mathbb{R}$, *exactly*.
- (BM3) *Strong additivity*: $\partial(\sum_{j \in J} x_j) = \sum_{j \in J} \partial x_j$ for every set-indexed summable family, and the family of derivatives is itself summable.
- (BM4) *Exponential compatibility*: $\partial(\exp x) = (\exp x)\partial x$, for the real surreal exponential.
- (BM5) *Normalization*: $\partial \omega = 1$.
- (BM6) *Surjectivity*: $\partial \mathbf{No} = \mathbf{No}$.
- (BM7) *H-field positivity*: $x > \mathbb{R}$ implies $\partial x > 0$.

This is the package supplied by [1, Theorems A/6.30 and B/7.7]; the theorem numbering is that of the linked preprint. Its proof is deep and is not reproduced here. Assigning a derivative to a few familiar monomials is not a construction of a strongly additive derivation on every surreal: log-atomic numbers require extra initial data, the construction specifies a pre-derivation there and extends it through summable path derivatives, and both the summability proof and the surjectivity proof are substantial inputs rather than consequences of the product rule. This report uses their conclusions, not a pretended implementation.

Warning 3.2 (Two clauses that are easy to drop, and one that is not in the list). Clause (BM3) is load-bearing for every Hahn-sum computation below, including the infinitesimal exponential and logarithm (Theorem 6.1), the Laurent resolvents (Theorem 18.6), and the coefficient derivation on common-domain families (Theorem 28.4); it is *not* supplied by the Leibniz rule. Clause (BM7) is load-bearing for Theorem 3.8 and hence for the entire obstruction theory, and for the ordering obstruction of Theorem 21.10. Clause (BM5) is indispensable for discussing a *named* equation such as $\partial y = iy$: every named-equation statement below is a statement about this derivation with this normalization, and none of them is derivation-agnostic.

What is deliberately *not* in the list is *smallness*, the assertion $\partial \mathfrak{m} \subseteq \mathfrak{m}$. Three of the seven source manuscripts import it as a clause of the Berarducci–Mantova package, and one of those flags it as specific to the chosen derivation rather than a general fact. The other four derive it from (BM7). Section 3.3 derives it, for two reasons: advertising smallness as part of an imported theorem overstates the import when it follows from an axiom already assumed; and the derived version additionally yields strict order reversal on the infinitesimals (Theorem 3.8), which the imported version does not give and which Theorem 7.5 uses.

Warning 3.3 (No uniqueness, and no other derivation is classified). Other surreal derivations exist. Theorem 3.1 is a usable interface, not a uniqueness characterization, and the elementary properties listed do not characterize every possible surreal derivation. Nothing below classifies the alternatives. The word “canonical”, where it survives from a source, names this distinguished construction and nothing more. Every intrinsic statement in this report is relative to this one fixed ∂ .

From now on we write $x' = \partial x$ interchangeably. The elementary rules follow at once from Theorem 3.1: for $x \neq 0$, $u > 0$ and $r \in \mathbb{R}$,

$$\partial(x^{-1}) = -x^{-2}\partial x, \quad \partial\left(\frac{x}{y}\right) = \frac{y\partial x - x\partial y}{y^2}, \quad \partial \log u = \frac{\partial u}{u}, \quad \partial(u^r) = r u^{r-1}\partial u, \quad (3.1)$$

where $u^r = \exp(r \log u)$; in particular $\partial \omega^r = r \omega^{r-1}$ and $\partial t^r = -r t^{r+1}$ for ordinary real r (see Theorem 2.2), $\partial t = -t^2$, and $\partial \log \omega = t$. With $\ell_0 = \omega$ and $\ell_{n+1} = \log \ell_n$, each ℓ_n is positive infinite and induction on equation (3.1) gives the iterated-logarithm derivatives

$$\partial \ell_n = \frac{1}{\ell_0 \ell_1 \cdots \ell_{n-1}} \quad (n \geq 1), \quad (3.2)$$

with the empty product understood as 1. A rational power rule $\partial(x^q) = qx^{q-1}\partial x$ for $q \in \mathbb{Q}$ and $x > 0$ is available purely algebraically, without the exponential.

For an algebraic element there is a corresponding implicit rule.

Proposition 3.4 (Algebraic implicit differentiation). *Let $F \subseteq \mathbf{No}[i]$ be a differential subfield and let α be algebraic over F with minimal polynomial $P = \sum_k a_k X^k \in F[X]$. Then*

$$\partial \alpha = -\frac{P^\partial(\alpha)}{P_X(\alpha)}, \quad P^\partial(X) = \sum_k (\partial a_k) X^k.$$

Proof. In characteristic zero an irreducible polynomial is separable, so $P_X(\alpha) \neq 0$. Differentiating the finite identity $P(\alpha) = 0$ gives $P^\partial(\alpha) + P_X(\alpha)\partial\alpha = 0$. $\square$

This is the derivative rule for an exact algebraic root object; it is not a choice of a numerical root branch.

3.2 The unique complexification

Proposition 3.5 (Unique extension to $\mathbf{No}[i]$). *There is exactly one derivation on $\mathbf{No}[i] = \mathbf{No}[i]$ extending ∂ , namely*

$$\partial(a + ib) = \partial a + i\partial b.$$

It commutes with conjugation, is strongly additive on set-indexed summable families in $\mathbf{No}[i]$, is surjective, and its constant field is exactly $\mathbb{C}$. Moreover, for $z \neq 0$ and $r = |z|$,

$$\frac{\partial r}{r} = \operatorname{Re}\left(\frac{\partial z}{z}\right). \quad (3.3)$$

Proof. Any extension satisfies $0 = \partial(i^2 + 1) = 2i\partial i$, hence $\partial i = 0$, which forces the displayed formula; direct expansion verifies the Leibniz rule. Conjugation commutes with it by inspection. For strong additivity, take real and imaginary parts: their derivative families are summable by (BM3), a finite union of well-ordered supports is well ordered, and each exponent retains finite multiplicity. The kernel consists of the elements both of whose components lie in $\ker \partial = \mathbb{R}$, that is exactly $\mathbb{C}$. Given $a + ib$, choose real primitives A, B by (BM6); then $\partial(A + iB) = a + ib$. Finally $r^2 = z\bar{z}$ gives $2r\partial r = (\partial z)\bar{z} + z\overline{\partial z}$; divide by $2r^2$ to obtain equation (3.3). $\square$

All seven manuscripts prove this proposition, by this four-line argument. Two consequences deserve emphasis. First, the constant field *changes* from $\mathbb{R}$ to $\mathbb{C}$ on complexification, and several later statements — the kernel of the logarithmic derivative, the fact that solution spaces are $\mathbb{C}$ -spans — depend on that. Second, ∂ is surjective on $\mathbf{No}[i]$ and $\mathbf{No}[i]$ is algebraically closed, and neither fact solves $\partial y = ay$ with $y \neq 0$: surjectivity solves $\partial y = f$, and algebraic closure supplies roots of polynomials in existing elements. Algebraic closure is not differential solution closure, and Section 8 measures the gap exactly.

Convention 3.6 (What “a differential equation over $\mathbf{No}[i]$ ” means). In Parts II and IV, a differential equation over $\mathbf{No}[i]$ is an algebraic equation in elements of $\mathbf{No}[i]$ and their iterated ∂ -images. It is *not* an equation for a class function of an independent variable. Part V treats the latter, under a different operator and with a different constant field, and Section 4 keeps them apart.

Every real coefficient is a logarithmic derivative in the real field.

Proposition 3.7 (Real exponential integration). *For every $p \in \mathbf{No}$, the element $s = \exp(\mathcal{I}_0 p)$ of Theorem 7.1 satisfies $\partial s = ps$ and $s > 0$. More generally, for $a, b \in \mathbf{No}$ the equation $\partial y = ay + b$ has surreal solutions, and with $h = \mathcal{I}_0 a$ all real solutions are $y = e^h(c + \mathcal{I}_0(e^{-h}b))$, $c \in \mathbb{R}$.*

Proof. Apply (BM4) to a primitive of p , respectively of a ; the product rule verifies the displayed formula, and the quotient of the difference of two solutions by e^h has derivative zero. $\square$

Notice the two distinct uses of structure: (BM6) supplies the primitive, and (BM4) supplies the exponential. Knowing only that ∂ is onto would not justify the second step. The difficulty in the complex case is therefore not additive integration; it is the absence of a suitable nonzero integrating factor, and Section 8 says exactly when one exists.

3.3 Smallness, derived, with strict inclusion

Lemma 3.8 (Order reversal on infinitesimals, and smallness). *If $0 < \varepsilon \in \mathfrak{m}$ then $\partial\varepsilon < 0$. Consequently ∂ is strictly order-reversing on $\mathfrak{m}$, and*

$$\partial\mathcal{O} = \partial\mathfrak{m} \subseteq \mathfrak{m}.$$

Proof. Since ε is a positive infinitesimal, $1/\varepsilon > \mathbb{R}$, so (BM7) and equation (3.1) give

$$0 < \partial(1/\varepsilon) = -\frac{\partial\varepsilon}{\varepsilon^2},$$

whence $\partial\varepsilon < 0$. For every positive integer n , the element $\omega + n\varepsilon$ is positive infinite, so (BM7) and (BM5) give $1 + n\partial\varepsilon > 0$. Therefore $-1/n < \partial\varepsilon < 0$ for every n , that is $\partial\varepsilon \in \mathfrak{m}$. Negative infinitesimals follow by additivity and 0 is immediate. Strict order reversal follows by applying the first assertion to the positive infinitesimal difference of two infinitesimals. Finally every finite element is $r + \varepsilon$ with $r \in \mathbb{R}$ and $\varepsilon \in \mathfrak{m}$, and $\partial r = 0$, so $\partial\mathcal{O} = \partial\mathfrak{m}$. $\square$

Remark 3.9 (An independent route to smallness, and a second route to the containment). One source manuscript instead *imports* the finite form $\partial\mathcal{O} \subseteq \mathcal{O}$ — which is what the argument needs — and strengthens it to $\partial\mathcal{O} = \partial\mathfrak{m} \subseteq \mathfrak{m}$ by a square-root trick: a positive $\varepsilon \in \mathfrak{m}$ is q^2 with $q = \sqrt{\varepsilon} \in \mathfrak{m}$, so $\partial q \in \mathcal{O}$ by the imported smallness and $\partial\varepsilon = 2q\partial q \in \mathfrak{m}$. That argument is correct and is recorded here as an alternative, but it starts from the containment rather than proving it, and it does not give the order reversal. A third formulation of the same one-line proof, used by three of the seven, applies (BM7) to $\omega \pm nf$ for $f \in \mathcal{O}$, obtaining $1 \pm n\partial f > 0$ directly; a fourth applies it to $\omega/n \pm f$. All are the same argument, and none uses a mean-value theorem, a topological limit, or an estimate about numerical rounding. Smallness is a structural hypothesis about this derivation.

Warning 3.10 (The containment is strict, and that gap is the whole subject). $\partial\mathfrak{m} \subseteq \mathfrak{m}$ is a *strict* inclusion. Indeed $\partial \log \omega = 1/\omega$, and every primitive of $1/\omega$ differs from $\log \omega$ by a real constant, so every primitive of $1/\omega$ is infinite and $1/\omega \notin \partial\mathfrak{m}$. On the other hand $\omega^{-2} = \partial(-\omega^{-1}) \in \partial\mathfrak{m}$. Testing only whether an imaginary coefficient is infinitesimal therefore loses the decisive information. The

entire obstruction theory of Part II is the measurement of this gap, and Theorem 5.4 states what goes wrong if it is collapsed.

A first, coarse consequence is already enough to kill a constant nonreal rate.

Proposition 3.11 (A preliminary obstruction). *For every $z \in \mathbf{No}[i]^\times$, the imaginary part of $\partial z/z$ is infinitesimal. Hence for $\lambda \in \mathbb{C} \setminus \mathbb{R}$,*

$$\partial z = \lambda z \implies z = 0.$$

Proof. Multiplying z by 1 or by i does not change $\partial z/z$, so we may write $z = u(1 + ih)$ with $u \in \mathbf{No}^\times$ and $h \in \mathcal{O}$, taking as u a nonzero coordinate of maximal absolute value. Then

$$\frac{\partial z}{z} = \frac{\partial u}{u} + \frac{i\partial h}{1 + ih} = \frac{\partial u}{u} + \frac{h\partial h}{1 + h^2} + i\frac{\partial h}{1 + h^2}.$$

By Theorem 3.8, ∂h is infinitesimal, and division by $1 + h^2 \geq 1$ preserves that. A nonzero ordinary imaginary part is not infinitesimal. $\square$

This already rules out a surcomplex value behaving as $e^{i\omega}$ under ∂ . It does *not* yet classify infinitesimal rates: the exact image is strictly smaller than $\mathbf{No} + i\mathbf{m}$, and identifying it needs the finite-angle structure of Section 6.

4 One derivation, and six other operators

Almost every apparent paradox in this subject is a type error. The seven manuscripts between them separate seven operators; the merged report keeps all seven distinct, with the symbols of Table 1. A formula can be correct for one and false for another with no contradiction, and several theorems below are true for one and false for another *on the same printed equation*.

Warning 4.1 (Seven separators, each of which has been mistaken for a paradox). (1) *t is a number, not a symbol.* With $t = \omega^{-1}$ one has $D_z(t^3) = 0$, $D_t(t^3) = 3t^2$ and $\partial(t^3) = -3t^4$, all correct. Treating t as a formal constant is a coefficient operation; substituting $t = \omega^{-1}$ and applying the number derivation gives $\partial t = -t^2$, not 0. Replacing D_t by ∂ in a displayed ordinary differential equation without changing the coefficient functions usually changes the equation. The notation D_t becomes related to ∂ only after an embedding is specified and the formal operations are verified to commute with its interpretation.

- (2) *The constant function is not the number.* $F(Z) = \omega$ has $\partial_Z F = 0$, while the number ω has $\partial\omega = 1$. For $f(Z) = e^Z$ at a real point r , $f'(r) = e^r$ but $\partial(f(r)) = \partial(e^r) = 0$.
- (3) *A fine derivative does not determine an intrinsic one.* For every $x \in \mathbf{No}$ and every finite h , exactly,

$$\text{fin}(x + h) = \text{fin}(x) + h, \quad \text{pp}(x + h) = \text{pp}(x),$$

so the fine derivatives of $x \mapsto \text{fin}(x)$ and $x \mapsto \text{pp}(x)$ are 1 and 0 everywhere, with exact difference quotients for all $0 < |h| < 1$ and no limit subtlety. Yet $\partial(\text{fin}(\omega)) = \partial 0 = 0$ while $\partial(\text{pp}(\omega)) = \partial\omega = 1$. Hence a blanket identity $\partial(F(x)) = F'_{\text{fine}}(x)\partial x$ is *false* for arbitrary fine-differentiable F ; the missing hypothesis is a compatibility structure for F , and fine differentiability is not one. The same example explains why a zero fine derivative need not force a class function to be constant, whereas $\partial f = 0$ for a *scalar* means exactly $f \in \mathbb{C}$.

Table 1: The seven operators. Only the first is “the derivation”.

Symbol	What it does	What it holds fixed	Sample value
∂	The intrinsic Berarducci–Mantova derivation on a <i>number</i> of No or No [i]	Exactly $\mathbb{R}$, or $\mathbb{C}$ after complexification	$\partial\omega = 1$, $\partial t = -t^2$, $\partial(t^3) = -3t^4$
∂_s	The rescaled derivation $\partial_s f = \partial f / \partial s$, for $\partial s \neq 0$	Same kernel; but not the same normalization	$\partial_s s = 1$; for $s = \log \omega$, $\partial s = t$
∂_Z	Formal differentiation of an indeterminate Z on No [i][[Z]]	Every coefficient in No [i], however infinite	$\partial_Z(\omega Z) = \omega$, $\partial_Z \omega = 0$
$\tilde{\partial}$	Coefficientwise <i>intrinsic</i> differentiation on No [i][[Z]], $\tilde{\partial} \sum a_n Z^n = \sum (\partial a_n) Z^n$	The formal indeterminate Z	$\tilde{\partial}(\omega Z) = Z$
D_z	Coordinate differentiation of the holomorphic coefficient functions of a common-domain Hahn family, in an ordinary complex z	Every Hahn monomial and every scalar parameter	$D_z(t^3) = 0$, $D_z \omega = 0$, $D_z z = 1$
∂_H	Intrinsic differentiation of the Hahn <i>monomials</i> of a common-domain family	The ordinary variable z	$\partial_H(\omega z) = z$, $\partial_H(t^2 e^z) = -2t^3 e^z$
D_t	Formal differentiation in a Hahn parameter treated as an indeterminate	The coefficients	$D_t(t^3) = 3t^2$
F'_{fine}	The pointwise fine order-field derivative of a class <i>function</i> , an ϵ - δ difference quotient in the actual field variable	Determined by the function and its topology; not a field derivation	$\text{fin}'_{\text{fine}} \equiv 1$, $\text{pp}'_{\text{fine}} \equiv 0$

- (4) *A second fine-derivative separator, with a phase.* Let $H(x) = \text{cis}(\text{fin } x)$, using the finite-angle map of Theorem 6.4 on finite angles only. Then $H(\omega) = 1$ and, for infinitesimal h , $H(\omega + h) = E(ih) = 1 + ih + O(h^2)$ as an ordered-field estimate, so $H'_{\text{fine}}(\omega) = i$. Nevertheless

$$\partial(H(\omega)) = \partial 1 = 0 \neq H'_{\text{fine}}(\omega) \partial \omega = i.$$

A locally valid Taylor rule grants no universal scalar chain rule for class functions.

- (5) *Omitting a variable, not using the wrong rule.* For $F(z, t) = tz$ at $z = \omega$, $t = \omega^{-1}$, the value is 1 and

$$\partial F = t \partial \omega + \omega \partial t = t - \omega t^2 = 0,$$

whereas the coordinate derivative $D_z F = t$ alone would return t . Here the ordinary multivariable chain rule is valid; what is wrong is dropping a variable. The same point in one variable: for $F(Z) = tZ$ and $x = \omega$, $\partial(F(\omega)) = (-t^2)\omega + t = 0$, as it must be.

- (6) *A coherent family is not a number.* For $F(z) = tz$ with z a fixed ordinary complex number, $D_z F = t$ while $\partial(F(z)) = -t^2 z$, because that fixed ordinary z is a constant of ∂ .
- (7) *The same printed equation, twice.* $D_z F = iF$, $F(0) = 1$ has the ordinary entire solution e^{iz} , a single-coefficient member of a common-domain Hahn ring. $\partial y = iy$ has no nonzero solution in **No**[i] (Theorem 8.8). There is no contradiction: one statement is about a holomorphic function of an ordinary coordinate, the other about a scalar element of a fixed differential field. The nonexistent bridge would be to evaluate the entire function at $z = \omega$, call the result a surcomplex scalar, and assume its scalar derivative comes from the coordinate chain rule. Neither the fixed-domain Hahn representation nor the ordinary initial-value theorem supplies that bridge.

Proposition 4.2 (Change of independent scale changes the solvability data). *Let $s \in \mathbf{No}$ with $\partial s \neq 0$. Then $\partial_s f = \partial f / \partial s$ is a derivation with $\partial_s s = 1$, and*

$$\partial_s y = qy \iff \partial y = (\partial s)qy.$$

Hence the phase test of Theorem 8.4 must be applied to $\text{Im}((\partial s)q)$, not to $\text{Im } q$. With $s = \log \omega$, so $\partial s = t$: the equation $\partial_s y = iy$ is the forbidden $\partial y = ity$, whereas $\partial_s y = ity$ becomes $\partial y = it^2 y$ and is solvable.

The rescaled operator preserves the field but not the normalization at ω , nor necessarily the same order conventions. A backend cannot change the independent scale while leaving differential solvability metadata untouched.

Proposition 4.3 (The two formal derivations commute; the valid polynomial chain rule). *On $\mathbf{No}[i][[Z]]$ the derivations ∂_Z and $\tilde{\partial}$ commute, and $\tilde{\partial} + u\partial_Z$ is a derivation for every coefficient scalar u ; commutation with that sum need not hold when $\partial u \neq 0$. For a polynomial $F(Z) = \sum_{k=0}^n a_k Z^k \in \mathbf{No}[i][Z]$ and $x \in \mathbf{No}[i]$,*

$$\partial(F(x)) = (\tilde{\partial}F)(x) + (\partial_Z F)(x) \partial x. \quad (4.1)$$

The same identity holds for rational functions at arguments where the denominator is nonzero.

Proof. Commutation and the derivation property are coefficientwise identities using only finite convolution sums at each degree. For equation (4.1), apply the finite product rule to each $a_k x^k$, obtaining $(\partial a_k)x^k + ka_k x^{k-1} \partial x$, and add. Both the coefficient and the variable derivative obey the quotient rule, which gives the rational case. $\square$

Example 4.4 (Two worked checks of equation (4.1)). For $F(Z) = \omega Z^2 + \omega^{-1}$ at $x = \omega$: $F(x) = \omega^3 + \omega^{-1}$ and $\partial(F(x)) = 3\omega^2 - \omega^{-2}$, whereas the variable derivative alone gives only $2\omega^2$; the missing term is the derivative of the coefficients after evaluation. For $F(Z) = \omega Z + tZ^2$ and $h = \omega^2$: $F(h) = 2\omega^3$ so $\partial(F(h)) = 6\omega^2$, while on the right $(\tilde{\partial}F)(h) = h - t^2 h^2 = 0$ and $(\partial_Z F)(h) \partial h = (\omega + 2th)(2\omega) = 6\omega^2$. The cancellation in the coefficient term is a feature of that example, not a licence to omit coefficient differentiation.

An infinite series needs more, and the extra hypothesis can be made minimal.

Proposition 4.5 (Two-derivative chain rule for series, with derived summability). *Let $F(Z) = \sum_{n \geq 0} a_n Z^n$ with $a_n \in \mathbf{No}[i]$, and let $h \in \mathbf{No}[i]$. If the evaluated family $(a_n h^n)_{n \geq 0}$ is summable, then the two families defining $(\tilde{\partial}F)(h)$ and $(\partial_Z F)(h)$ are summable as well, and*

$$\partial(F(h)) = (\tilde{\partial}F)(h) + (\partial_Z F)(h) \partial h.$$

Proof. Assume first $h \neq 0$. Multiplying each member of a summable family by its ordinary integer index preserves summability, since it enlarges no support; multiplying the whole family by the fixed scalar h^{-1} preserves Hahn summability. Hence $(na_n h^{n-1})_{n \geq 1} = (nh^{-1}(a_n h^n))_{n \geq 1}$ is summable, and so is its multiple by the fixed ∂h . By (BM3) the family $(\partial(a_n h^n))_n$ is summable. Subtracting the already controlled variable-derivative family shows $((\partial a_n)h^n)_n$ is summable. The termwise product rule may now be summed. If $h = 0$, the evaluated series and both formal-derivative evaluations have at most one nonzero term each, and the identity reduces to $\partial a_0 = \partial a_0 + a_1 \partial 0$. $\square$

Remark 4.6 (Why this is stronger than three summability certificates). An arbitrary sum of two families can be summable through cancellation even when neither summand family is (Theorem 2.7). The special power-series form is what lets the variable-derivative family be controlled *first*, after which the split is justified. So Theorem 4.5 assumes one certificate and derives two, which is strictly stronger than assuming three independent ones. It still does not assert that an arbitrary formal series is evaluable at an arbitrary scalar; see Section 28 for the relevant separations and for the counterexample of Theorem 28.11.

Remark 4.7 (No general composition is assumed). No global operation “replace ω by x in every surreal” is ever used. Berarducci and Mantova construct a composition $f \circ x$ for f in the ω -series class and positive infinite x , identify $(\partial f) \circ x$ with the fine derivative of $x \mapsto f \circ x$, give a compatible chain rule inside that setting, and prove a Taylor theorem there [2, Cor. 7.6–7.7, Thms 7.14, 8.2]. They also prove that the simplest derivation on *all* of $\mathbf{No}$ admits no composition satisfying their composition and compatibility axioms [2, Thm 8.4]. Section 22 records both results with their hypotheses intact and explains the exact sense in which the two derivatives then agree. All differentiation in this report is either the fixed ∂ , finite algebra, or a summable infinitesimal formal substitution whose validity is proved.

5 The central hazard: three objects called “phase”

5.1 Three names, three symbols

Three different objects occur in the source manuscripts under the single word “phase”. Two of them are angles; the third is not an angle of anything, and does not even live in $\mathcal{O}$. Merging them — in a sentence, in a word, or in a symbol — produces false statements. This report therefore fixes the following vocabulary once and uses it without exception.

- Convention 5.1** (The three phases). (a) The *finite angle* θ . The map cis of Theorem 6.4 sends the *finite* surreals $\mathcal{O}$ onto the unit circle $\mathbb{T}(\mathbf{No})$ with kernel $2\pi\mathbb{Z}$ (Theorem 6.5). Its argument θ is the whole angle, and it is always finite: $\theta \in \mathcal{O}$. No value of a circular function at an infinite angle is ever used.
- (b) The *infinitesimal phase residue* η . Every unit $u \in \mathbb{T}(\mathbf{No})$ factors *uniquely* as $u = cE(i\eta)$ with c an ordinary unit complex number and $\eta \in \mathfrak{m}$ (Theorem 6.3). The constant c carries all of the ordinary phase, $\partial c = 0$, and η carries all of the variable intrinsic phase: $u^\dagger = i\partial\eta$. The two are related to (a) by $\theta = \text{st}(\theta) + \eta$ and $c = e^{i\text{st}(\theta)}$, so (b) is the *infinitesimal part* of (a), not a synonym for it.
- (c) The *accumulated phase* B . For a real coefficient b , an accumulated phase is any $B \in \mathbf{No}$ with $\partial B = b$; the normalized choice is $\mathcal{I}_0(b)$. Any two differ by a real constant. B is an arbitrary surreal. It need not be finite, it is not the angle of any surcomplex number, and when it is infinite it lies outside $\mathcal{O}$ entirely, with its purely infinite part $\text{pp}(B) \in \mathcal{P}$ — the *obstruction* — nonzero. The whole theory of Part II is the statement that $\partial y = (a + ib)y$ is solvable exactly when the accumulated phase of b is finite, that is $B \in \mathcal{O}$. Only then does B acquire an angular reading at all: a finite B is the argument of $\text{cis } B$ in sense (a), with $\eta = B - \text{st}(B)$ its infinitesimal residue in sense (b). An infinite B is the argument of nothing. So (c) is not a third kind of angle; it is the object whose *finiteness* is what sense (a) requires, and the criterion is exactly the assertion that it can fail.

- (d) **A missing oscillatory generator.** In Section 21.4 and in the Picard–Vessiot sections the phrase *independent phases* counts something else again: the number of transcendental generators T that must be adjoined *outside* $\mathbf{No}[i]$ before a diagonal system has a full solution basis. That count is $d = m - \text{rank } \Lambda$, a rank of a lattice of obstructions, and it is not a count of angles, residues or accumulated phases. Wherever this report writes “independent phases” or “phase generators”, that fourth sense is meant, and it is never any of (a)–(c).
- (e) The *differential phase spectrum* $\text{Ph}(A)$. For a finite matrix system $\partial y = Ay$ over $\mathbf{No}[i]$, Theorem 15.9 attaches a *multiset* $\text{Ph}(A) = \{P_1, \dots, P_n\}$ of purely infinite surreals: the accumulated obstructions of the diagonal entries of its normal form. Its members are objects of sense (c) — more precisely, values of $\mathfrak{o}$ — so Ph adds no new kind of phase; what is new is that there are n of them at once and that the multiset, not its sum, is the complete invariant (Theorem 15.14). The eighth source manuscript writes Ψ for the map $b \mapsto \text{pp}(B)$ with $\partial B = b$. That map is exactly $\mathfrak{o}$ of Theorem 7.4, so this report keeps the single symbol $\mathfrak{o}$ for it and does not adopt Ψ , in accordance with Theorem 5.6. The adoption would in any case have collided: the letter Ψ already occurs once in this report, in Theorem 11.4, as a local placeholder for a putative derivation-compatible exponential $\mathbf{No}[i] \rightarrow \mathbf{No}[i]^\times$ — a map that is there proved not to exist, and that is not a phase map of any of the senses (a)–(e).

Warning 5.2 (“Spectrum” names three inequivalent things across the collection). The differential phase spectrum $\text{Ph}(A)$ of sense (e) is a multiset of *purely infinite surreal numbers*. It is not a list of eigenvalues: the companion report *Surcomplex Spectral Theory* [21] uses “spectrum” for the algebraic eigenvalue and singular-value lists of a finite matrix and their valuations, which are ordinary surcomplex numbers of the field, and Theorem 16.5 is precisely the theorem that relates the two — by integrating and taking purely infinite parts, not by identifying them. Theorem 16.10 shows that for a nonnormal coefficient the two are unrelated. Nor is $\text{Ph}(A)$ a set of angles: no member of a phase spectrum is ever routed through the finite-angle map cis of sense (a) or through the finite-angle constructions of [17], since a nonzero member of $\mathcal{P}$ is infinite and cis is defined on $\mathcal{O}$ only. And it is not a prime spectrum of a ring: where the collection’s analytic-geometry report discusses Spec of a coefficient ring, that is the ring-theoretic object and is written out as *prime spectrum* there.

Table 2: The phase objects, and where each source’s notation lands.

Name here	Symbol	Lives in	What it is	Source notations
Finite angle	θ	$\mathcal{O}$	The argument of cis ; the <i>whole</i> angle of a unit, finite by theorem	θ, y
Infinitesimal phase residue	η	$\mathfrak{m}$	What is left of a unit after factoring out the constant ordinary unit complex number c	$\eta, \theta, \varepsilon, \delta$
Accumulated phase	B	$\mathbf{No}$	A primitive of the imaginary coefficient; <i>not</i> an angle; obstructed exactly when $\text{pp}(B) \neq 0$	$B; \mathcal{I}_0(b)$; and four different letters for the primitive operator
Differential phase spectrum	$\text{Ph}(A)$	multisets in $\mathcal{P}$	The n accumulated obstructions of a finite matrix system, as a multiset; the complete invariant of differential equivalence	$\text{Ph}; \Psi(\lambda_j)$, where the map Ψ is this report’s $\mathfrak{o}$

Warning 5.3 (Where the senses collide in the sources). Two of the first seven manuscripts use “phase” for sense (a), four use it for sense (b), and all seven use “accumulated phase” for sense (c); the eighth introduces sense (e) and, with it, a fourth letter for the primitive obstruction. One of them prints (a) and (b) twenty lines apart, both called the phase. Sense (c) is the one that carries the obstruction, and it is the one that is *not* an angle: $\mathfrak{o}(i) = \omega$ and $\mathfrak{o}(i/\omega) = \log \omega$ are purely infinite surreals, not arguments of anything. A merged sentence such as “the phase of the solution is $\log \omega$ ” is meaningless; the correct sentence is “the accumulated phase of the coefficient i/ω is $\log \omega$, which is infinite, so there is no solution”. Similarly, “every surcomplex number has a finite phase” is a theorem about (a); “the variable phase of a unit is infinitesimal” is a theorem about (b); and neither implies anything about (c) without Theorem 8.4.

5.2 Two load-bearing classes that differ by one operator

The second half of the hazard is not a word but a symbol. Four of the seven manuscripts print, within a few pages of each other, the two classes

$$\mathbf{No} + i\mathcal{A} \quad \text{and} \quad \mathcal{S} = \mathbf{No} + i\mathcal{O},$$

where $\mathcal{A} = \partial\mathbf{m}$. The first is the exact image of the logarithmic derivative (Theorem 8.1); the second is the maximal domain of a derivation-compatible exponential (Theorem 11.2). They are different classes, and the difference is one application of ∂ .

Warning 5.4 (Do not collapse $\partial\mathcal{O}$ to $\mathcal{O}$). $\mathcal{A} = \partial\mathbf{m}$ is a *strict* subclass of $\mathbf{m}$ (Theorem 3.10), and $\mathbf{m}$ is a strict subclass of $\mathcal{O}$. If a merge collapses $\partial\mathcal{O}$ to $\mathcal{O}$, the rank-one criterion becomes “ b infinitesimal”, which would make $\partial y = i\omega^{-1}y$ solvable. Every one of the seven manuscripts explicitly refutes exactly that, and the counterexample pair of Theorem 10.4 — $i/(\omega \log \omega)$ insoluble against $i/(\omega(\log \omega)^2)$ soluble — exists precisely to block it. The strip $\mathcal{S} = \mathbf{No} + i\mathcal{O}$ is ∂ -stable *because* $\partial\mathcal{O} \subseteq \mathbf{m}$; that is a consequence of the difference between the two classes, not evidence that they are the same.

Warning 5.5 ($\mathbf{No} + i\partial\mathcal{O}$ and $\mathbf{No} + i\partial\mathbf{m}$ are equal only after a lemma). The sources print the logarithmic-derivative image in both forms. They agree, but only because $\partial\mathcal{O} = \partial\mathbf{m}$, which is a *theorem* (Theorem 3.8, with the independent square-root route of Theorem 3.9). The two forms must never be presented as notational variants: one is the other after a proved step. This report uses the single symbol $\mathcal{A}$ for the common value and states the lemma before using it.

5.3 The letter J , and the word “obstruction”

Across the source batch the letter J denotes three unrelated objects: in one manuscript it is the normalized intrinsic primitive on $\mathbf{No}$; in another it is not an operator at all but the representative *class* of purely infinite surreals; and in a companion report that remains standalone — the rank-one Berkovich manuscript, on analytic functions on surcomplex disks and annuli — it is the zero-constant *coordinate* primitive of a convergent power series over a fixed coefficient field, whose “complete primitive obstruction” is an annular Laurent residue and has nothing to do with the phase obstruction here. Two further manuscripts write J for the inverse of $\partial|_{\mathbf{m}}$.

Convention 5.6 (No letter J ; one symbol per object). This report uses no symbol J for any operator or class. The letter occurs only as a neutral index set for set-indexed families (Theorem 2.4 and Theorem 3.1), where no confusion with the objects below is possible. The normalized intrinsic primitive is $\mathcal{I}_0$ (Theorem 7.1); its target class of purely infinite surreals is $\mathcal{P}$; the inverse of $\partial|_{\mathbf{m}}$ is written $\mathcal{I}_0|_{\mathcal{A}}$ and named the *unique infinitesimal primitive*. The obstruction on $\mathbf{No}$ is $\mathfrak{o}$ and

its surcomplex companion is Φ (Theorem 7.4). Cross-references to companion reports name the operator rather than the letter.

Remark 5.7 (The obstruction does not depend on the normalization). Adding a real constant to a primitive does not change its purely infinite part, so $\mathfrak{o}$ is insensitive to the normalization of $\mathcal{I}_0$. That insensitivity is what makes $\mathfrak{o}$ well defined, and it is why the normalization is a convenience rather than a hypothesis. It is also why a “primitive” defined only up to $\mathbb{R}$, or selected by a choice function, is a *different* object: the first would not give a well-defined primitive operator, and the second would not give a class function without a choice principle. $\mathcal{I}_0$ avoids both by singling out a unique element; see Theorem 7.1 and Theorem 7.2.

Part II

The scalar theory over $\text{No}[i]$

6 Infinitesimal exponential, finite angle, polar form

6.1 The infinitesimal exponential and logarithm

For $h \in \mathfrak{m}_{\mathbb{C}}$ define the Hahn sums

$$E(h) = \sum_{n \geq 0} \frac{h^n}{n!}, \quad L(1+h) = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} h^n. \quad (6.1)$$

Theorem 2.6 makes both families summable, because the support of a nonzero infinitesimal is positive and well ordered. These are Hahn sums, not limits of their partial sums in any topology. Their domain is *infinitesimal arguments only*: no attempt is made to sum the ordinary exponential Taylor series at a nonzero ordinary constant, still less at an infinite argument.

Lemma 6.1 (Formal identities survive infinitesimal evaluation). *The maps in equation (6.1) are inverse group isomorphisms*

$$(\mathfrak{m}_{\mathbb{C}}, +) \xrightleftharpoons[L]{E} (1 + \mathfrak{m}_{\mathbb{C}}, \cdot),$$

they commute with conjugation, and

$$\partial E(h) = E(h) \partial h, \quad \partial L(w) = \frac{\partial w}{w} \quad (w \in 1 + \mathfrak{m}_{\mathbb{C}}). \quad (6.2)$$

Moreover E is injective on $\mathfrak{m}_{\mathbb{C}}$: for $h \in \mathfrak{m}_{\mathbb{C}}$,

$$E(h) = 1 \iff h = 0. \quad (6.3)$$

More generally, for any $f(X) = \sum_{n \geq 0} c_n X^n$ with ordinary constant coefficients $c_n \in \mathbb{C}$ and any $h \in \mathfrak{m}_{\mathbb{C}}$, the family $(c_n h^n)_n$ is summable, its sum is denoted $f(h)$, and

$$\partial(f(h)) = f'_{\text{formal}}(h) \partial h. \quad (6.4)$$

Proof. The formal identities $\exp(U+V) = \exp(U)\exp(V)$, $\log(\exp U) = U$ and $\exp(\log(1+U)) = 1+U$ hold over the ordinary characteristic-zero coefficient field. After substitution at infinitesimals,

Theorem 2.6 makes every coefficient calculation a finite one at each fixed exponent, which justifies the products and regroupings; this is a finite check per exponent, not an appeal to a limit. Conjugation acts on coefficients and preserves supports.

For equation (6.2) and equation (6.4), the finite product rule gives $\partial(h^n) = nh^{n-1}\partial h$, and (BM3) permits termwise differentiation, including the assertion that the differentiated family is summable. The family $(nc_n h^{n-1})_{n \geq 1}$ is separately summable by Theorem 2.6, and multiplication by the single fixed scalar ∂h preserves summability; so the factorisation

$$\partial E(h) = \partial h \sum_{m \geq 0} \frac{h^m}{m!}$$

and, for the logarithm, $\partial h \sum_{n \geq 1} (-1)^{n+1} h^{n-1} = \partial h / (1 + h)$, are both legitimate. For equation (6.3): if $h \neq 0$, every power h^n with $n \geq 2$ has valuation strictly larger than that of h , so the leading term of $E(h) - 1$ is the leading term of h . $\square$

Warning 6.2 (Three limits on Theorem 6.1). (i) The injectivity equation (6.3) is *local* to $\mathfrak{m}_{\mathbb{C}}$ and asserts nothing about an injective exponential on the whole class $\mathbf{No}[i]$. (ii) The argument does *not* extend to a Taylor series at an arbitrary infinite argument: the original family need not be summable there, and Theorem 11.9 exhibits the failure. (iii) The evaluation rule $f(h)$ requires *constant* coefficients. A formal series with arbitrary coefficients in $\mathbf{No}[i]$ need not be summable at any given nonzero infinitesimal: $\sum_{n \geq 0} \omega^{n^2} X^n$ is not summable at $X = \omega^{-1}$, because its monomials ω^{n^2-n} are not reverse well ordered. The infinitesimal chain rule must not be exported to arbitrary $\mathbf{No}[i]$ -coefficient streams without a joint support certificate (Theorem 28.11). No lower bound on an ordinary radius of convergence is needed, however: factorially growing complex coefficients remain coefficients at the same Hahn monomials.

6.2 The finite angle

Let

$$\mathbb{T}(\mathbf{No}) = \{u \in \mathbf{No}[i] : u\bar{u} = 1\}, \quad \mathbb{T}(\mathbb{C}) = \{c \in \mathbb{C} : |c| = 1\}.$$

Theorem 6.3 (Unique infinitesimal phase residue). *Every $u \in \mathbb{T}(\mathbf{No})$ has a unique representation*

$$u = c E(i\eta), \quad c \in \mathbb{T}(\mathbb{C}), \quad \eta \in \mathfrak{m}, \quad (6.5)$$

namely $c = \text{st}(u)$ and $i\eta = L(u/c)$; and

$$u^\dagger := \frac{\partial u}{u} = i \partial \eta. \quad (6.6)$$

Consequently $\mathbb{T}(\mathbf{No}) \cong \mathbb{T}(\mathbb{C}) \times (\mathfrak{m}, +)$ as abelian groups.

Proof. Write $u = a + ib$ with $a^2 + b^2 = 1$; both coordinates are finite, so $c = \text{st}(u)$ exists and $c\bar{c} = 1$, whence $c \neq 0$. Then $w = u/c$ lies in $1 + \mathfrak{m}_{\mathbb{C}}$ and satisfies $w\bar{w} = 1$, that is $\bar{w} = w^{-1}$. Put $q = L(w)$. By Theorem 6.1,

$$\bar{q} = L(\bar{w}) = L(w^{-1}) = -q,$$

so $q = i\eta$ with $\eta \in \mathfrak{m}$ real, and $w = E(i\eta)$. Conversely every expression equation (6.5) has modulus one, its standard part recovers c , and the injective local logarithm recovers η . Finally $\partial c = 0$, so equation (6.2) gives equation (6.6). Multiplication multiplies the c 's and adds the η 's. $\square$

The proof chooses no global value of $\sin \omega$, no branch of a global logarithm and no global circular character. The ordinary phase is stored as the *discrete datum* c , not as a selected real argument, so there is no ambiguity by multiples of 2π : the branch ambiguity of an angle is an ordinary constant $2\pi k$, which ∂ kills, and the infinitesimal residue in equation (6.5) is branch-free outright.

Definition 6.4 (The finite-angle map). For a *finite* real surreal $\theta = r + \eta$, with $r = \text{st}(\theta) \in \mathbb{R}$ and $\eta \in \mathfrak{m}$, set

$$\text{cis}(\theta) = e^{ir} E(i\eta),$$

where $e^{ir} \in \mathbb{C}$ is the ordinary complex exponential, a differential constant. The real and imaginary coordinates of cis are the canonical finite-angle cosine and sine, and they agree with their Taylor construction. The domain is $\mathcal{O}$ only; no value of a circular function at an infinite angle is used.

Theorem 6.5 (Finite-angle phase group, with its differential identity). *The map $\text{cis} : (\mathcal{O}, +) \rightarrow (\mathbb{T}(\mathbf{No}), \cdot)$ is a surjective group homomorphism with kernel exactly $2\pi\mathbb{Z}$, and*

$$\partial \text{cis}(\theta) = i(\partial\theta) \text{cis}(\theta), \quad \text{equivalently} \quad \frac{\partial \text{cis}(\theta)}{\text{cis}(\theta)} = i \partial\theta. \quad (6.7)$$

Consequently every nonzero $y \in \mathbf{No}[i]$ is $y = \rho \text{cis}(\theta)$ with $\rho > 0$ in $\mathbf{No}$ and θ finite.

Proof. The group law follows from Theorem 6.1 and the ordinary phase group law; conjugation inverts $E(i\eta)$, so the image lies in $\mathbb{T}(\mathbf{No})$. For surjectivity, let $u \in \mathbb{T}(\mathbf{No})$ and take c, η from Theorem 6.3; choosing an ordinary real r with $e^{ir} = c$ gives $u = \text{cis}(r + \eta)$. If $\text{cis}(r + \eta) = 1$, standard parts give $e^{ir} = 1$, so $r \in 2\pi\mathbb{Z}$, and then equation (6.3) gives $\eta = 0$; conversely these elements lie in the kernel. Since r and e^{ir} are differential constants, equation (6.2) gives equation (6.7). Applying surjectivity to $y/|y|$ gives the last assertion. $\square$

The surjectivity and kernel content of Theorem 6.5 is the finite-angle phenomenon already central to the repository's trigonometry report [17]; it is reproved here because the merged development needs it with the derivative identity attached. The differential clause equation (6.7) is the part the present theory adds, and it is where the obstruction originates: the factor $\partial\theta$ is indispensable, and removing it would replace an infinitesimal angular rate by the forbidden constant rate 1. In particular, for an infinitesimal ε the finite-angle sine and cosine satisfy $\partial \sin \varepsilon = (\cos \varepsilon) \partial \varepsilon$ and $\partial \cos \varepsilon = -(\sin \varepsilon) \partial \varepsilon$, so there is no conflict between the existence of finite-angle trigonometry and the nonexistence of an intrinsic oscillator (Theorem 8.8).

6.3 Polar decomposition and its differential content

Definition 6.6 (Logarithmic derivative). For $z \in \mathbf{No}[i]^\times$ write $z^\dagger = \partial z / z$. The notation does not presuppose the existence of a complex logarithm. It is a group homomorphism $(\mathbf{No}[i]^\times, \cdot) \rightarrow (\mathbf{No}[i], +)$: $(zw)^\dagger = z^\dagger + w^\dagger$, $(z^{-1})^\dagger = -z^\dagger$, and $\ker(\dagger) = \mathbb{C}^\times$ by Theorem 3.5.

Corollary 6.7 (Branch-free polar form). *Every $z \in \mathbf{No}[i]^\times$ has a unique decomposition*

$$z = \rho c E(i\eta), \quad \rho = |z| > 0, \quad c = \text{st}(z/|z|) \in \mathbb{T}(\mathbb{C}), \quad \eta \in \mathfrak{m}, \quad (6.8)$$

and

$$z^\dagger = \partial \log \rho + i \partial \eta = \frac{\partial \rho}{\rho} + i \partial \eta. \quad (6.9)$$

In coordinates $z = u + iv$,

$$\operatorname{Re}(z^\dagger) = \frac{uu' + vv'}{u^2 + v^2}, \quad \operatorname{Im}(z^\dagger) = \frac{uv' - vu'}{u^2 + v^2}. \quad (6.10)$$

Equivalently $z = \rho \operatorname{cis}(\theta)$ with $\theta \in \mathcal{O}$, and then $\operatorname{Re}(z^\dagger) = \partial|z|/|z|$ and $\operatorname{Im}(z^\dagger) = \partial\theta$.

Proof. Apply Theorem 6.3 to $z/|z|$ and use $\partial \log \rho = \partial \rho / \rho$ together with equation (6.6); the coordinate formulas are direct computation. The cis form is Theorem 6.5, and $\partial\theta = \partial\eta$ because $\theta - \eta = \operatorname{st}(\theta) \in \mathbb{R}$. $\square$

The first summand of equation (6.9) is a radial logarithmic rate and is unrestricted; the second is an angular rate, and its allowed primitives are constrained to finite angles. An enormous modulus does not create an enormous circular phase: the direction of $\omega + i$, for example, is a small perturbation of 1. Identity equation (6.9) is precisely the reason that an arbitrary radial logarithmic derivative is possible while an arbitrary imaginary one is not.

Warning 6.8 (A unit cannot hide an infinite accumulated phase). A unit u may be denoted by an expression involving a large parameter. As an element of $\mathbf{No}[i]$ it nevertheless has a finite angle θ , and hence $u^\dagger = i\partial\theta \in i\mathcal{A}$. A notation such as $e^{i \log \omega}$ cannot, by itself, turn an infinitesimal coefficient into a permissible logarithmic derivative. This is not a restriction on ordinary complex exponentiation, where the variable is a coordinate, and it is not a claim that global phase characters are impossible; it is a restriction on compatibility with the specified scalar derivation.

7 Primitives, accumulated phase, and the bounded-primitive ideal

7.1 The normalized primitive

Proposition 7.1 (Normalized primitive). *For each $b \in \mathbf{No}$ there is a unique $\mathcal{I}_0(b) \in \mathbf{No}$ with*

$$\partial \mathcal{I}_0(b) = b, \quad \operatorname{ct}(\mathcal{I}_0(b)) = 0.$$

The map $\mathcal{I}_0$ is $\mathbb{R}$ -linear and satisfies $\partial \circ \mathcal{I}_0 = \operatorname{id}$ and $\mathcal{I}_0 \circ \partial = \operatorname{id} - \operatorname{ct}$. Its componentwise extension $\mathcal{I}_0(a + ib) = \mathcal{I}_0(a) + i\mathcal{I}_0(b)$ is the unique $\mathbb{C}$ -linear map $\mathbf{No}[i] \rightarrow \mathbf{No}[i]$ with $\partial \mathcal{I}_0 = \operatorname{id}$ and $\operatorname{ct} \circ \mathcal{I}_0 = 0$, and it satisfies $\mathcal{I}_0(\partial f) = f - \operatorname{ct}(f)$.

Proof. By (BM6) choose any primitive F of b and put $\mathcal{I}_0(b) = F - \operatorname{ct}(F)$. Any two primitives differ by an element of $\ker \partial = \mathbb{R}$, and if both have vanishing ω^0 coefficient that difference is zero; this gives existence, uniqueness and independence of the choice of F . Linearity follows from uniqueness, since a sum or real multiple of normalized primitives has the required derivative and normalization. The identity $\mathcal{I}_0(\partial f) = f - \operatorname{ct}(f)$ holds because $f - \operatorname{ct}(f)$ is a primitive of ∂f with zero constant coefficient. The complex statements follow componentwise from Theorem 3.5. $\square$

Warning 7.2 ($\mathcal{I}_0$ is a normalization, not an algorithm). The definition is by a *uniqueness* property and makes no assertion about an effective primitive-finding procedure. The normalization refers to the coefficient of ω^0 , *not* to evaluation at an initial point; $\mathcal{I}_0$ is not asserted to be multiplicative, not asserted to preserve arbitrary Hahn sums, and not asserted to be computable from a supplied symbolic representation. It does not require a choice function on primitive families — the normalization singles out a unique element, so no choice principle is involved. Its value is semantic precision. Sample values: $\mathcal{I}_0(1) = \omega$, and $\mathcal{I}_0(\omega^{-1}) = \log \omega - \operatorname{ct}(\log \omega)$, which is infinite regardless of its real constant term.

The integration constant is not evaluation at an endpoint, and one familiar algebraic identity accordingly acquires a correction term.

Proposition 7.3 (Integration by parts, and a Rota–Baxter correction). *For $f, g \in \mathbf{No}[i]$,*

$$\mathcal{I}_0(f\partial g + g\partial f) = fg - \text{ct}(fg), \quad (7.1)$$

whose final term is not in general $\text{ct}(f)\text{ct}(g)$; and

$$\mathcal{I}_0(f)\mathcal{I}_0(g) = \mathcal{I}_0(f\mathcal{I}_0(g) + \mathcal{I}_0(f)g) + \text{ct}(\mathcal{I}_0(f)\mathcal{I}_0(g)). \quad (7.2)$$

The correction term in equation (7.2) cannot in general be omitted.

Proof. Both identities follow by differentiating the left-hand sides and applying $\mathcal{I}_0 \circ \partial = \text{id} - \text{ct}$. For the last assertion take $f = 1$ and $g = t^2$: then $\mathcal{I}_0(f) = \omega$, $\mathcal{I}_0(g) = -t$, so $\mathcal{I}_0(f)\mathcal{I}_0(g) = -1$, while $f\mathcal{I}_0(g) + \mathcal{I}_0(f)g = -t + \omega t^2 = 0$. Without the correction the identity would assert $-1 = 0$. $\square$

In the terminology sometimes used for integral operators, $\mathcal{I}_0$ is therefore *not* a weight-zero Rota–Baxter operator on all of $\mathbf{No}[i]$, and equation (7.2) is the precise replacement. No such terminology is needed to use the formula; the point is that the selected constant coefficient is linear but not multiplicative (Theorem 2.3), so a “zero constant of integration” convention is not automatically an integral based at a common evaluation point.

7.2 Accumulated phase and its obstruction

Definition 7.4 (Accumulated phase and obstruction). For $b \in \mathbf{No}$, an *accumulated phase* of b is any $B \in \mathbf{No}$ with $\partial B = b$; the normalized choice is $B = \mathcal{I}_0(b)$. Put

$$\mathcal{A} := \partial \mathbf{m} = \partial \mathcal{O}, \quad \mathfrak{o}(b) := \text{pp}(\mathcal{I}_0(b)) \in \mathcal{P},$$

and for $q \in \mathbf{No}[i]$ put

$$\Phi(q) := \mathfrak{o}(\text{Im } q) \in \mathcal{P}.$$

We call $\mathcal{A}$ the *bounded-primitive ideal* (equivalently, the class of *admissible angular rates*), and $\mathfrak{o}(b)$, $\Phi(q)$ the *phase obstruction*.

By Theorem 5.7 the value $\mathfrak{o}(b) = \text{pp}(B)$ is the same for every accumulated phase B of b , so $\mathfrak{o}$ is well defined without the normalization. It has an actual surreal representative, not an ambiguous antiderivative choice. Sample values:

$$\mathfrak{o}(1) = \omega, \quad \mathfrak{o}(\omega^{-1}) = \log \omega, \quad \mathfrak{o}(\omega^{-2}) = 0. \quad (7.3)$$

The obstruction measures accumulated phase, not the instantaneous size of a coefficient: a real coefficient has zero obstruction however large it is, and an infinitesimal coefficient may have a nonzero one.

Theorem 7.5 (Structure of the bounded-primitive ideal). (i) $\mathcal{A} = \partial \mathbf{m} = \partial \mathcal{O}$, and $\partial : \mathbf{m} \rightarrow \mathcal{A}$ is an $\mathbb{R}$ -linear bijection that reverses strict order.

(ii) For $b \in \mathbf{No}$ the following are equivalent: $b \in \mathcal{A}$; $\mathcal{I}_0(b)$ is infinitesimal; $\mathcal{I}_0(b)$ is finite; some primitive of b is finite; every primitive of b is finite; $\mathfrak{o}(b) = 0$.

- (iii) $\mathcal{A}$ is convex in $\mathbf{No}$, is an additive subgroup, and is an $\mathcal{O}$ -submodule of $(\mathbf{No}, +)$, that is, an ideal of the valuation ring $\mathcal{O}$.
- (iv) $0 \neq \mathcal{A} \subsetneq \mathfrak{m} \subsetneq \mathcal{O}$, with witnesses $\omega^{-2} = \partial(-\omega^{-1}) \in \mathcal{A}$ and $\omega^{-1} \in \mathfrak{m} \setminus \mathcal{A}$.
- (v) There is an exact cut description

$$\mathcal{A} = \{b \in \mathbf{No} : |b| < \partial f \text{ for every } f > \mathbb{R}\}. \quad (7.4)$$

- (vi) $\mathfrak{o} : \mathbf{No} \rightarrow \mathcal{P}$ is $\mathbb{R}$ -linear and surjective with kernel exactly $\mathcal{A}$; indeed $\mathcal{I}_0(\partial P) = P$ and $\mathfrak{o}(\partial P) = P$ for $P \in \mathcal{P}$.
- (vii) As additive real vector classes,

$$\mathbf{No} = \partial\mathcal{P} \oplus \mathcal{A} = \partial\mathcal{P} \oplus \partial\mathfrak{m}, \quad (7.5)$$

with projection $b \mapsto \partial \text{pp} \mathcal{I}_0(b)$ onto the first summand; ∂ is injective on each of $\mathcal{P}$ and $\mathfrak{m}$ separately, and $\partial : \mathcal{P} \rightarrow \partial\mathcal{P}$ is an order-preserving $\mathbb{R}$ -linear bijection.

- (viii) ∂ induces an order-preserving $\mathbb{R}$ -linear isomorphism of ordered additive quotients

$$\mathbf{No}/\mathcal{O} \longrightarrow \mathbf{No}/\mathcal{A}, \quad x + \mathcal{O} \longmapsto \partial x + \mathcal{A}, \quad (7.6)$$

and in particular $\partial^{-1}(\mathcal{A}) = \mathcal{O}$.

Proof. (i) The equality $\partial\mathcal{O} = \partial\mathfrak{m}$ and the order reversal are Theorem 3.8; the kernel of $\partial|_{\mathfrak{m}}$ is $\mathfrak{m} \cap \mathbb{R} = \{0\}$, giving injectivity, and surjectivity onto $\mathcal{A}$ is the definition.

(ii) A primitive in $\mathfrak{m}$ is finite; conversely a finite primitive becomes infinitesimal after subtracting its standard part. All primitives differ by a real constant, so either all are finite or none is. Since $\text{ct}(\mathcal{I}_0(b)) = 0$, the splitting equation (2.3) makes $\mathcal{I}_0(b)$ finite exactly when it is infinitesimal, exactly when $\text{pp} \mathcal{I}_0(b) = 0$.

(iii) Suppose $0 < b \leq d$ with $d \in \mathcal{A}$. Write $d = -\partial\varepsilon$ with $0 \leq \varepsilon \in \mathfrak{m}$, using (i), and let B be a primitive of b . If B were negative infinite then $-B$ would be positive infinite and (BM7) would give $b = \partial B < 0$, a contradiction. If B were positive infinite then $B + \varepsilon$ would be positive infinite, whence $\partial(B + \varepsilon) = b - d > 0$, contradicting $b \leq d$. So B is finite and $b \in \mathcal{A}$ by (ii). Sign symmetry and additivity give convexity in general. For $a \in \mathcal{O}$ choose a positive integer $n > |a|$; then $|ab| \leq n|b|$ for $b \in \mathcal{A}$, and convexity with real linearity gives $ab \in \mathcal{A}$.

(iv) $\mathcal{A} \subseteq \mathfrak{m}$ is Theorem 3.8. Nonzero and strict: the displayed witnesses, using Theorem 3.10.

(v) If $b = \partial\varepsilon$ with $\varepsilon \in \mathfrak{m}$, then for every $f > \mathbb{R}$ both $f + \varepsilon$ and $f - \varepsilon$ are positive infinite, so their derivatives are positive and $\partial f > |b|$. Conversely let $b \notin \mathcal{A}$ and let B be a primitive; then B is infinite. Take $f = |B|/2 > \mathbb{R}$; then $\partial f = |b|/2$ by (BM7), so the required inequality $|b| < \partial f$ fails.

(vi) Linearity is inherited from $\mathcal{I}_0$ and pp. If $P \in \mathcal{P}$ then $\text{ct}(P) = 0$, so P is the normalized primitive of ∂P and $\mathfrak{o}(\partial P) = \text{pp}(P) = P$; this gives surjectivity. The kernel is (ii).

(vii) Decompose $\mathcal{I}_0(b) = P + \varepsilon$ with $P = \text{pp} \mathcal{I}_0(b) \in \mathcal{P}$ and $\varepsilon \in \mathfrak{m}$, and differentiate: $b = \partial P + \partial\varepsilon$. If $\partial P = \partial\varepsilon$ with $P \in \mathcal{P}$, $\varepsilon \in \mathfrak{m}$, then $P - \varepsilon \in \mathbb{R}$ and uniqueness of equation (2.3) forces $P = \varepsilon = 0$; the same argument with one term zero gives the injectivity assertions. A nonzero positive member of $\mathcal{P}$ is positive infinite, so its derivative is positive by (BM7); applying this to differences gives the order assertion.

(viii) Surjectivity is inherited from (BM6), and $\partial^{-1}(\mathcal{A}) = \mathcal{O}$ is (ii) applied to ∂x . If $x + \mathcal{O}$ is positive then x is positive infinite, and for every $d = \partial\varepsilon \in \mathcal{A}$ the element $x - \varepsilon$ is positive infinite, so $\partial x > d$; hence the induced map preserves positivity, and the converse follows by bijectivity and sign symmetry. $\square$

Definition 7.6 (The unique infinitesimal primitive). By Theorem 7.5(i) the restriction $\partial|_{\mathfrak{m}}$ is a bijection onto $\mathcal{A}$; its inverse is $\mathcal{I}_0|_{\mathcal{A}} : \mathcal{A} \rightarrow \mathfrak{m}$, the *unique infinitesimal primitive*. Unlike a general primitive, it carries no residual additive-constant ambiguity. (This is the object that two source manuscripts call J ; see Theorem 5.6.)

Remark 7.7 (The represented quotient, concretely). Equation (7.6) is the represented form of the familiar isomorphism “ $\mathbf{No}/\mathcal{O} \cong \mathbf{No}/\mathcal{A}$ ”. Its actual representatives are the classes $\mathcal{P}$ and $\partial\mathcal{P}$, in accordance with Theorem 2.1; no proper-class coset is made into an object. Inside a set-sized field (Section 20) the corresponding ordinary set quotients are available.

8 The finite-primitive criterion

8.1 The exact image of the logarithmic derivative

Theorem 8.1 (Exact logarithmic-derivative image). *For the fixed derivation on $\mathbf{No}[i]$,*

$$\{z^\dagger : z \in \mathbf{No}[i]^\times\} = \mathbf{No} + i\mathcal{A} = \mathbf{No} + i\partial\mathfrak{m} = \mathbf{No} + i\partial\mathcal{O}. \quad (8.1)$$

The kernel of $\dagger$ is $\mathbb{C}^\times$, and $\partial : \mathfrak{m} \rightarrow \mathcal{A}$ is injective. The image is a proper subclass of $\mathbf{No}[i]$: $1 \in \mathbf{No}$ is in the image, being $(e^\omega)^\dagger$, while i is not.

Proof. If $z \neq 0$, Theorem 6.7 gives $z^\dagger = \partial \log|z| + i\partial\eta$ with $\eta \in \mathfrak{m}$, so the imaginary part lies in $\partial\mathfrak{m} = \mathcal{A}$; this is the inclusion from left to right.

For the reverse inclusion, let $a \in \mathbf{No}$ and $b \in \mathcal{A}$. Put $A = \mathcal{I}_0(a)$ and let $\eta \in \mathfrak{m}$ be the unique infinitesimal primitive of b (Theorem 7.6). Then

$$y_0 = \exp(A) E(i\eta) \neq 0 \quad (8.2)$$

and, by (BM4) and equation (6.2), $y_0^\dagger = \partial A + i\partial\eta = a + ib$. The kernel assertion is Theorem 6.6; injectivity of ∂ on $\mathfrak{m}$ is Theorem 7.5(i). Properness: $1 \notin \mathcal{A}$ by Theorem 3.8, since 1 is not infinitesimal. $\square$

Warning 8.2 (Two products of two different exponentials). The witness equation (8.2) is a product of *two* exponential constructions with explicitly different declared domains: $\exp$ is the real surreal exponential on $\mathbf{No}$, and E is the infinitesimal Hahn exponential of equation (6.1) on $\mathfrak{m}_{\mathbb{C}}$. The product is not an abbreviation for an unspecified global surcomplex exponential, and the two are never glued. Section 11 determines exactly how far a single compatible exponential can reach, and Theorem 11.4 shows it cannot reach all of $\mathbf{No}[i]$.

Warning 8.3 (The image is real-linear, not complex-linear). $\mathbf{No} + i\mathcal{A}$ is closed under addition and under multiplication by real constants, but not under multiplication by i : every real coefficient has zero obstruction, while multiplying one by i may not. An implementation must not model this image as a complex vector subspace.

8.2 The criterion, and all solutions

Theorem 8.4 (Finite-primitive criterion for rank-one equations). *Let $q = a + ib \in \mathbf{No}[i]$ with $a, b \in \mathbf{No}$, and let $A, B \in \mathbf{No}$ satisfy $\partial A = a$ and $\partial B = b$. The following are equivalent.*

- (i) $\partial y = qy$ has a nonzero solution in $\mathbf{No}[i]$.

- (ii) $b \in \mathcal{A}$, that is, $b = \partial\eta$ for some $\eta \in \mathfrak{m}$.
- (iii) b has a finite accumulated phase: $B \in \mathcal{O}$.
- (iv) Every accumulated phase of b is finite.
- (v) $\mathcal{I}_0(b)$ is finite, equivalently infinitesimal.
- (vi) $\text{pp}(B) = 0$, equivalently $\mathfrak{o}(b) = 0$, equivalently $\Phi(q) = 0$.

When these hold, the infinitesimal η with $\partial\eta = b$ is unique, namely $\eta = \mathcal{I}_0(b) = B - \text{st}(B)$, and all solutions are

$$y = c \exp(\mathcal{I}_0(a)) E(i\mathcal{I}_0(b)), \quad c \in \mathbb{C}, \quad (8.3)$$

the nonzero ones being those with $c \in \mathbb{C}^\times$. When they fail, the only solution is $y = 0$.

Proof. (i) $\Leftrightarrow$ (ii) is Theorem 8.1, since for $y \neq 0$ the equation says $y^\dagger = q$. Alternatively and directly: if $y \neq 0$ solves it, write $y = \rho \text{cis}(\theta)$ by Theorem 6.5; then equation (6.9) gives $\partial\rho/\rho = a$ and $\partial\theta = b$ with θ finite, which is (iii). The equivalences among (ii)–(vi) are Theorem 7.5(ii). The displayed family consists of solutions by equation (8.2); if y is any solution and y_0 is the nonzero one, the quotient rule gives $\partial(y/y_0) = 0$, so $y/y_0 \in \mathbb{C}$ by Theorem 3.5. The case $c = 0$ is the zero solution. $\square$

This is the theorem that all seven source manuscripts prove, in seven notations. Three points about it deserve separate statements.

Warning 8.5 (Integrated, not pointwise). The criterion is *not* “ b must be infinitesimal”. That condition is necessary and strictly weaker. The three equations

$$\partial y = iy, \quad \partial y = i\omega^{-1}y, \quad \partial y = i\omega^{-2}y$$

have, respectively: only the zero solution (the accumulated phase of 1 is ω); only the zero solution, although ω^{-1} is infinitesimal (its accumulated phase is $\log \omega$, infinite); and the solutions $c E(-i\omega^{-1})$, $c \in \mathbb{C}$. A sibling statement that “ $b \in \mathfrak{m}$ is necessary” is a strictly weaker and different result.

Remark 8.6 (The real coefficient never obstructs). Adding any real coefficient a introduces no further obstruction in the full class: multiply the solution by $\exp(\mathcal{I}_0(a))$. Thus

$$\partial y = \left(\omega^3 + \frac{i}{\omega^2} \right) y \quad \text{has solutions} \quad y = c \exp(\omega^4/4) E(-i/\omega),$$

while replacing i/ω^2 by i/ω leaves only $y = 0$; and $\partial y = (\omega + i\omega^{-2})y$ has solutions $c \exp(\omega^2/2) E(-i/\omega)$. The real growth rate may be arbitrarily large. That assertion can fail in a *restricted workspace*, because A or e^A may leave the workspace; Section 19 computes that failure exactly.

Remark 8.7 (A one-dimensional equation with a zero-dimensional solution space). A one-dimensional linear equation need not have a one-dimensional nonzero solution space *inside this field*. The familiar dimension count is an upper bound until a sufficiently large differential extension is supplied; Section 21 supplies one.

8.3 No intrinsic oscillator: three independent proofs

Corollary 8.8 (The missing oscillator). *For $\lambda \in \mathbb{C}$, the equation $\partial y = \lambda y$ has a nonzero solution in $\mathbf{No}[i]$ if and only if $\lambda \in \mathbb{R}$, in which case the solutions are exactly $c e^{\lambda\omega}$ with $c \in \mathbb{C}$. In particular $\partial y = iy$ and $\partial y = -iy$ have only the zero solution, and for every ordinary real $\lambda \neq 0$ so do $\partial y = i\lambda y$ and $\partial^2 y + \lambda^2 y = 0$. In particular $\partial^2 y + y = 0$ has only $y = 0$ in $\mathbf{No}[i]$, while $\partial^2 y - y = 0$ has all solutions $A e^\omega + B e^{-\omega}$ with $A, B \in \mathbb{C}$.*

First proof, from the criterion. Write $\lambda = \alpha + i\beta$. An accumulated phase of β is $\beta\omega$, which is finite exactly when $\beta = 0$; apply Theorem 8.4. When $\beta = 0$, dividing a solution by $e^{\alpha\omega}$ gives derivative zero. For the second-order equation, factor $\partial^2 + \lambda^2 = (\partial - i\lambda)(\partial + i\lambda)$ over $\mathbf{No}[i]$; both factors are injective by the first part, so their composite is. $\square$

Second proof, from smallness alone. Suppose $\partial y = i\beta y$ with $\beta \in \mathbb{R} \setminus \{0\}$ and $y \neq 0$. Conjugation gives $\partial \bar{y} = -i\beta \bar{y}$, so $\partial(y\bar{y}) = 0$ and $y\bar{y}$ is a positive real constant. Dividing by its positive square root gives $u + iv$ with

$$u^2 + v^2 = 1, \quad \partial u = -\beta v, \quad \partial v = \beta u.$$

Both u and v are finite, so both derivatives are infinitesimal by Theorem 3.8; dividing by the nonzero real constant β then makes both u and v infinitesimal, so $u^2 + v^2$ is infinitesimal and cannot equal 1. $\square$

Third proof, an energy argument for the second-order equation. First let $f \in \mathbf{No}$ be real with $\partial^2 f + f = 0$. The energy $E = f^2 + (\partial f)^2$ satisfies $\partial E = 2\partial f(f + \partial^2 f) = 0$, so E is a nonnegative real constant; hence f and ∂f are finite. By Theorem 3.8 both ∂f and $\partial^2 f$ are infinitesimal, and since $f = -\partial^2 f$ the element f is infinitesimal too. Then E is both real and infinitesimal, so $E = 0$, and positivity gives $f = 0$. For complex f , apply this to its real and imaginary parts. $\square$

The three proofs are genuinely different and all three survive. The first uses the whole criterion. The second uses only the constant field, the H -field positivity clause and the normalization $\partial\omega = 1$ — not the finite-angle parametrization, not surjectivity, and no logarithm — so the obstruction is provably not an artefact of a branch choice or of a chosen phase convention. The third uses only conservation of energy plus smallness, and isolates the role of smallness in the second-order equation; the factorization proof, however, scales more efficiently to general operators (Theorem 13.2).

Warning 8.9 (What the nonexistence statements are about). All of these are statements about surcomplex *number* solutions under the fixed ∂ . They do not deny ordinary oscillatory functions, do not say that the coordinate functions $\sin z$, $\cos z$ fail their ordinary differential equations, and do not deny algebraically defined global surcomplex phases. They do say that $\sin \omega$ and $\cos \omega$ are not Berarducci–Mantova differential elements of $\mathbf{No}$ in the sense of Theorem 3.6: the system $c^2 + s^2 = 1$, $c' = -s$, $s' = c$ has no solution in $\mathbf{No}$. See Theorem 8.10 and Section 11.3.

Corollary 8.10 (The unit-circle version). *For $x \in \mathbf{No}$, the system*

$$c^2 + s^2 = 1, \quad \partial c = -(\partial x)s, \quad \partial s = (\partial x)c$$

has a solution $c, s \in \mathbf{No}$ if and only if x is finite.

Proof. If c, s solve it then $z = c + is \neq 0$ satisfies $\partial z = i(\partial x)z$, so by Theorem 8.4 ∂x has a finite accumulated phase; all those are $x + r$ with $r \in \mathbb{R}$, so x is finite. Conversely, for finite x take the coordinates of $\text{cis}(x)$ and apply equation (6.7). $\square$

This concerns the derivatives of the *values* c, s . An externally differentiable sine function is a different object, and its derivative with respect to its argument must not be substituted for ∂ without a proved chain rule (Theorem 4.1).

Corollary 8.11 (A Riccati transfer). *The only solutions in $\mathbf{No}[i]$ of $\partial u = 1 + u^2$ are $u = i$ and $u = -i$; in particular there is none in $\mathbf{No}$.*

Proof. Both constants are solutions. For any other solution, $v = (u - i)/(u + i)$ is defined and nonzero, and

$$v^\dagger = \frac{\partial u}{u - i} - \frac{\partial u}{u + i} = \frac{2i \partial u}{u^2 + 1} = 2i,$$

contradicting Theorem 8.8. $\square$

This is not a general nonlinear solvability theorem. It is a direct way to transfer the first-order phase obstruction to a familiar nonlinear equation.

Remark 8.12 (Algebraic closure is not differential closure). $\mathbf{No}[i]$ is algebraically closed and ∂ is additively surjective, yet $\partial y = iy$ has no nonzero solution. These are three different closure properties: $X^2 + 1 = 0$ has a solution, $\partial y = f$ has a solution, and $\partial y = iy$ with $y \neq 0$ does not. Adjoining an algebraic element cannot repair the last, since algebraic closedness leaves no proper algebraic extension; and taking an algebraic root cannot cancel a nonzero obstruction, because every nonzero integer multiple of a nonzero element of $\mathcal{P}$ is nonzero (Theorem 9.4). A genuine differential extension must introduce a transcendental element, and Section 21 constructs several. The nonzero requirement can itself be written differentially, by adding an unknown v and the equation $yv = 1$; so this is an explicit failure of differential closedness, not a complaint about an undefined expression such as $e^{i\omega}$.

9 Gauge normal form and the obstruction sequence

9.1 Gauge equivalence and the unique normal form

A change of unknown $y = gz$ with $g \in \mathbf{No}[i]^\times$ sends

$$\partial y = qy \quad \text{to} \quad \partial z = (q - g^\dagger)z. \quad (9.1)$$

Call $q, r \in \mathbf{No}[i]$ *gauge equivalent* when $q - r = g^\dagger$ for some $g \in \mathbf{No}[i]^\times$. A gauge transformation changes the coefficient by a logarithmic derivative, not by an arbitrary primitive, and it does not change whether a nonzero solution exists in the base field. No differential-module terminology is needed to use it.

Theorem 9.1 (Purely infinite phase normal form, with uniqueness). *The map $\Phi : \mathbf{No}[i] \rightarrow \mathcal{P}$, $\Phi(q) = \mathfrak{o}(\text{Im } q)$, is additive and $\mathbb{R}$ -linear, is surjective, and satisfies*

$$\Phi(q) = 0 \iff q \text{ is a logarithmic derivative.}$$

Two coefficients are gauge equivalent if and only if their obstructions agree. Explicitly, writing $q = a + ib$ and $\mathcal{I}_0(b) = P + \eta$ with $P = \Phi(q) \in \mathcal{P}$ and $\eta \in \mathfrak{m}$, the gauge

$$g = \exp(\mathcal{I}_0(a)) E(i\eta)$$

transforms $\partial y = qy$ into

$$\partial z = i(\partial P)z. \quad (9.2)$$

Every equation $\partial y = qy$ is therefore gauge equivalent to exactly one equation of the form equation (9.2) with $P \in \mathcal{P}$, and the canonical parameter is $P = \Phi(q)$.

Proof. Linearity and surjectivity are Theorem 7.5(vi) composed with Im ; the kernel description is Theorem 8.4. The logarithmic derivative of the displayed g is $a + i\partial\eta$, and $b = \partial P + \partial\eta$, so equation (9.1) gives equation (9.2). If $q_1 - q_2 = g^\dagger$ then $\text{Im}(q_1 - q_2) \in \mathcal{A}$ and linearity of $\mathfrak{o}$ makes the obstructions equal; conversely equal obstructions put $\text{Im}(q_1 - q_2)$ in $\mathcal{A}$, and Theorem 8.1 supplies a g with $g^\dagger = q_1 - q_2$. Uniqueness: $\Phi(i\partial P) = \mathfrak{o}(\partial P) = P$ by Theorem 7.5(vi), so distinct P represent distinct classes. $\square$

The normal form says more than “no solution”. It isolates the precise accumulated phase that cannot be represented by a surcomplex direction: the real growth factor and the infinitesimal phase residue are exactly the removable parts, and the purely infinite part is exactly the irremovable one. The parameter P is *not* an angular value assigned to an infinite input; it records the part of an accumulated phase that no existing surcomplex phase can absorb, the absorbed factor carrying only the infinitesimal residue η (Theorem 5.1).

Corollary 9.2 (Obstruction exact sequence). *The following is an exact sequence of class groups, multiplicative before $\dagger$ and additive after it:*

$$1 \longrightarrow \mathbb{C}^\times \longrightarrow \mathbf{No}[i]^\times \xrightarrow{\dagger} (\mathbf{No}[i], +) \xrightarrow{\Phi} (\mathcal{P}, +) \longrightarrow 0. \quad (9.3)$$

Restricted to $\mathbf{No}$ it reads $0 \rightarrow \mathcal{A} \rightarrow \mathbf{No} \xrightarrow{\mathfrak{o}} \mathcal{P} \rightarrow 0$.

Proof. $\ker(\dagger) = \mathbb{C}^\times$ and $\text{im}(\dagger) = \mathbf{No} + i\mathcal{A}$ by Theorem 8.1; that class is $\ker \Phi$ by Theorem 8.4; and Φ is onto by Theorem 9.1. $\square$

Warning 9.3 (The sequence is elementwise). The word “class” in equation (9.3) is intentional, and the sequence is shorthand for the displayed kernel and image identities *about elements*. No set or class of all equivalence classes is formed, and $\mathcal{P}$ appears precisely so that no coset of $\mathcal{O}$ has to be an object (Theorem 2.1). Every map in the sequence acts on set-sized elements, and Φ together with the representative $i\partial P$ carries all the content.

Corollary 9.4 (Tensor, dual, and no finite-order obstruction). *Under multiplication of scalar solution equations — equivalently, tensor product of the corresponding rank-one objects, represented by addition of coefficients — the obstructions add; under inversion, that is dualization, they negate. For an integer $n \neq 0$,*

$$\Phi(nq) = n\Phi(q), \quad \Phi(nq) = 0 \iff \Phi(q) = 0.$$

Every nonzero gauge class has infinite order. Hence raising a putative solution of an obstructed equation to a finite power does not remove the obstruction.

Proof. The coefficient of a product of nonzero solutions is the sum of their logarithmic derivatives, and of an inverse is the negative. The rest is $\mathbb{R}$ -linearity of Φ together with the absence of additive torsion in the real vector class $\mathcal{P}$. $\square$

The integer linearity just recorded — rather than real linearity — is what drives the differential Galois arguments of Sections 21 and 21.4: rational functions in solutions involve integral monomial exponents, so the relations that matter are integer relations. This rank-one classification is a differential gauge classification; it is not the holomorphic line-bundle or divisor classification of the repository’s global analytic reports, and no theory of higher-rank differential Galois groups is needed to state it.

10 Sharp thresholds and worked scales

10.1 The support threshold over a real-exponent Hahn field

Fix $t = \omega^{-1}$ and, for an additive subgroup $\Gamma \subseteq \mathbb{R}$, embed the Hahn fields

$$H_\Gamma = \mathbb{R}((t^\Gamma)), \quad K_\Gamma = \mathbb{C}((t^\Gamma)) = H_\Gamma[i]$$

into **No** and **No**[i] by Conway monomials. On these fields the real-exponent rule of equation (3.1) applies, so $\partial(t^\alpha) = -\alpha t^{\alpha+1}$ for ordinary real α and $\partial = -t^2 d/dt$ (Theorem 2.2 applies: this formula is for real exponents only). The case $\Gamma = \mathbb{R}$ gives the largest real-exponent field; the cases $\Gamma = \mathbb{Z}$ and $\Gamma = \mathbb{Q}$ recur in Part IV.

Theorem 10.1 (Sharp support threshold). *Let $\Gamma \subseteq \mathbb{R}$ be an additive subgroup and let $b = \sum_{\alpha \in S} b_\alpha t^\alpha \in H_\Gamma$ with S well ordered. A primitive of b in **No** is*

$$B = \sum_{\substack{\alpha \in S \\ \alpha \neq 1}} \frac{b_\alpha}{1-\alpha} t^{\alpha-1} + b_1 \log \omega, \quad (10.1)$$

and for every $p \in \mathbf{No}$ the equation $\partial y = (p + ib)y$ has a nonzero solution in **No**[i] if and only if

$$b = 0 \quad \text{or} \quad \min \text{supp}(b) > 1. \quad (10.2)$$

Equivalently, for real $f \in H_\Gamma$ one has $f \in \mathcal{A}$ if and only if $f = 0$ or $v(f) > 1$, where v is the least-exponent valuation. In the soluble case the sum in equation (10.1) is the unique infinitesimal accumulated phase of b .

Proof. Shifting a well-ordered support by -1 preserves well ordering, and omitting one exponent while adding one logarithmic term preserves summability, so equation (10.1) is a legitimate element of **No**; the real-exponent rule and (BM3) give $\partial B = b$.

If every exponent exceeds 1, all powers in B have positive exponent, there is no logarithmic term, and B is infinitesimal. If the least exponent α_0 is less than 1, the corresponding primitive term has negative exponent and nonzero coefficient; it dominates the remaining power terms and also $\log \omega$, since $\log \omega \prec \omega^\delta$ for every real $\delta > 0$. So B is infinite. If the least exponent is exactly 1, the logarithmic term has nonzero coefficient while all remaining power terms are infinitesimal, so B is again infinite. Adding a real constant changes none of these conclusions. Apply Theorem 8.4. $\square$

The logarithmic comparison used here is a standard consequence of the surreal exponential; equivalently it follows from the normal-form description of $\log \omega$ [1, 7]. The threshold is a *strict* inequality at exponent 1, and the strictness is not an artefact of the statement: it is the $\log \omega$ term.

Corollary 10.2 (Power threshold). *For $p \in \mathbb{R}$, the equation $\partial y = i\omega^{-p}y$ has a nonzero surcomplex solution exactly when $p > 1$, and then all solutions are*

$$y = c E\left(\frac{i\omega^{1-p}}{1-p}\right), \quad c \in \mathbb{C}.$$

Equivalently $\omega^{-p} \in \mathcal{A} \iff p > 1$.

Proof. For $p \neq 1$ a primitive of ω^{-p} is $\omega^{1-p}/(1-p)$, which is infinitesimal for $p > 1$ and infinite for $p < 1$; for $p = 1$ a primitive is $\log \omega$, infinite. Apply Theorems 8.4 and 10.1. $\square$

Corollary 10.3 (Iterated-logarithm thresholds at every finite depth). *With $\ell_0 = \omega$, $\ell_{n+1} = \log \ell_n$ and, for an ordinary integer $n \geq 0$ and real p ,*

$$b_{n,p} = \frac{1}{\ell_0 \ell_1 \cdots \ell_{n-1} \ell_n^p}$$

(the prefix product empty for $n = 0$), the equation $\partial y = ib_{n,p}y$ has a nonzero solution in $\mathbf{No}[i]$ exactly when $p > 1$. An accumulated phase is $\ell_n^{1-p}/(1-p)$ for $p \neq 1$ and ℓ_{n+1} for $p = 1$.

Proof. By equation (3.2), $\partial \ell_n = 1/(\ell_0 \cdots \ell_{n-1})$; differentiate the displayed expressions using the real power and logarithm rules of equation (3.1). For $p > 1$ the accumulated phase is infinitesimal; for $p \leq 1$ it is infinite, each ℓ_n being positive infinite. Apply Theorem 8.4. $\square$

These are finite iterated constructions: no infinite product and no transfinite logarithm tower is used. The tests resemble familiar convergence thresholds for improper integrals, but their proof here is algebraic differentiation and comparison of surreal sizes. No improper real integral, no limiting process and no integration to a numerical endpoint occurs anywhere.

10.2 Why the criterion cannot be globalized by valuation

Example 10.4 (The separating pair). Put $L = \log \omega$. Then

$$\partial(\log L) = \frac{1}{\omega L}, \quad \partial(-L^{-1}) = \frac{1}{\omega L^2}.$$

The first accumulated phase, $\log \log \omega$, is infinite; the second, $-1/\log \omega$, is infinitesimal. Therefore

$$\partial y = \frac{i}{\omega \log \omega} y \text{ has no nonzero solution,} \quad \partial y = \frac{i}{\omega (\log \omega)^2} y \text{ does.} \quad (10.3)$$

Both coefficients have $v(\operatorname{Im} a) > 1$ in the full Hahn valuation normalized by $v(t) = 1$: indeed $v(L) < 0$, so $v(t/L) = 1 - v(L) > 1$.

Warning 10.5 (The threshold theorem is restricted, and must stay restricted). Theorem 10.1 is stated over a real-exponent Hahn field and the restriction is substantive. By Theorem 10.4, even $v(\operatorname{Im} a) > 1$ is *not* sufficient over all of $\mathbf{No}$: the missing information lies in the slower logarithmic scale, which a real-exponent workspace cannot see. A computer-algebra backend must not apply Theorem 10.1 outside its declared coefficient domain, and the support threshold must not be merged into the general- Γ resolvent of Theorem 18.6, whose hypotheses are different. A primitive-based representation handles all of these scales uniformly; a single power cutoff does not.

10.3 An exact decision procedure for rational coefficients

The obstruction map is a semantic classification, not a universal effective procedure. For rational functions over the ordinary constants the condition is nevertheless decidable by finite algebra.

Theorem 10.6 (Rational phase criterion). *Let $b(X) \in \mathbb{R}(X)$ be nonzero, in reduced form $b = P/Q$. Then*

$$b(\omega) \in \mathcal{A} \iff \deg Q - \deg P \geq 2. \quad (10.4)$$

Hence for $q(\omega) = a(\omega) + ib(\omega)$ with $a, b \in \mathbb{R}(X)$ this is exactly the criterion for a nonzero solution of $\partial y = q(\omega)y$. The case $b = 0$ is always admissible.

Table 3: Distinguishable situations at two and three scales. $L = \log \omega$.

Imaginary coefficient b	An accumulated phase B	Nonzero solution of $\partial y = iby$?
1	ω	No
$t = \omega^{-1}$	L	No, although b is infinitesimal
$t^2 = \omega^{-2}$	$-t$	Yes: $E(-it)$
$t^{3/2}$	$-2t^{1/2}$	Yes: $E(-2it^{1/2})$
$1/(\omega L)$	$\log L$	No: a slower divergence still obstructs
$1/(\omega L^2)$	$-1/L$	Yes: $E(-i/L)$
$\partial \eta$ with $\eta \in \mathfrak{m}$	η	Yes: $E(i\eta)$
$\omega^3 + i\omega^{-2}$ (whole coefficient)	—	Yes: $c \exp(\omega^4/4)E(-i/\omega)$

Proof. Let $p = \deg Q - \deg P$. Expanding at infinity gives $b(\omega) = ct^p +$ (higher integer powers of t) with $c \in \mathbb{R}^\times$. If $p \geq 2$, write $b(\omega) = \sum_{n \geq 2} b_n t^n$; then $\eta = -\sum_{n \geq 2} b_n t^{n-1}/(n-1)$ is infinitesimal and differentiates strongly to $b(\omega)$, so $b(\omega) \in \mathcal{A}$. If $p \leq 1$, the leading term gives $|b(\omega)| \geq |c|t^p/2$; were $b(\omega) \in \mathcal{A}$, real linearity and the convexity of Theorem 7.5(iii) would force $t^p \in \mathcal{A}$, contradicting Theorem 10.2. The assertion for q is Theorem 8.4. $\square$

Table 4: The rational criterion on examples.

$b(\omega)$	Finite accumulated phase?	Reason, or the phase
0	yes	0
1	no	ω is infinite
$1/\omega$	no	$\log \omega$ is infinite
$1/\omega^2$	yes	$-1/\omega$
$\omega/(1 + \omega^2)$	no	degree gap 1
$1/(1 + \omega^2)$	yes	degree gap 2
$(1 - \omega^2)/(1 + \omega^2)^2$	yes	$\omega/(1 + \omega^2)$

Warning 10.7 (The theorem is over $\mathbb{R}(X)$; the implementation is over $\mathbb{Q}(X)$). Theorem 10.6 is an algebraic statement over $\mathbb{R}(X)$. The delivered executable restricts its input to exact rational coefficients so that polynomial arithmetic and degree comparison are effective, and it *raises* on inputs outside $\mathbb{Q}(\text{variable})$ — other functions, nonrational coefficients, unbound parameters, non-numeric tokens — rather than reporting them as mathematical nonexistence. No algorithm for an arbitrary real coefficient is asserted. See Sections D and 29.1.

Example 10.8 (An algebraic solution with a small angular rate). For $b = 1/(1 + \omega^2)$ the unit-modulus *algebraic* number

$$y = \frac{1 + i\omega}{\sqrt{1 + \omega^2}}$$

satisfies $\partial y = iby$, since

$$y^\dagger = \frac{i}{1 + i\omega} - \frac{\omega}{1 + \omega^2} = \frac{i}{1 + \omega^2}.$$

This example needs neither a named inverse trigonometric function nor an infinite-phase exponential.

Example 10.9 (A rational unit-circle (Cayley) coordinate). With $t = \omega^{-1}$ put $u = (1 + it)/(1 - it)$. Then $u\bar{u} = 1$, $\text{st}(u) = 1$, and

$$u^\dagger = \frac{-2it^2}{1 + t^2}.$$

Its infinitesimal phase residue is $2 \arctan(t)$, understood as the infinitesimal Taylor evaluation of Theorem 6.1; indeed $\partial(2 \arctan t) = -2t^2/(1 + t^2)$. So an admissible solution need not be stored as an infinite sum: an exact rational expression can already carry the required phase, and its full expansion is optional. The coefficient domain here is Gaussian rationals in t .

Example 10.10 (A solvable coefficient with an infinite Hahn expansion). For $q = i/\omega^2 = it^2$ the normalized solution is

$$y = E(-it) = 1 - it - \frac{t^2}{2} + \frac{it^3}{6} + \frac{t^4}{24} - \cdots,$$

whose support lies in $\mathbb{N}$, so the family is summable. The identity $\partial y = it^2 y$ is exact, not asymptotic. A finite truncation $E_N(t) = \sum_{n \leq N} (-i)^n t^n / n!$ has the exact residual

$$\partial E_N - it^2 E_N = -i \frac{(-i)^N}{N!} t^{N+2}, \quad (10.5)$$

which is what a finite verifier can check; it is not the whole solution.

11 The maximal derivation-compatible exponential strip

11.1 A surjective partial exponential

Set

$$\mathcal{S} := \mathbf{No} + i\mathcal{O} \subset \mathbf{No}[i].$$

This is an additive subgroup and a real vector subclass, and it is ∂ -stable precisely because $\partial\mathcal{O} \subseteq \mathfrak{m}$ (Theorem 3.8). It is *not* a subring: $\omega \in \mathcal{S}$ and $i \in \mathcal{S}$, while $i\omega \notin \mathcal{S}$. Note carefully that $\mathcal{S}$ involves $\mathcal{O}$, whereas the logarithmic-derivative image of Theorem 8.1 involves $\partial\mathcal{O} = \mathcal{A}$; see Theorem 5.4. Define

$$E_{\mathcal{S}}(x + i\theta) = \exp(x) \text{cis}(\theta) = \exp(x) e^{i \text{st}(\theta)} E(i(\theta - \text{st}(\theta))), \quad x \in \mathbf{No}, \theta \in \mathcal{O}. \quad (11.1)$$

Theorem 11.1 (The strip exponential). $E_{\mathcal{S}} : (\mathcal{S}, +) \rightarrow (\mathbf{No}[i]^\times, \cdot)$ is a surjective group homomorphism with kernel exactly $2\pi i\mathbb{Z}$, it extends the ordinary complex exponential, and it satisfies

$$\partial E_{\mathcal{S}}(w) = E_{\mathcal{S}}(w) \partial w \quad (w \in \mathcal{S}). \quad (11.2)$$

Hence $\mathcal{S}/(2\pi i\mathbb{Z}) \cong \mathbf{No}[i]^\times$ as abelian groups. Moreover $\partial\mathcal{S} = \mathbf{No} + i\mathcal{A}$.

Proof. Multiplicativity follows from the real exponential law, the additivity of standard part on finite elements and the group law of Theorem 6.5; the derivative identity follows from (BM4) and equation (6.7), noting that $\text{st}(\theta)$ and $e^{i \text{st}(\theta)}$ are differential constants. If $E_{\mathcal{S}}(x + i\theta) = 1$, taking moduli gives $\exp x = 1$, so $x = 0$, and then $\text{cis} \theta = 1$ gives $\theta \in 2\pi\mathbb{Z}$; conversely these lie in the kernel. For surjectivity, write $z = \rho c E(i\eta)$ by Theorem 6.7 and choose an ordinary real r with $e^{ir} = c$; then $z = E_{\mathcal{S}}(\log \rho + i(r + \eta))$. This uses the ordinary circle only, not a branch on the whole surcomplex plane. Finally $\partial\mathbf{No} = \mathbf{No}$ and $\partial\mathcal{O} = \mathcal{A}$ give the image assertion. $\square$

So $E_{\mathcal{S}}$ reaches every nonzero surcomplex number while restricting the allowable *exponents*: surjectivity does not require an exponent for every element of $\mathbf{No}[i]$. Its kernel consists of ordinary integer periods, not of a surreal integer part. Every nonzero z has strip logarithms, all differing by an ordinary period $2\pi in$, and all with the *same* intrinsic derivative $z^\dagger$ — a conclusion that is canonical even when no choice of logarithm branch is.

11.2 Maximality is pointwise

Theorem 11.2 (Exact domain of the exponential differential equation). *For $w = x + iB \in \mathbf{No}[i]$ with $x, B \in \mathbf{No}$, the following are equivalent.*

- (1) *There exists $y \in \mathbf{No}[i]^\times$ with $y^\dagger = \partial w$.*
- (2) *B is finite.*
- (3) *$w \in \mathcal{S}$.*

Consequently no map taking nonzero surcomplex values and satisfying equation (11.2) at every point of its domain can extend $E_{\mathcal{S}}$ to even one exponent outside $\mathcal{S}$.

Theorem 11.3 (Maximality restated). *Equivalently, $\partial^{-1}(\mathcal{A}) = \mathcal{O}$: the condition in Theorem 11.2(1) is $\partial B \in \mathcal{A}$, and Theorem 7.5(viii) identifies its solution class as $\mathcal{O}$.*

Proof of Theorems 11.2 and 11.3. By Theorem 8.1, condition (1) is equivalent to $\text{Im}(\partial w) = \partial B \in \mathcal{A}$. Every accumulated phase of ∂B is $B + r$ with $r \in \mathbb{R}$, so by Theorem 8.4 such a phase is finite exactly when B is finite; equivalently $\partial^{-1}(\mathcal{A}) = \mathcal{O}$ by Theorem 7.5(viii). The remaining equivalence is the definition of $\mathcal{S}$, and Theorem 11.1 supplies a value at every point of $\mathcal{S}$. The reasoning is pointwise: it excludes even an isolated proposed value outside the strip, irrespective of any group law, homomorphism axiom, or regularity imposed on the map. $\square$

The claimed strength is exactly this pointwise form. It is stronger than the failure of one candidate formula: at an excluded point there is *no* admissible nonzero value at all. On the other hand the differential law alone is not claimed to pin down every normalization on $\mathcal{S}$: ratios of admissible values are differential constants, and equation (11.1) is the specified canonical construction with the ordinary complex normalization. A symbolic system should therefore attach the domain predicate $\text{Im } w \in \mathcal{O}$ to $E_{\mathcal{S}}$ rather than advertise an exponential *operation* on a differential subfield.

11.3 No globally compatible exponential, in three strengths

Corollary 11.4 (Phase-independent nonexistence). *There is no map $\Psi : \mathbf{No}[i] \rightarrow \mathbf{No}[i]^\times$ with*

$$\partial \Psi(z) = \Psi(z) \partial z \quad \text{for every } z \in \mathbf{No}[i]. \quad (11.3)$$

No homomorphism, continuity, analyticity, conjugation or normalization hypothesis is needed. Likewise there is no map $E : \mathbf{No} \rightarrow \mathbf{No}[i]^\times$ with $\partial(E(x)) = iE(x)\partial x$ for every $x \in \mathbf{No}$, and no extension of the finite-angle cis to all real surreals can satisfy $\partial \text{cis}(x) = i(\partial x) \text{cis}(x)$ everywhere.

Proof. Evaluate equation (11.3) at $z = i\omega$: the value $y = \Psi(i\omega)$ would be nonzero with $\partial y = iy$, contradicting Theorem 8.8 (or Theorem 3.11, or Theorem 11.2). For the second statement evaluate at $x = \omega$. $\square$

Because no structural hypothesis is used, this cannot be repaired by a cleverer choice of infinite circular phase. It is stronger in its phase-independence than an obstruction proved for one chosen canonical exponential, and it is compatible with the repository's existing derivation-obstruction theorem [18], which it strengthens rather than contradicts.

The defect can also be quantified, uniformly over all conventions of one natural shape.

Proposition 11.5 (Character-independent invariant defect). *Let $\chi : (\mathcal{P}, +) \rightarrow \mathbb{T}(\mathbf{No})$ be any group homomorphism, and for $y = P + r + \eta$ with $P \in \mathcal{P}$, $r \in \mathbb{R}$, $\eta \in \mathfrak{m}$ define the phase extension*

$$E_\chi(x + iy) = \exp(x) e^{ir} E(i\eta) \chi(P).$$

Put $\delta_\chi(z) = E_\chi(z)^\dagger - \partial z$ for $z = x + iy$. Then

$$\mathfrak{o}(\text{Im } \delta_\chi(z)) = -\text{pp}(y). \quad (11.4)$$

Proof. By Theorem 6.3 the individual value $\chi(P)$ is $c_P E(i\eta_P)$ with $c_P \in \mathbb{T}(\mathbb{C})$ and $\eta_P \in \mathfrak{m}$; we differentiate that value only, and never differentiate χ as a function. Its logarithmic derivative is $i\partial\eta_P$, so $E_\chi(z)^\dagger = \partial x + i\partial\eta + i\partial\eta_P$ and $\delta_\chi(z) = i(\partial\eta_P - \partial P)$. Apply $\mathfrak{o}$, using $\partial\eta_P \in \mathcal{A}$ and $\mathfrak{o}(\partial P) = P$ from Theorem 7.5(vi). $\square$

So no choice of character repairs the obstruction at a purely infinite imaginary exponent: the defect may change with χ , but its purely infinite class cannot. The trivial character is one admissible convention; nothing here assumes that every conceivable global exponential has this form.

Warning 11.6 (What is not being denied). Ehrlich and Kaplan construct canonical global surcomplex exponential and circular functions using an initial integer part, with kernel described by $2\pi i$ times that integer part [8, §11]; the repository carries that construction alongside its phase-extension discussion [17]. These are algebraic and structural choices, not assertions of compatibility with equation (11.2). The no-go theorems above do not invalidate those functions, do not say that a symbol such as $\text{Exp}(i\omega)$ cannot be assigned a surcomplex value, and do not settle their separate robustness questions. They say only that *whatever* nonzero value is assigned, its Berarducci–Mantova derivative cannot equal i times itself. The trigonometry report's global phase constructions likewise do not assert the scalar chain rule and are not refuted.

Example 11.7 (A phase convention is not a differential solution). A convention that discards the purely infinite part of an angle may assign phase 1 to ω . That value is algebraically meaningful under its own convention, but its intrinsic derivative is $\partial 1 = 0$, whereas differential exponential compatibility at $i\omega$ would require $\partial 1 = i$. The constant-value convention makes the distinction especially transparent: a nonzero ordinary constant has derivative zero, not i times itself. So a phase convention and a differential solution are different objects even when both are written $e^{i\omega}$.

Remark 11.8 (There is no finite clock). Ordinary sine and cosine lifts on finite halos obey the chain rule with the factor ∂z , and for finite z that factor is infinitesimal: there is no finite surreal z with $\partial z = 1$. The forbidden oscillator would require a genuinely infinite intrinsic clock.

Remark 11.9 (The formal exponential series is not summable at ω). The formal series $F(Z) = \sum_{n \geq 0} (iZ)^n / n! \in \mathbb{C}[[Z]]$ satisfies $\partial_Z F = iF$, an ordinary formal differential equation with no bearing on $\partial y = iy$ until one evaluates. At $Z = \omega$ the terms carry monomials $\omega^n = t^{-n}$, whose additive valuation support $\{0, -1, -2, \dots\}$ is *not* well ordered; the family therefore has no Hahn sum and does not define an element of $\mathbf{No}[i]$. The same holds for $\sum_{n \geq 0} (i\omega)^n / n! = \sum_n (i^n / n!) t^{-n}$, and choosing a larger exponent group does not repair that support defect. At an infinitesimal argument the series *is* summable, but then the chain rule involves the derivative of that infinitesimal argument rather than the value 1. This is how formal analytic solutions are reconciled with the obstruction.

12 Inhomogeneous equations: existence is a separate question

Theorem 12.1 (Variation of constants). *Suppose $y_0 \neq 0$ satisfies $\partial y_0 = qy_0$ in $\mathbf{No}[i]$ — equivalently, by Theorem 8.4, suppose $\Phi(q) = 0$ and take $y_0 = \exp(\mathcal{I}_0 a)E(i\mathcal{I}_0 b)$ for $q = a + ib$. Then for every $f \in \mathbf{No}[i]$ all solutions of $\partial y - qy = f$ are*

$$y = y_0(\mathcal{I}_0(f/y_0) + c), \quad c \in \mathbb{C}. \quad (12.1)$$

In particular $\partial - q$ is surjective on $\mathbf{No}[i]$ in this case. If the homogeneous equation has no nonzero solution, then for a given f the inhomogeneous equation has at most one solution in $\mathbf{No}[i]$.

Proof. Put $y = y_0 u$; since $\partial y_0 = qy_0$ the equation becomes $y_0 \partial u = f$, and (BM6) with Theorem 7.1 gives $u = \mathcal{I}_0(f/y_0) + c$ with $c \in \mathbb{C}$, these being all primitives by Theorem 3.5. Substitution verifies equation (12.1). For the last clause, the difference of two solutions would solve the obstructed homogeneous equation. $\square$

Warning 12.2 (“At most one” must not be replaced by “none”). The absence of a nonzero homogeneous solution does *not* imply inhomogeneous nonexistence. Particular solutions may exist and then be unique:

$$\begin{aligned} (\partial - i)i &= 1, & (\partial - i)(i\omega + 1) &= \omega, \\ \partial y - iy &= 1 \text{ has the unique } y = i, & \partial y = iy - i &\text{ has the solution } y = 1, \\ \partial^2 y + y &= \omega^2 \text{ has the unique } y = \omega^2 - 2, & \partial^2 y + y &= \omega^3 \text{ has the unique } y = \omega^3 - 6\omega, \end{aligned}$$

existence verified by differentiation and uniqueness by Theorem 8.8. An integrating-factor expression using an undefined value $e^{i\omega}$ is unnecessary and would obscure the correct reasoning; the integrating factor simply does not exist in $\mathbf{No}[i]$, so variation of constants yields no existence conclusion there. No general inhomogeneous existence theorem is claimed outside the admissible locus $\Phi(q) = 0$; see Section 31. Note also that having a trivial kernel is not the same as failing to be surjective: Theorem 18.6 exhibits an operator that is bijective on a workspace although its homogeneous equation has no nonzero solution anywhere in $\mathbf{No}[i]$.

Proposition 12.3 (Polynomial forcing, and a terminating inverse). *Let $f \in \mathbb{C}[\omega]$ of degree N and $\lambda \in \mathbb{C}^\times$. Then*

$$y = - \sum_{k=0}^N \frac{\partial^k f}{\lambda^{k+1}} \quad (12.2)$$

solves $(\partial - \lambda)y = f$; for $f = 0$ read the sum as 0. If λ is nonreal this is the unique solution in $\mathbf{No}[i]$; if λ is real, all solutions are obtained by adding $c \exp(\lambda\omega)$, $c \in \mathbb{C}$. In particular, with $\lambda = i$,

$$(\partial - i)^{-1} f = i \sum_{k=0}^N (-i)^k \partial^k f. \quad (12.3)$$

More generally, for real μ and $P \in \mathbb{C}[X]$, $P(\partial)(e^{\mu\omega} q(\omega)) = e^{\mu\omega} P(\mu + D)q(\omega)$ where D is ordinary polynomial differentiation; writing $P(\mu + X) = X^r Q(X)$ with $Q(0) \neq 0$, the operator $Q(D)^{-1}$ is a finite nilpotent expansion on a finite-dimensional polynomial space and repeated polynomial integration solves the remaining D^r equation. Hence every forcing term that is a finite sum of real-exponent exponential polynomials has a particular solution of the same kind.

Proof. Applying $\partial - \lambda$ to the finite sum equation (12.2) makes all intermediate terms cancel, and $\partial^{N+1}f = 0$ leaves f . For equation (12.3) use $\partial - i = -i(1 + i\partial)$ together with $\partial^{N+1}f = 0$, so the finite geometric identity applies: this is a true terminating formula, with no infinite summation and no asymptotic truncation. On ordinary polynomials in ω the operator ∂ is ordinary polynomial differentiation, since $\partial\omega = 1$; the displayed conjugation identity then follows from (BM4), and Theorems 8.8 and 13.2 give the uniqueness statements. $\square$

13 All homogeneous constant-coefficient equations

Every ordinary complex number is a differential constant, so for $P(T) \in \mathbb{C}[T]$ the substitution $T \mapsto \partial$ gives a well-defined $\mathbb{C}$ -linear operator on $\mathbf{No}[i]$.

Lemma 13.1 (Kernels of powers and real shifts). *For every ordinary integer $m \geq 1$,*

$$\ker(\partial^m) = \text{span}_{\mathbb{C}}\{1, \omega, \dots, \omega^{m-1}\} = \{P(\omega) : P \in \mathbb{C}[X], \deg P < m\},$$

with unique coefficients. For $\lambda \in \mathbb{R}$, $\ker(\partial - \lambda)^m = e^{\lambda\omega} \ker(\partial^m)$. For $\lambda \in \mathbb{C} \setminus \mathbb{R}$, every power of $\partial - \lambda$ is injective.

Proof. The case $m = 1$ is the constant-field statement. Inductively, if $\partial^m y = 0$ then $\partial^{m-1}y = c \in \mathbb{C}$, and $y - c\omega^{m-1}/(m-1)!$ is killed by ∂^{m-1} , hence is a polynomial of degree at most $m-2$; the converse is repeated differentiation, using $\partial\omega = 1$. Coefficients are unique because a nonzero ordinary-coefficient polynomial cannot vanish at the infinite ω , its highest-degree term dominating all lower ones; equivalently, ω is transcendental over the ordinary constants. For real λ , (BM4) gives

$$(\partial - \lambda)(e^{\lambda\omega}f) = e^{\lambda\omega}\partial f, \quad (13.1)$$

and iteration proves the second assertion. For nonreal λ , Theorem 3.11 or Theorem 8.8 makes $\partial - \lambda$ injective — alternatively multiply by $e^{-\alpha\omega}$ to reduce to $\partial v = i\beta v$ — and a finite power of an injective operator is injective. $\square$

Theorem 13.2 (Constant-coefficient solution classification). *Let $P \in \mathbb{C}[T]$ be nonzero and let m_λ be the multiplicity of a root λ . The complete solution space in $\mathbf{No}[i]$ of $P(\partial)y = 0$ is*

$$\ker P(\partial) = \bigoplus_{\substack{\lambda \in \mathbb{R} \\ P(\lambda)=0}} e^{\lambda\omega} \text{span}_{\mathbb{C}}\{1, \omega, \dots, \omega^{m_\lambda-1}\}, \quad (13.2)$$

of complex dimension $\sum_{\lambda \in \mathbb{R}, P(\lambda)=0} m_\lambda$. Nonreal characteristic roots contribute nothing; a nonzero constant polynomial gives the zero space. In particular an operator of positive order may have zero kernel, and the solution dimension may be strictly less than $\deg P$.

Proof. Factor P over $\mathbb{C}$ into pairwise coprime primary factors $p_\lambda(T) = (T - \lambda)^{m_\lambda}$; a nonzero scalar factor does not change the kernel. For pairwise coprime $f, g \in \mathbb{C}[T]$ choose A, B with $Af + Bg = 1$; then for $v \in \ker(f(\partial)g(\partial))$,

$$v = A(\partial)f(\partial)v + B(\partial)g(\partial)v,$$

whose first summand is killed by $g(\partial)$ and second by $f(\partial)$, and whose two kernels intersect in 0 by the same identity. This coprime-kernel decomposition is valid for any linear operator on any vector space; no finite-dimensionality is needed. Induction over the primary factors, or equivalently the Chinese-remainder idempotents modulo P , decomposes $\ker P(\partial)$ as the direct sum of the primary

kernels. Apply Theorem 13.1: nonreal-root summands vanish and real-root summands are those displayed. (Equivalently, factor $P = cQR$ with Q the product of the real-root factors and R that of the nonreal ones; $R(\partial)$ is injective and commutes with $Q(\partial)$, so $\ker P(\partial) = \ker Q(\partial)$.) Each displayed summand has dimension m_λ , and directness gives the count. $\square$

Example 13.3 (Order four, dimension two; order five, dimension three). In $\mathbf{No}[i]$:

$$\begin{aligned} \partial^2 y + y &= 0 \implies y = 0, \\ \partial^2 y - y &= 0 \iff y = Ae^\omega + Be^{-\omega}, \\ (\partial - 1)^2(\partial^2 + 1)y &= 0 \iff y = (A + B\omega)e^\omega, \\ (\partial - 2)^2(\partial^2 + 1)y &= 0 \iff y = (A + B\omega)e^{2\omega}, \\ (\partial - 2)^3(\partial + 1)(\partial^2 + 1)y &= 0 \iff y = e^{2\omega}(A + B\omega + C\omega^2) + De^{-\omega}, \\ (\partial - 2)^2(\partial + 1)(\partial^2 + 1)y &= 0 \iff y = (A + B\omega)e^{2\omega} + Ce^{-\omega}, \\ \partial(\partial - 1)^2(\partial^2 + 1)y &= 0 \iff y = A + e^\omega(B + C\omega), \end{aligned}$$

with all constants in $\mathbb{C}$. The missing modes are in every case exactly the oscillatory ones. There is no contradiction with the ordinary theorem that an order- n linear equation has an n -dimensional local solution space: that theorem concerns functions of a coordinate, or solutions in a sufficiently large differential extension, not solutions in an arbitrary fixed differential field.

Remark 13.4 (Real solutions, and the coefficient hypothesis). If $P \in \mathbb{R}[T]$ and real surreal solutions are wanted, the constants in equation (13.2) must be taken in $\mathbb{R}$: the imaginary part is a linear combination of the same independent real basis elements and vanishes only when all its coefficients do. The hypothesis that the coefficients of P are *ordinary complex constants* is essential: for $a \in \mathbf{No}[i] \setminus \mathbb{C}$ the operators ∂ and multiplication by a do not commute, so polynomial factorization in a commuting symbol cannot be transported to differential operators. Theorem 13.2 says nothing about variable-coefficient operators.

Corollary 13.5 (Constant matrix systems). *Let $A \in M_n(\mathbb{C})$ and consider $\partial Y = AY$ with $Y \in \mathbf{No}[i]^n$. The solution space has complex dimension equal to the total algebraic multiplicity of the real eigenvalues of A . On a Jordan block $\lambda I + N$ of size m with $\lambda \in \mathbb{R}$, all solutions are*

$$Y = e^{\lambda\omega} \left(\sum_{k=0}^{m-1} \frac{\omega^k N^k}{k!} \right) c, \quad c \in \mathbb{C}^m, \quad (13.3)$$

a finite polynomial in the nilpotent N , not an infinite matrix exponential requiring convergence; its inverse replaces ω by $-\omega$. On a block with nonreal eigenvalue only $Y = 0$ occurs. An invertible fundamental matrix in $\mathrm{GL}_n(\mathbf{No}[i])$ exists if and only if every eigenvalue of A is real.

Proof. An ordinary complex similarity is constant for ∂ , so pass to Jordan form. On a block with nonreal λ the last coordinate satisfies $(\partial - \lambda)y_m = 0$ and is zero by Theorem 13.1; upward substitution through the block makes every coordinate zero. On a real block, the displayed finite matrix polynomial B satisfies $\partial B = NB$ and is invertible, so multiplication by $e^{\lambda\omega}B$ carries constant vectors to solutions and its inverse carries solutions to constant vectors. Add the block dimensions. $\square$

For the real rotation matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ the solution space is zero, while a unipotent Jordan block has a fundamental matrix that is a polynomial in ω . The contrast isolates oscillation, not general complexity, as the obstruction, and it already rules out a universal matrix-exponential constructor in the intrinsic field.

14 Triangular systems, and the limits of a determinant test

Let $A = (a_{jk})$ be an upper-triangular $n \times n$ matrix over $\mathbf{No}[i]$, with n an ordinary positive integer. A *fundamental matrix* is $Y \in \mathrm{GL}_n(\mathbf{No}[i])$ with $\partial Y = AY$, entrywise.

Theorem 14.1 (Triangular fundamental-matrix criterion). *A fundamental matrix in $\mathrm{GL}_n(\mathbf{No}[i])$ exists if and only if*

$$\Phi(a_{jj}) = 0 \quad (1 \leq j \leq n). \quad (14.1)$$

When it does, an upper-triangular one can be constructed by finitely many first-order homogeneous integrations and additive primitives.

Proof. Sufficiency. Choose $s_j \neq 0$ with $\partial s_j = a_{jj}s_j$ by Theorem 8.4, put $Y_{jj} = s_j$ and $Y_{ij} = 0$ for $i > j$. For each column j , construct the entries above the diagonal in *descending* order of i by

$$Y_{ij} = s_i \mathcal{I}_0 \left(s_i^{-1} \sum_{k=i+1}^j a_{ik} Y_{kj} \right), \quad i < j. \quad (14.2)$$

The entries on the right are already constructed. The product rule gives $\partial Y_{ij} = a_{ii}Y_{ij} + \sum_{k=i+1}^j a_{ik}Y_{kj}$, so $\partial Y = AY$, and $\det Y = \prod_j s_j \neq 0$.

Necessity, by induction on n . The last row of any fundamental matrix satisfies $\partial Y_{nj} = a_{nn}Y_{nj}$, and at least one entry is nonzero, so $\Phi(a_{nn}) = 0$. All last-row entries are complex constant multiples of one nonzero solution, so a change of columns by an invertible constant matrix makes the last row $(0, \dots, 0, s_n)$; the result is still fundamental and has block form $\begin{pmatrix} Z & v \\ 0 & s_n \end{pmatrix}$ with $\det Z \neq 0$, whose upper-left block satisfies the system for the leading $(n-1) \times (n-1)$ block of A . Induction gives the remaining conditions. $\square$

In particular every strictly upper-triangular system has a unipotent fundamental matrix over $\mathbf{No}[i]$: this is a finite iterated-integration fact, and no infinite exponential series for an arbitrary matrix is needed.

Proposition 14.2 (The trace test is necessary and not sufficient). *For any finite matrix system, triangular or not, Jacobi's formula gives $\partial \det Y = (\mathrm{tr} A) \det Y$ for a fundamental Y , so*

$$\Phi(\mathrm{tr} A) = 0 \quad (14.3)$$

is necessary. It is not sufficient: for $A = \mathrm{diag}(i, -i)$ the trace vanishes, yet neither diagonal equation has a nonzero solution and the vector system has no nonzero solution at all.

Opposite phase obstructions can cancel in a determinant without disappearing from the individual modes, so a determinant-only test discards essential differential information. For a general matrix system the trace condition is only the first invariant. The complete one is the *multiset* of phases, and Part III supplies it: Theorem 15.9 shows that every finite system over $\mathbf{No}[i]$ has a well-defined differential phase spectrum $\mathrm{Ph}(A) \subset \mathcal{P}$ which is a complete invariant of gauge equivalence, and equation (15.5) recovers equation (14.3) from it as the sum

$$\Phi(\mathrm{tr} A) = \sum_P m_P(A) P,$$

whose failure to determine the multiset is exactly why the trace test is insufficient. Two of the phase spectrum's consequences belong beside the trace test. First, by Theorem 15.11 a fundamental matrix exists in $\mathrm{GL}_n(\mathbf{No}[i])$ if and only if every member of $\mathrm{Ph}(A)$ is zero, which extends Theorem 14.1 beyond the triangular case. Second, by Theorem 16.7, perturbing every entry of a Hermitian H by a coefficient with a finite primitive leaves the entire differential unitary class of $\partial y = iHy$ unchanged, with *no* spectral-gap hypothesis — a statement the trace invariant cannot see at all, and one that Theorem 16.9 shows is sharp. For non-Hermitian coefficients no such spectral formula holds, and Theorem 16.10 is the counterexample that marks the line; Section 31.5 records what remains unclassified.

Part III

Finite matrix systems over $\mathbf{No}[i]$

15 The differential phase spectrum of a finite system

15.1 From one necessary invariant to a complete one

Theorem 14.2 leaves a finite matrix system with exactly one invariant, $\Phi(\mathrm{tr} A)$, and with a proof that it is not enough: $\mathrm{diag}(i, -i)$ has vanishing trace obstruction and no nonzero solution. The missing information is a *multiset*, not a sum. This part supplies it for every finite system, and computes it from ordinary algebraic eigenvalues when the coefficient is Hermitian.

The eighth source manuscript is the origin of this part (Section C). Its own audit names the gap in this report's terms — general nontriangular systems and higher-rank differential modules left outside the classification — and states that the rank-one phase criterion, the bounded-primitive ideal and the diagonal Galois tori of Part II and Section 21 are prior coverage rather than discoveries of its own. That is how they are used below: nothing already proved in Part II is reproved here, and every statement of this part that coincides with one already in the report is cited, not restated.

Two agreements are worth printing as agreements before anything new is proved.

Remark 15.1 (The eighth manuscript refutes the same collapse). Theorem 5.4 is load-bearing: $\mathcal{A} = \partial \mathbf{m} = \partial \mathcal{O}$ is a *strict* subclass of $\mathbf{m}$, so the criterion is never “the coefficient is infinitesimal”. The eighth manuscript opens with exactly that point and exactly that witness: i/ω is obstructed because its imaginary part has the infinite primitive $\log \omega$. Its structure lemma for the ideal — order-convexity, the $\mathcal{O}$ -module property, containment in $\mathbf{m}$, and the separating pair $\omega^{-2} \in \mathcal{A}$, $\omega^{-1} \notin \mathcal{A}$ — is Theorem 7.5(iii),(iv), proved there by the same order-theoretic route from (BM7). Nothing in this part weakens that warning, and Theorem 16.9 below shows that the strictness is sharp for matrices as well as for scalars.

Remark 15.2 (The rank-one criterion is imported from Part II). The eighth manuscript reproves the exact logarithmic-derivative image equation (8.1) by polar decomposition and a formal Hahn logarithm, and credits it to this collection. It is Theorem 8.1 and Theorem 8.4, with the same proof idea as Theorem 6.7; it is not reproved here. Every uniqueness argument below rests on the following special case of Theorem 8.4, which is used often enough to deserve a number.

Lemma 15.3 (The rank-one separation test). *Let $P, Q \in \mathcal{P}$ and let $x \in \mathbf{No}[i]$ satisfy $\partial x = i \partial(P - Q)x$. If $x \neq 0$ then $P = Q$ and $x \in \mathbb{C}^\times$.*

Proof. By Theorem 8.4 a nonzero solution forces $\Phi(i\partial(P - Q)) = \mathfrak{o}(\partial(P - Q)) = 0$, which is $P - Q = 0$ by Theorem 7.5(vi). Then $\partial x = 0$, so $x \in \mathbb{C}^\times$ by the constant-field statement of Theorem 3.1. $\square$

We also need the order behaviour of $\mathfrak{o}$ itself, which Theorem 7.5 contains but does not display.

Corollary 15.4 (The obstruction preserves order). *The map $\mathfrak{o} : \mathbf{No} \rightarrow \mathcal{P}$ is order preserving: $b \leq c$ implies $\mathfrak{o}(b) \leq \mathfrak{o}(c)$. Consequently, for $P, Q \in \mathcal{P}$,*

$$P < Q \iff \partial P < \partial Q,$$

and an ordered tuple of real coefficients has an ordered tuple of obstructions.

Proof. The displayed equivalence is Theorem 7.5(vii), which states that $\partial : \mathcal{P} \rightarrow \partial\mathcal{P}$ is an order-preserving bijection. For the first assertion, suppose $b \leq c$ and put $P = \mathfrak{o}(b)$, $Q = \mathfrak{o}(c)$. If $Q < P$ then $\partial Q < \partial P$, and $b - \partial P \in \mathcal{A}$, $c - \partial Q \in \mathcal{A}$ by Theorem 7.5(vi),(vii); so $b - c = (\partial P - \partial Q) + d$ with $d \in \mathcal{A}$ and $\partial P - \partial Q > 0$. Since $\partial P - \partial Q = \partial(P - Q)$ with $P - Q \in \mathcal{P} \setminus \{0\}$, it does not lie in $\mathcal{A}$, and convexity of $\mathcal{A}$ (Theorem 7.5(iii)) then gives $b - c > 0$, contradicting $b \leq c$. $\square$

Convention 15.5 (The alphabet of this part). Throughout Part III, n is an ordinary positive integer and all matrices are $n \times n$ over $\mathbf{No}[i]$ unless said otherwise; ∂ acts entrywise, A^* is the conjugate transpose for the conjugation of Theorem 3.5, and I is the identity matrix.

- A is a coefficient matrix, as in Section 14; Y is a fundamental matrix.
- H is a *Hermitian* matrix, $H^* = H$. In this report the letter H otherwise occurs only in the hyphenated adjective “ H -field” and in the subscripted coefficient fields $H_\Gamma = \mathbb{R}((t^\Gamma))$; a bare H in this part is always a matrix and never either of those.
- $G \in \mathrm{GL}_n(\mathbf{No}[i])$ is a gauge matrix — the matrix upgrade of the scalar gauge g of equation (9.1) — and W is a gauge that is *unitary*, $W^*W = I$. The letter U is reserved for an ordinary complex domain (Theorem B.1) and is not used for a unitary matrix; $\mathrm{U}_n(\mathbf{No}[i])$ denotes the unitary group.
- $\mathcal{M}$ is a horizontal Hermitian metric (Theorem 15.15), and $\mathcal{R} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ is the rotation generator. The sources write J for the latter; this report uses no letter J (Theorem 5.6).
- $\mathcal{T}$ is the differential field of transseries, which occurs only in Theorem 15.6; it is never the unit circle $\mathbb{T}(\mathbf{No})$ and never a transcendental generator $\mathbb{T}$.
- Π is the additive *phase lattice* generated inside $\mathcal{P}$ by the phases of a system, and $\Lambda \subseteq \mathbb{Z}^n$ remains the integer *relation* lattice of equation (21.6). The two are different objects over different groups and are never written with the same letter.

15.2 Matrix gauge equivalence and the complete phase normal form

A change of unknown $y = Gz$ with $G \in \mathrm{GL}_n(\mathbf{No}[i])$ carries $\partial y = Ay$ to $\partial z = A^G z$, where

$$A^G = G^{-1}AG - G^{-1}\partial G. \tag{15.1}$$

Call A, B *differentially equivalent* when $B = A^G$ for some $G \in \mathrm{GL}_n(\mathbf{No}[i])$, and *differentially unitarily equivalent* when G may be taken unitary. The connection term $G^{-1}\partial G$ is not optional:

similarity of matrices alone is not differential equivalence, and Theorem 16.10 below turns on exactly that.

Two inputs are needed that Part II does not contain. Both are imported, and the import is flagged here and in Section 31.9.

Theorem 15.6 (Imported: splitting and first-order surjectivity, transferred). *The following two properties hold over $\mathcal{T}[i]$, where $\mathcal{T}$ is the field of transseries, and are recorded in the asymptotic-differential-algebra literature [4, Lemma 2.4.19]: every monic scalar differential operator of positive order splits into first-order factors, and every nonzero scalar linear differential operator is surjective. Both transfer to $\mathbf{No}[i]$. In particular, for all $a, f \in \mathbf{No}[i]$ there is $u \in \mathbf{No}[i]$ with*

$$\partial u - au = f. \quad (15.2)$$

Proof. For each fixed order the two assertions are first-order sentences of the ordered differential-field language: expanding a product of first-order operators uses only the finite rule $\partial a = a\partial + \partial(a)$, so the coefficient identities are polynomial identities in finitely many coefficients and their derivatives, and surjectivity of a given order is a first-order sentence with a nonzero leading coefficient as hypothesis. A complex coefficient is an ordered pair of real ones, and complex addition, multiplication and differentiation are definable in that interpretation, so the sentences about $\mathcal{T}[i]$ are sentences about $\mathcal{T}$. The elementary embedding $\mathcal{T} \preceq \mathbf{No}$ of [3] transfers them to $\mathbf{No}$, hence to $\mathbf{No}[i]$, for arbitrary surreal coefficients and not merely for coefficients in the embedded copy of $\mathcal{T}$. $\square$

Warning 15.7 (What this import does and does not add). Theorem 15.6 is an *additional imported input*, on the same footing as Theorem 3.1 and not a consequence of anything proved in Part II. It must not be read as making $\mathbf{No}[i]$ differentially closed, and it must not be read as a homogeneous existence theorem: “linearly closed” there means the splitting property, not that every homogeneous equation has a full solution set in the base field. Reading it the second way would make $\partial y = iy$ appear solvable and contradict Theorem 8.8. It also yields no algorithm: no effective factorization of an arbitrary surreal-coefficient operator follows from a first-order transfer (Section 31.4). What it does supply is *inhomogeneous existence* for first-order scalar equations, which Theorem 12.1 deliberately did not assert outside the admissible locus $\Phi(q) = 0$; see Theorem 15.11 and the amended item N22.

Lemma 15.8 (Cyclic vector, and a rank-one filtration). *Let M be a differential module over $\mathbf{No}[i]$ of ordinary finite dimension $n \geq 1$, that is, a free $\mathbf{No}[i]$ -module of rank n with an additive ∇ satisfying $\nabla(fv) = (\partial f)v + f\nabla v$. Then M has a vector v for which $v, \nabla v, \dots, \nabla^{n-1}v$ is a basis, and M has a filtration by differential submodules whose successive quotients have dimension one.*

Proof. Fix a basis $e_0, \dots, e_{n-1}$ and extend ∇ to $M[T]$ by giving the indeterminate derivative 1. Define $v_0 = e_0$ and $v_k = e_k - \sum_{j < k} \binom{k}{j} \nabla^{k-j} v_j$, and put $v(T) = \sum_{j < n} T^j v_j / j!$. The Leibniz rule gives $\nabla^k v(T)|_{T=0} = e_k$ for $k < n$, so the determinant of the columns $v(T), \dots, \nabla^{n-1}v(T)$ is a polynomial in T with constant term 1, hence nonzero. Substituting $T = \omega - c$ for a real c respects differentiation because $\partial(\omega - c) = 1$, and a nonzero polynomial has only finitely many roots while the elements $\omega - c$ are pairwise distinct; choose c outside that finite exceptional set to get a cyclic vector.

For the filtration, a cyclic vector presents M as $\mathbf{No}[i][\partial]/\mathbf{No}[i][\partial]L$ with L monic of order n . By Theorem 15.6 write $L = (\partial - b)R$ with R a product of $n - 1$ first-order factors. The class of R is nonzero, spans a one-dimensional differential submodule and satisfies $\partial R \equiv bR$ modulo L ; the quotient is the module with annihilator R , and induction on n finishes. The dimension counts follow from division by a monic operator, whose remainder has strictly smaller order. $\square$

Theorem 15.9 (Complete phase normal form for finite systems). *For every $A \in \text{Mat}_n(\mathbf{No}[i])$ there are $G \in \text{GL}_n(\mathbf{No}[i])$ and $P_1, \dots, P_n \in \mathcal{P}$ with*

$$A^G = i \text{diag}(\partial P_1, \dots, \partial P_n). \quad (15.3)$$

The multiset

$$\text{Ph}(A) := \{P_1, \dots, P_n\} \subset \mathcal{P}$$

does not depend on G , and it is a complete invariant of differential equivalence: two finite systems of the same size are differentially equivalent if and only if their multisets agree. We call $\text{Ph}(A)$ the differential phase spectrum of A , in the sense of Theorem 5.1(e).

Proof. Apply Theorem 15.8 to the connection $\nabla = \partial - A$ on $\mathbf{No}[i]^n$; an adapted basis makes the system upper triangular. Suppose inductively that the first $n-1$ coordinates have been diagonalized, leaving

$$A = \begin{pmatrix} D & b \\ 0 & a \end{pmatrix}, \quad D = \text{diag}(d_1, \dots, d_{n-1}).$$

For $S = \begin{pmatrix} I & u \\ 0 & 1 \end{pmatrix}$ the off-diagonal column of A^S is $b + Du - ua - \partial u$, and each scalar equation $\partial u_j - (d_j - a)u_j = b_j$ has a solution by equation (15.2). This kills the column, with no hypothesis on repetitions among the diagonal entries: when $d_j = a$ the equation is ordinary intrinsic integration, available by (BM6). Induction gives a diagonal form, and Theorem 9.1 applied to each diagonal entry supplies a further diagonal gauge bringing it to equation (15.3) with $P_j = \Phi(a_j)$.

For independence and completeness, suppose $D_P = i \text{diag}(\partial P_j)$ and $D_Q = i \text{diag}(\partial Q_k)$ satisfy $D_Q = (D_P)^\Xi$ for an invertible Ξ . Then $\partial \Xi = D_P \Xi - \Xi D_Q$, that is

$$\partial \Xi_{jk} = i \partial(P_j - Q_k) \Xi_{jk} \quad (15.4)$$

entrywise. By Theorem 15.3 an entry can be nonzero only when $P_j = Q_k$, and is then a complex constant. Invertibility of Ξ forces the two multisets to have the same multiplicities. Conversely equal multisets give the same normal form after a constant permutation. $\square$

Remark 15.10 (This is the matrix upgrade of Theorem 9.1, and of nothing else). For $n = 1$, Theorem 15.9 is Theorem 9.1. For $n > 1$ it is a genuinely stronger statement, and it is the first classification in this report of a system that is neither triangular nor constant. It is still *not* a module-theoretic filtration theorem with computable data: the normal form is obtained from an existence theorem, the phases are semantic values of $\mathfrak{o}$, and Section 31.9 records what remains open.

15.3 Solution spaces, projectors, and the refined trace identity

For $P \in \mathcal{P}$ write $m_P(A)$ for the multiplicity of P in $\text{Ph}(A)$, zero if P does not occur.

Corollary 15.11 (Existence for every forcing, and the exact kernel dimension). *For every $A \in \text{Mat}_n(\mathbf{No}[i])$ and every $f \in \mathbf{No}[i]^n$ the equation $\partial y - Ay = f$ has a solution in $\mathbf{No}[i]^n$, and its solution set is an affine space over $\mathbb{C}$ of dimension $m_0(A)$. In particular*

$$\dim_{\mathbb{C}} \ker(\partial - A)|_{\mathbf{No}[i]^n} = m_0(A),$$

and $\partial - A$ is bijective on $\mathbf{No}[i]^n$ exactly when $0 \notin \text{Ph}(A)$.

Proof. Transport by the gauge of Theorem 15.9 and solve each scalar inhomogeneous equation by equation (15.2). In the homogeneous equation a coordinate can be nonzero only at phase 0, where it is a complex constant, by Theorem 15.3. The difference of two particular solutions is homogeneous. $\square$

Warning 15.12 (What Theorem 15.11 changes, and what it does not). Theorem 12.1 proves surjectivity of $\partial - q$ only on the admissible locus $\Phi(q) = 0$, and Theorem 12.2 states the correct weaker conclusion elsewhere: *at most one*, not none. Theorem 15.11 upgrades “at most one” to “exactly one” in the obstructed case, and it does so *only* by invoking the separately imported first-order surjectivity of Theorem 15.6. It does not follow from the rank-one criterion, it does not follow from Theorem 12.1, and it does not make the homogeneous equation solvable. The examples of Theorem 12.2 are unaffected and remain correct; what changes is that their uniqueness is now known to be accompanied by existence. Item N22 of Section 31 is amended accordingly and not deleted.

Proposition 15.13 (Canonical phase projectors). *Let E_P be the coordinate projector onto the phase- P block of a normal form and put $\Pi_P = GE_PG^{-1}$ for a normalizing gauge G . Then Π_P is independent of the choice of G , and*

$$\Pi_P \Pi_Q = \delta_{PQ} \Pi_P, \quad \sum_P \Pi_P = I, \quad \partial \Pi_P = A \Pi_P - \Pi_P A.$$

The algebra of horizontal endomorphisms of the system — matrices X with $\partial X = AX - XA$ — is isomorphic to $\prod_P \text{Mat}_{m_P(A)}(\mathbb{C})$, and for two systems

$$\dim_{\mathbb{C}} \text{Hom}_{\partial}(A, B) = \sum_P m_P(A) m_P(B).$$

Direct sums unite phase multisets, tensor products add phases pairwise, and duals negate them.

Proof. By equation (15.4) any change between normalizing gauges is a constant block-diagonal matrix, so it commutes with every E_P ; the displayed identities follow by multiplying and differentiating $\Pi_P = GE_PG^{-1}$ and using $\partial G = AG - GD_P$. The same entrywise calculation, applied to $\partial X = AX - XB$ in normal-form coordinates, identifies the horizontal homomorphisms with constant matrices supported on matching phase blocks. The last three statements are the corresponding computations for $A \oplus B$, $A \otimes I + I \otimes B$ and $-A^T$, in each case reduced to the scalar addition rule of Theorem 9.4. $\square$

Corollary 15.14 (The determinant sees only the phase sum). *For every finite A ,*

$$\Phi(\text{tr } A) = \sum_P m_P(A) P. \tag{15.5}$$

Proof. Take traces in $A = (\partial G)G^{-1} + GD_PG^{-1}$ and use Jacobi’s identity $\text{tr}((\partial G)G^{-1}) = (\det G)^\dagger$, whose obstruction is zero by Theorem 8.4. $\square$

Equation (15.5) contains Theorem 14.2 and explains it: a fundamental matrix exists only if every phase is zero, so $\Phi(\text{tr } A) = 0$ is necessary, and it fails to be sufficient precisely because a multiset is not recoverable from its sum. The witness $\text{diag}(i, -i)$ of Theorem 14.2 has $\text{Ph} = \{\omega, -\omega\}$, which sums to zero while $\{0, 0\}$ does too; the two systems are not differentially equivalent, and Theorem 15.9 says so. The trace condition is the first invariant; the phase multiset is the complete one.

15.4 Horizontal metrics and the unitary normal form

Definition 15.15 (Horizontal metric). A Hermitian $\mathcal{M} \in \text{Mat}_n(\mathbf{No}[i])$ is *horizontal* for A when

$$\partial \mathcal{M} + A^* \mathcal{M} + \mathcal{M} A = 0, \quad (15.6)$$

and *positive definite* when $v^* \mathcal{M} v > 0$ for every nonzero $v \in \mathbf{No}[i]^n$.

Equation (15.6) says formally that $y^* \mathcal{M} y$ is constant along a solution, in any differential extension carrying the compatible conjugation. It does not assert that the system has nonzero solutions in $\mathbf{No}[i]^n$; by Theorem 15.11 it usually does not.

Theorem 15.16 (The complete metric cone). *Let G normalize A as in Theorem 15.9. Every horizontal Hermitian matrix is uniquely*

$$\mathcal{M} = G^{-*} \left(\bigoplus_P C_P \right) G^{-1}, \quad C_P \in \text{Herm}_{m_P(A)}(\mathbb{C}), \quad (15.7)$$

and is positive definite exactly when every C_P is. In particular every finite system has a positive horizontal metric, and the real dimension of its space of horizontal Hermitian forms is $\sum_P m_P(A)^2$.

Proof. Put $B = G^* \mathcal{M} G$. Since $\partial G = AG - GD_P$, differentiating gives $\partial B = D_P B - B D_P$, that is $\partial B_{jk} = i \partial(P_j - P_k) B_{jk}$, so Theorem 15.3 makes every off-phase entry zero and every within-phase entry a complex constant. Hermitian symmetry leaves exactly the blocks C_P . Congruence preserves definiteness, and for a *constant* Hermitian matrix positivity over $\mathbf{No}[i]$ is ordinary complex positive definiteness: diagonalize by an ordinary constant unitary matrix and read off its real eigenvalues. Taking every $C_P = I$ gives existence, and $\dim_{\mathbb{R}} \text{Herm}_m(\mathbb{C}) = m^2$. $\square$

Warning 15.17 (A positive horizontal metric is not a stability statement). For

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad G = e^\omega \begin{pmatrix} 1 & \omega \\ 0 & 1 \end{pmatrix},$$

one has $\partial G = AG$, so both phases are zero, and

$$\mathcal{M} = e^{-2\omega} \begin{pmatrix} 1 & -\omega \\ -\omega & \omega^2 + 1 \end{pmatrix}$$

is horizontal with $\det \mathcal{M} = e^{-4\omega} > 0$. The ordinary differential equation with the same printed coefficients has growing solutions. There is no contradiction, and no Lyapunov or uniform-equivalence claim is made anywhere in this part: the invariant metric itself has a strongly varying scale, here $e^{-2\omega}$, and Theorem 15.16 has no hypothesis relating it to a fixed norm. This is the matrix form of the separation already made in Theorem 4.1(7) between a printed equation read in $\mathbf{No}[i]$ and the same equation read in an ordinary coordinate.

Theorem 15.18 (Unitary phase normal form). *If $A^* = -A$, the normalizing gauge of Theorem 15.9 can be taken unitary. For a fixed ordering of the phases, any two unitary normalizing gauges differ by a constant block unitary matrix, one block per repeated phase, and the projectors of Theorem 15.13 are then Hermitian orthogonal projectors.*

Proof. The identity matrix is horizontal for a skew-Hermitian A , so Theorem 15.16 applied to any normalizing G makes $G^*G = C$ a constant positive definite block Hermitian matrix. Let $R = C^{-1/2}$, computed in the ordinary finite-dimensional complex block matrices; then $\partial R = 0$ and R commutes with D_P , so $W = GR$ is still normalizing and $W^*W = I$. Uniqueness is equation (15.4) again, with unitarity making the constant diagonal blocks unitary, and WE_PW^* is Hermitian and idempotent. $\square$

The square root in that proof is of a *constant complex* matrix. No analytic functional calculus on a class-sized space is used, and the theorem is about differential diagonalization: the term $W^*\partial W$ has not been removed and reappears, essentially, in equation (16.5).

16 The spectral–integral theorem for Hermitian coefficients

16.1 Finite Hermitian algebra over a real closed field

The two lemmas of this subsection hold over any real closed field and its complexification, and use no property of ∂ at all. Related finite spectral material for $\mathbf{No}[i]$ is developed in the companion report *Surcomplex Spectral Theory* [21], whose “spectrum” is the algebraic eigenvalue list; see Theorem 5.1(e) for why that is not the object named in Theorem 16.5.

Lemma 16.1 (Finite spectral theorem with attained min–max). *Every Hermitian $H \in \text{Mat}_n(\mathbf{No}[i])$ has a unitary eigenbasis and eigenvalues in $\mathbf{No}$. Writing $\lambda_1(H) \leq \dots \leq \lambda_n(H)$,*

$$\lambda_k(H) = \min_{\dim S=k} \max_{v \in S, v^*v=1} v^*Hv, \quad (16.1)$$

and both extrema are attained.

Proof. $\mathbf{No}[i]$ is algebraically closed, so there is an eigenvector $v \neq 0$ with eigenvalue λ ; since $v^*v > 0$ and v^*Hv is fixed by conjugation, $\lambda = v^*Hv/v^*v \in \mathbf{No}$. Normalize v by the positive square root of v^*v , available because $\mathbf{No}$ is real closed; the orthogonal complement is H -invariant, and finite induction gives a unitary eigenbasis. For a fixed k -dimensional S the compression of H to an orthonormal basis of S is again finite Hermitian, so its largest eigenvalue attains the inner maximum; the span of the first k eigenvectors attains the value $\lambda_k(H)$, and every k -dimensional S meets the span of the last $n - k + 1$ eigenvectors nontrivially, where a unit vector has Rayleigh quotient at least $\lambda_k(H)$. No compactness and no completeness is used; the argument is finite algebra over an ordered field. $\square$

Lemma 16.2 (An entrywise Weyl bound, and its ideal form). *For Hermitian $H, E \in \text{Mat}_n(\mathbf{No}[i])$ put $m = \max_{j,k} |E_{jk}|$. Then*

$$|\lambda_j(H + E) - \lambda_j(H)| \leq nm \quad (1 \leq j \leq n). \quad (16.2)$$

If every entry of E lies in $\mathcal{A}_{\mathbb{C}} := \mathcal{A} + i\mathcal{A}$, then every eigenvalue difference in equation (16.2) lies in $\mathcal{A}$.

Proof. For $v^*v = 1$ the triangle and Cauchy–Schwarz inequalities over the real closed field $\mathbf{No}$ give $|v^*Ev| \leq m(\sum_j |v_j|)^2 \leq nm$. Using the span of the first j eigenvectors of H in equation (16.1) gives the upper bound, and interchanging H and $H + E$ the lower one. If $E_{jk} = a + ib$ with $a, b \in \mathcal{A}$ then $|E_{jk}| \leq |a| + |b| \in \mathcal{A}$, so convexity of $\mathcal{A}$ (Theorem 7.5(iii)) puts $|E_{jk}|$ in $\mathcal{A}$; a finite maximum and the integer multiple nm stay in $\mathcal{A}$, and one more application of convexity finishes the proof. $\square$

16.2 Bounded gauges have integrable connection terms

Lemma 16.3 (Bounded-gauge estimate). *Let $W \in U_n(\mathbf{No}[i])$. Then every entry of W and of W^* is finite, every entry of ∂W lies in $\mathcal{A}_{\mathbb{C}}$, every entry of $W^*\partial W$ lies in $\mathcal{A}_{\mathbb{C}}$, and $W^*\partial W$ is skew-Hermitian.*

Proof. A unitary column has squared moduli summing to 1, so each entry has modulus at most 1 and is finite. Differentiating a finite real or imaginary coordinate lands in $\partial\mathcal{O} = \mathcal{A}$ by Theorem 7.5(i). Since $\mathcal{A}$ is an $\mathcal{O}$ -module (Theorem 7.5(iii)), finite sums of products of entries of W^* with entries of ∂W stay in $\mathcal{A}_{\mathbb{C}}$. Differentiating $W^*W = I$ gives $(\partial W)^*W + W^*\partial W = 0$. $\square$

Warning 16.4 (The conclusion is about primitives, not about size). It is the conclusion $W^*\partial W \in \text{Mat}_n(\mathcal{A}_{\mathbb{C}})$, and not the weaker statement that its entries are infinitesimal, that makes Theorem 16.5 true. The two are different by Theorem 5.4: an infinitesimal entry may accumulate an infinite phase, as ω^{-1} does. This is the precise point at which the strictness $\mathcal{A} \subsetneq \mathfrak{m}$ is load-bearing for the matrix theory.

16.3 The theorem

Theorem 16.5 (Spectral–integral classification of Hermitian systems). *Let $H = H^* \in \text{Mat}_n(\mathbf{No}[i])$ with algebraic eigenvalues ordered $\lambda_1 \leq \dots \leq \lambda_n$ in $\mathbf{No}$, and put*

$$P_j := \mathfrak{o}(\lambda_j) \in \mathcal{P}.$$

Then $P_1 \leq \dots \leq P_n$ and

$$\text{Ph}(iH) = \{P_1, \dots, P_n\}. \quad (16.3)$$

There is a unitary $W \in U_n(\mathbf{No}[i])$ with

$$W^*(iH)W - W^*\partial W = i \text{diag}(\partial P_1, \dots, \partial P_n), \quad (16.4)$$

and two Hermitian coefficients give differentially unitarily equivalent systems $\partial y = iH_1 y$ and $\partial y = iH_2 y$ if and only if their ordered algebraic eigenvalues have identical images under $\mathfrak{o}$, counted with multiplicity. Neither commuting coefficients nor a spectral gap is assumed.

Proof. The matrix iH is skew-Hermitian, so Theorem 15.18 supplies a unitary normalizing gauge W with phases $Q_1 \leq \dots \leq Q_n$, ordered by Theorem 15.4. Rearranging its defining identity gives

$$W^*HW = \text{diag}(\partial Q_1, \dots, \partial Q_n) + E, \quad E := -iW^*\partial W. \quad (16.5)$$

By Theorem 16.3 the matrix E is Hermitian with every entry in $\mathcal{A}_{\mathbb{C}}$. Conjugation by a unitary matrix does not change eigenvalues, and $\partial Q_1 \leq \dots \leq \partial Q_n$ by Theorem 15.4; so Theorem 16.2, applied to the diagonal matrix and the perturbation E , gives $\lambda_j(H) - \partial Q_j \in \mathcal{A}$ for every j . Apply $\mathfrak{o}$: its kernel is exactly $\mathcal{A}$ by Theorem 7.5(vi), and $\mathfrak{o}(\partial Q_j) = Q_j$ by the same item, so $\mathfrak{o}(\lambda_j(H)) = Q_j$. This is equations (16.3) and (16.4), and the ordering of the P_j is Theorem 15.4. The equivalence statement is Theorem 15.9 together with Theorem 15.18, which makes the normalizing gauges of two skew-Hermitian systems unitary, so that the equivalence between them is unitary as well. $\square$

Warning 16.6 (The connection term is not removed). Theorem 16.5 does *not* say that pointwise diagonalization commutes with differentiation; it does not. The term $W^*\partial W$ in equation (16.5) is present, and the theorem says only that it is invisible in the ordered additive quotient by $\mathcal{A}$

described in Theorem 7.5(viii). Nor is the classification the assertion that every phase grade has a horizontal vector in $\mathbf{No}[i]^n$: by Theorem 15.11 only the grade $P = 0$ does. The formula is exact for arbitrary surreal scales and arbitrary ordinary finite multiplicities; there is no hypothesis that the λ_j be distinct, that gaps dominate a derivative, or that H commute with its derivative or with any of its primitives.

16.4 Gap-free perturbation, the exact kernel count, and sharpness

Corollary 16.7 (Finite-primitive perturbation invariance, with no gap hypothesis). *If H_1, H_2 are Hermitian and every entry of $H_1 - H_2$ lies in $\mathcal{A}_{\mathbb{C}}$, then $\partial y = iH_1 y$ and $\partial y = iH_2 y$ are differentially unitarily equivalent.*

Proof. Theorem 16.2 puts corresponding ordered eigenvalue differences in $\mathcal{A} = \ker \mathfrak{o}$; apply Theorem 16.5. $\square$

Corollary 16.8 (Exact kernel count). *For Hermitian H ,*

$$\dim_{\mathbb{C}}\{y \in \mathbf{No}[i]^n : \partial y = iHy\} = \#\{j : \lambda_j(H) \in \mathcal{A}\}, \quad (16.6)$$

counted with algebraic multiplicity. The forced equation $\partial y - iHy = f$ is solvable for every $f \in \mathbf{No}[i]^n$, and uniquely so exactly when no eigenvalue of H lies in $\mathcal{A}$.

Proof. Combine Theorems 15.11 and 16.5 with $\ker \mathfrak{o} = \mathcal{A}$. $\square$

For $H = \text{diag}(\omega^{-2}, \omega^{-1}, 1)$ the kernel has complex dimension one, because $\omega^{-2} \in \mathcal{A}$ while $\omega^{-1} \notin \mathcal{A}$ and $1 \notin \mathcal{A}$ (equation (7.3)). Replacing H by any Hermitian matrix differing from it entrywise by $\mathcal{A}_{\mathbb{C}}$ preserves that answer, even when the perturbation is far larger than a gap at some more remote infinitesimal scale — which is exactly what a gap hypothesis would forbid.

Proposition 16.9 (Sharpness of the perturbation class). *For real $h, k \in \mathbf{No}$ the equations $\partial y = ihy$ and $\partial y = iky$ are differentially unitarily equivalent exactly when $h - k \in \mathcal{A}$. Hence no strictly larger additive class of scalar perturbations preserves the phase of every Hermitian system.*

Proof. A scalar unitary gauge equivalence is $i(h - k) = u^\dagger$ with $u\bar{u} = 1$; Theorem 8.1 together with Theorem 6.3 gives necessity and constructs a unit solution for sufficiency. If $b \notin \mathcal{A}$ the one-dimensional systems with coefficients 0 and b have obstructions 0 and $\mathfrak{o}(b) \neq 0$. $\square$

“Infinitesimal perturbation” therefore cannot replace “finite-primitive perturbation”: the scalar change $0 \mapsto \omega^{-1}$ already moves the phase from 0 to $\log \omega$. This is Theorem 5.4 again, now in its sharpest form: $\mathcal{A}$ is not merely a convenient subclass of $\mathfrak{m}$, it is the exact class for which Theorem 16.7 is true.

16.5 The two boundaries, exhibited

Example 16.10 (A nonnormal system whose eigenvalues predict the wrong phases). Let $x = \omega$ and

$$G = \begin{pmatrix} 1 - x^2 & x \\ -x & 1 \end{pmatrix}, \quad A = \begin{pmatrix} -x & 1 + x^2 \\ -1 & x \end{pmatrix}.$$

Then $\det G = 1$ and $\partial G = AG$, so A has a fundamental matrix in $\mathrm{GL}_2(\mathbf{No}[i])$ and, by Theorem 15.11, $\mathrm{Ph}(A) = \{0, 0\}$. Yet

$$\det(\zeta I - A) = \zeta^2 + 1,$$

so the instantaneous algebraic eigenvalues of A are i and $-i$, and taking obstructions of their imaginary parts would predict $\{\omega, -\omega\}$. The prediction is wrong by Theorem 15.9.

Warning 16.11 (What the counterexample rules out, and what it marks). Theorem 16.10 is not a curiosity; it is the boundary of Theorem 16.5. It shows that the Hermitian hypothesis cannot be dropped and that the main formula does not extend to arbitrary matrices by extracting imaginary parts of eigenvalues. The mechanism is exactly the one Theorem 16.3 controls in the Hermitian case: the similarity gauges needed for a general nonnormal matrix need not be entrywise finite, so their connection term need not lie in $\mathrm{Mat}_n(\mathcal{A}_{\mathbb{C}})$ and can carry a nonzero phase of its own. Here G has the infinite entry $1 - \omega^2$.

The example also agrees with, rather than contradicts, the hedge already recorded in Section 31.10: a classification of nontriangular systems by module-theoretic data cannot be replaced by the eigenvalues of an instantaneous coefficient matrix. Theorem 16.5 does not do that either; it replaces them by the eigenvalues *after* applying $\mathfrak{o}$, and only under Hermitian symmetry. The non-Hermitian case remains open, and Theorem 16.10 is what marks the line.

Example 16.12 (A genuinely coupled unitary system with two independent phases). Let $x = \omega$ and

$$a = \frac{1 - x^2}{1 + x^2}, \quad b = \frac{2x}{1 + x^2}, \quad W = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}, \quad D = i \operatorname{diag}(x, x^{-1}),$$

so that W is a real orthogonal matrix over $\mathbf{No}$ and $A := (\partial W)W^{\top} + WDW^{\top}$ satisfies $A^* = -A$. Explicitly

$$A = \begin{pmatrix} i(a^2x + b^2x^{-1}) & -\frac{2}{1 + x^2} + iab(x - x^{-1}) \\ \frac{2}{1 + x^2} + iab(x - x^{-1}) & i(b^2x + a^2x^{-1}) \end{pmatrix},$$

which is neither diagonal nor a constant-matrix system. Its exact normal form is D , so

$$\mathrm{Ph}(A) = \left\{ \frac{1}{2}\omega^2, \log \omega \right\},$$

using $\mathfrak{o}(\omega) = \frac{1}{2}\omega^2$ and $\mathfrak{o}(\omega^{-1}) = \log \omega$ (equation (7.3)). Writing $H = -iA$, one computes

$$\operatorname{tr} H = x + x^{-1}, \quad \det H = 1 - \frac{4}{(1 + x^2)^2},$$

so the instantaneous eigenvalues of H are *not* x and x^{-1} . Their discrepancies lie in $\mathcal{A}$, so Theorem 16.5 recovers the same two phases from them. Both phases are nonzero, so by Theorem 15.11 every forced system $\partial y - Ay = f$ has a unique solution in $\mathbf{No}[i]^2$; and the two phases are $\mathbb{Q}$ -linearly independent, which Section 17.6 uses.

17 Consequences of the matrix normal form

17.1 Normalized non-Abelian integration

Write

$$\mathfrak{u}_n(\mathcal{A}_{\mathbb{C}}) = \{A \in \mathrm{Mat}_n(\mathcal{A}_{\mathbb{C}}) : A^* = -A\},$$

and let $U_n(1 + \mathfrak{m}_{\mathbb{C}})$ be the unitary matrices W with $W - I \in \text{Mat}_n(\mathfrak{m}_{\mathbb{C}})$. The latter is a congruence subgroup, not the requirement that off-diagonal entries be near 1.

Theorem 17.1 (Normalized compact integration). *The logarithmic-derivative map*

$$U_n(1 + \mathfrak{m}_{\mathbb{C}}) \longrightarrow \mathfrak{u}_n(\mathcal{A}_{\mathbb{C}}), \quad W \longmapsto (\partial W)W^*, \quad (17.1)$$

is a bijection of classes. Equivalently, for skew-Hermitian A the following are equivalent:

- (i) *every entry of A has finite real and imaginary primitives, that is $A \in \text{Mat}_n(\mathcal{A}_{\mathbb{C}})$;*
- (ii) *$\partial y = Ay$ has a fundamental matrix in $\text{GL}_n(\mathbf{No}[i])$;*
- (iii) *$\partial W = AW$ has a unitary solution with $\text{st}(W) = I$ entrywise.*

The solution in (iii) is unique.

Proof. Theorem 16.3 makes equation (17.1) well defined and gives (iii) $\Rightarrow$ (i). Assume (i) and put $H = -iA$, Hermitian with entries in $\mathcal{A}_{\mathbb{C}}$. Comparing H with the zero matrix, Theorem 16.2 places every eigenvalue of H in $\mathcal{A}$, so Theorem 16.5 gives $\text{Ph}(A) = \{0, \dots, 0\}$ and a unitary V with $A^V = 0$, that is $\partial V = AV$. Taking standard parts entrywise in $V^*V = I$ — legitimate because every entry is finite by Theorem 16.3 — shows that $C = \text{st}(V)$ is an ordinary complex unitary matrix, and $W = VC^{-1}$ has $\partial W = AW$ and $\text{st}(W) = I$.

If W_1, W_2 are two normalized solutions then $\partial(W_1^{-1}W_2) = 0$, so $W_1^{-1}W_2$ is a constant complex matrix with standard part I , hence I . That is uniqueness and injectivity of equation (17.1). Clearly (iii) $\Rightarrow$ (ii); conversely a fundamental matrix makes every phase zero by Theorem 15.11, and for a skew-Hermitian system Theorem 15.18 then supplies a unitary fundamental matrix, which after multiplication by the inverse of its standard part is normalized. $\square$

Warning 17.2 (This is an existence theorem, not an exponential formula). Theorem 17.1 is not the assertion that $W = \exp(\int A)$, and there is no ordering prescription, no chosen contour, and no claimed convergence of Picard iteration. Its normalization is residue data, $\text{st}(W) = I$, not an initial value at a real or surreal time — the same distinction Theorem 7.2 makes for $\mathcal{I}_0$. For the unique solution W_A the ordinary product rule gives

$$(\partial(W_A W_B))(W_A W_B)^* = A + W_A B W_A^*, \quad (17.2)$$

whose right-hand side again lies in $\mathfrak{u}_n(\mathcal{A}_{\mathbb{C}})$. So the bijection transports the group law to $A \star B = A + W_A B W_A^*$, which is *not* addition unless the matrices commute.

Example 17.3 (An exact noncommuting Hahn solution). Let $t = \omega^{-1}$, so $\partial t = -t^2$, and take the constant skew-Hermitian matrices

$$X = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}.$$

For $A = -t^2 X - 2t^3 Z$ — every entry of which lies in $\mathcal{A}_{\mathbb{C}}$ — the normalized solution of Theorem 17.1 is the Hahn series $W = \sum_{n \geq 0} W_n t^n$ determined by

$$W_0 = I, \quad (n+1)W_{n+1} = XW_n + 2ZW_{n-1}, \quad W_{-1} = 0, \quad (17.3)$$

whose first terms are

$$W = I + tX + t^2\left(\frac{1}{2}X^2 + Z\right) + t^3\left(\frac{1}{6}X^3 + \frac{1}{3}XZ + \frac{2}{3}ZX\right) + \dots$$

This differs from $\exp(tX + t^2Z)$ at order t^3 by $(ZX - XZ)/6 \neq 0$. The support is contained in $\mathbb{N}$, so the series is strongly summable by Theorem 2.6 and the recurrence proves the equation exactly; $(\partial(W^*W)) = 0$ with standard part I makes it unitary, so it is the unique solution. This is a computation in a restricted Hahn workspace and is not a claim that such a recurrence is effectively available for a general coefficient matrix.

Theorem 17.3 is the same phenomenon as the two-scale computation of Section 27, which refutes the naive exponential for the *coordinate* derivative D_z ; the two live in different categories (Table 7) and neither is evidence for the other. What they share is the mechanism: a noncommuting pair of constant matrices produces an exact third-order defect that no exponential of an integral can absorb.

17.2 Real systems: rotation blocks and metric signatures

For a real system, conjugation sends phase P to phase $-P$, so the two multiplicities are equal.

Theorem 17.4 (Real rotation-block classification). *For every $A \in \text{Mat}_n(\mathbf{No})$ there is $G \in \text{GL}_n(\mathbf{No})$ with*

$$A^G = 0_{m_0} \oplus \bigoplus_{P>0} ((\partial P)\mathcal{R})^{\oplus m_P}, \quad n = m_0 + 2 \sum_{P>0} m_P, \quad (17.4)$$

where $\mathcal{R} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. The positive phases $P \in \mathcal{P}$ and their multiplicities are unique. Every finite real differential module over $\mathbf{No}$ is a direct sum of trivial one-dimensional modules and nontrivial two-dimensional rotation modules, and if $A^\top = -A$ then G can be chosen orthogonal.

Proof. Uniqueness of the complex phase subspaces (Theorem 15.13) makes conjugation interchange the P and $-P$ subspaces. On the zero-phase subspace the horizontal complex space is conjugation stable, so a real basis of its fixed part is a complex basis after complexification. For $P > 0$ take normalizing columns $v_1, \dots, v_{m_P}$ of phase P ; their conjugates span phase $-P$. Write $v_j = a_j + ib_j$. From $\partial v_j = Av_j - i(\partial P)v_j$,

$$\partial a_j = Aa_j + (\partial P)b_j, \quad \partial b_j = Ab_j - (\partial P)a_j,$$

which is the block $(\partial P)\mathcal{R}$. The real columns are independent because the change between $(v_j, \bar{v}_j)$ and (a_j, b_j) is invertible over $\mathbb{C}$. The only invariant real subspaces inside one nonzero phase pair are 0 and the whole pair, since a real subspace complexifies to a conjugation-stable sum of two distinct phase lines. For skew-symmetric A the identity metric is horizontal, so $G^\top G$ is a positive constant matrix commuting with the rotation blocks, by Theorem 17.5 below; its positive constant square root commutes with them too, and multiplying G by that inverse square root makes it orthogonal without changing the normal form. $\square$

Theorem 17.5 (The real metric cone and its parity). *Every real finite system has a positive symmetric horizontal metric. With the multiplicities of equation (17.4), the cone of positive symmetric horizontal metrics is isomorphic to*

$$\text{Sym}_{m_0}^+(\mathbb{R}) \times \prod_{P>0} \text{Herm}_{m_P}^+(\mathbb{C}), \quad (17.5)$$

a product over a finite set, empty products being a point. The real vector space of all horizontal symmetric forms has dimension $\frac{1}{2}m_0(m_0 + 1) + \sum_{P>0} m_P^2$, and on each nonzero real phase block both indices of a nondegenerate horizontal symmetric form are even.

Proof. Pull a real horizontal form back by the gauge of Theorem 17.4. Complex phase separation (Theorem 15.16) eliminates all terms between different positive phase pairs and between such a pair and phase zero, and leaves an arbitrary constant real symmetric matrix on the zero block. Group a repeated positive phase as $(\partial P)\mathcal{R}_m$ with $\mathcal{R}_m = \begin{pmatrix} 0 & I_m \\ -I_m & 0 \end{pmatrix}$; the real symmetric horizontal matrices are then the constant matrices commuting with $\mathcal{R}_m$, that is

$$C = \begin{pmatrix} S & T \\ -T & S \end{pmatrix}, \quad S^\top = S, \quad T^\top = -T,$$

which is the realification of the complex Hermitian matrix $S - iT$. A constant complex unitary diagonalization of $S - iT$ becomes a real orthogonal diagonalization of C with every eigenvalue repeated twice; this gives positivity, the cone formula and the parity statement. Congruence by G^{-1} returns to the original coordinates. $\square$

There is no integration over phases in equation (17.5): the products are over a finite multiset of constant-matrix cones. In particular a real irreducible two-dimensional system has a one-dimensional space of horizontal symmetric forms, hence a single positive ray. That fact is the source of the uniqueness in Theorem 17.6.

17.3 A positive Ermakov–Pinney dichotomy

For $q \in \mathbf{No}$ consider the intrinsic equations

$$\partial^2 y + qy = 0 \tag{17.6}$$

and

$$\partial^2 \rho + q\rho = \rho^{-3}, \quad \rho > 0. \tag{17.7}$$

Both derivatives are the fixed ∂ . The positive sign on the right of equation (17.7) is part of the statement; the sign convention in the classical literature varies [6].

Theorem 17.6 (Positive Pinney existence and uniqueness). *For every $q \in \mathbf{No}$, equation equation (17.7) has a positive solution in $\mathbf{No}$, and exactly one of the following holds.*

- (i) *Equation (17.6) has no nonzero solution in $\mathbf{No}$. Then equation (17.7) has exactly one positive solution in $\mathbf{No}$, and $\partial^2 y + qy = f$ has a unique solution in $\mathbf{No}$ for every $f \in \mathbf{No}$.*
- (ii) *Equation (17.6) has a real fundamental pair in $\mathbf{No}$. Then the positive solutions of equation (17.7) form the classical positive quadratic-form family equation (17.10), with two real parameters after determinant normalization.*

Proof. Let $A_q = \begin{pmatrix} 0 & 1 \\ -q & 0 \end{pmatrix}$. By Theorem 17.5 it has a positive symmetric horizontal metric $\mathcal{M}$, and equation (15.6) gives $\partial \det \mathcal{M} = -2(\operatorname{tr} A_q) \det \mathcal{M} = 0$, so $\det \mathcal{M}$ is a positive real constant; scale it to 1. Writing $\mathcal{M} = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$, the horizontality equations are

$$\partial a = 2qb, \quad \partial b = qc - a, \quad \partial c = -2b. \tag{17.8}$$

Since $c > 0$ and $\mathbf{No}$ is real closed, put $\rho = \sqrt{c}$. The last equation gives $b = -\rho \partial \rho$ and $\det \mathcal{M} = 1$ gives $a = (\partial \rho)^2 + \rho^{-2}$; the middle equation becomes exactly equation (17.7). Conversely a positive solution of equation (17.7) produces the positive determinant-one horizontal metric

$$\mathcal{M}_\rho = \begin{pmatrix} (\partial \rho)^2 + \rho^{-2} & -\rho \partial \rho \\ -\rho \partial \rho & \rho^2 \end{pmatrix}, \tag{17.9}$$

positive because its lower diagonal entry is positive and its determinant is 1. So positive Pinney solutions correspond bijectively to positive determinant-one horizontal metrics.

In dimension two the real normal form equation (17.4) has either two zero phases or one nonzero conjugate pair. If equation (17.6) has no nonzero solution in $\mathbf{No}$ it must be the second case, the positive metric cone is a single ray by equation (17.5), and the determinant-one condition selects one point. In the zero-phase case a real fundamental matrix exists and the cone is the three-dimensional positive symmetric constant-matrix cone, whose determinant-one slice is the asserted two-parameter family. In the nonzero-phase case Theorem 15.11 gives a unique solution in $\mathbf{No}[i]^2$ of every forced system, and conjugating shows it is real when f is.

There is no intermediate one-dimensional case: if $u \in \mathbf{No}^\times$ solves equation (17.6), choose B with $\partial B = u^{-2}$ by (BM6) and put $v = uB$; then $u\partial v - v\partial u = 1$, so u, v are an independent real fundamental pair. $\square$

With u, v a real fundamental pair of constant Wronskian $\text{Wr} = u\partial v - v\partial u \neq 0$, alternative (ii) reads

$$\rho = \sqrt{\alpha u^2 + 2\beta uv + \gamma v^2}, \quad \alpha, \gamma > 0, \quad \alpha\gamma - \beta^2 = \text{Wr}^{-2}, \quad \alpha, \beta, \gamma \in \mathbb{R}. \quad (17.10)$$

This is the classical Pinney parametrization [6], here read in an ordered differential field; no priority is claimed for it, nor for the abstract Schwarz-closed-field existence principle that underlies it [4]. What is new to this field is that alternative (i) can occur, and that it collapses the positive family to a single point. For recent existence and sharp uniqueness results for amplitude–phase representations over Hardy fields see [5]; no solution of the named Boshernitzan conjecture discussed there is claimed here.

For $q = 1$ alternative (i) holds by Theorem 8.8, and the unique positive Pinney solution is $\rho = 1$. For $q = -1$ the linear equation has the fundamental pair $e^\omega, e^{-\omega}$ and alternative (ii) gives many positive amplitudes. A unique positive solution of a second-order equation is not a contradiction to ordinary initial-value theory: intrinsic differentiation has no two-real-parameter initial-data interpretation (Theorem 4.1).

Theorem 17.7 (Canonical amplitude–phase reduction). *Call $q \in \mathbf{No}$ intrinsically oscillatory when alternative (i) of Theorem 17.6 holds; this means only that the nonzero homogeneous oscillations are missing from $\mathbf{No}$, and says nothing about oscillation of a function of a real variable. Let ρ be the unique positive Pinney solution, $h = \rho^{-2}$, and $B = \mathcal{I}_0(h) = P + \eta$ with $P \in \mathcal{P}$ and $\eta \in \mathfrak{m}$ (equation (2.3)). Then $P > 0$, and with*

$$G_\rho = \begin{pmatrix} \rho & 0 \\ \partial\rho & \rho^{-1} \end{pmatrix}, \quad R(\eta) = \text{Re}(\mathcal{E}(i\eta)) I + \text{Im}(\mathcal{E}(i\eta)) \mathcal{R},$$

the matrix $S_q = G_\rho R(\eta) \in \text{SL}_2(\mathbf{No})$ satisfies

$$A_q^{S_q} = (\partial P) \mathcal{R}. \quad (17.11)$$

The amplitude ρ , the normalized primitive B , the phase P and this amplitude-times-infinitesimal-rotation gauge are canonical; no uniqueness among all symplectic gauges is asserted.

Proof. A direct multiplication using equation (17.7) gives $A_q^{G_\rho} = \rho^{-2} \mathcal{R} = h \mathcal{R}$. The rotation $R(\eta)$ is a well-defined real matrix with entries the infinitesimal cosine and sine of Theorem 6.3; it has determinant one, commutes with $\mathcal{R}$, and satisfies $R(\eta)^{-1} \partial R(\eta) = (\partial \eta) \mathcal{R}$. Subtracting gives equation (17.11). If $P = 0$ then $B = \eta$ is finite and the same calculation reduces the system to

zero, producing a real fundamental matrix and contradicting alternative (i). Since $\partial B = h > 0$, B cannot be negative infinite by (BM7), so its nonzero purely infinite part is positive. The uniqueness assertions are Theorem 17.6 and the normalization $\text{ct}(B) = 0$ of Theorem 7.1. $\square$

The conserved quadratic expression attached to ρ is

$$\mathcal{E}_\rho(y, \partial y) = (\rho \partial y - y \partial \rho)^2 + (y/\rho)^2, \quad (17.12)$$

whose derivative vanishes formally along equation (17.6) by equation (17.9). It is meaningful in an oscillatory extension of the kind constructed in Section 21 even when the only solution in $\mathbf{No}$ is zero. It is an exact invariant, not an averaged energy.

17.4 The oscillatory Airy equation as an exact Hahn example

Let $x = \omega$ and consider

$$\partial^2 y + xy = 0. \quad (17.13)$$

Put $P = \frac{2}{3}x^{3/2}$, so $\partial P = x^{1/2}$ by the real-power rule (Theorem 2.2, applied at the ordinary real exponent $\frac{3}{2}$). Let T_P denote a formal oscillatory symbol with $\partial \mathsf{T}_P = i(\partial P)\mathsf{T}_P$ in the sense of Theorem 21.1; it is a new differential element and never a surcomplex number. Seek a solution $a\mathsf{T}_P$ with

$$a = x^{-1/4} \sum_{n \geq 0} c_n x^{-3n/2}, \quad c_0 = 1. \quad (17.14)$$

Substituting into equation (17.13) gives

$$\partial^2 a + 2ix^{1/2}\partial a + \frac{i}{2x^{1/2}}a = 0, \quad (17.15)$$

and with $\alpha_n = -\frac{1}{4} - \frac{3}{2}n$ coefficient comparison yields

$$c_{n+1} = \frac{\alpha_n(\alpha_n - 1)}{3i(n+1)}c_n = -i \frac{(6n+1)(6n+5)}{48(n+1)}c_n, \quad c_n = (-i)^n \left(\frac{3}{4}\right)^n \frac{(1/6)_n(5/6)_n}{n!}. \quad (17.16)$$

The exponents in equation (17.14) decrease strictly to $-\infty$, so the displayed expression is a legitimate surcomplex Hahn sum however fast the ordinary coefficients grow, and its termwise differentiation is justified by (BM3). This is the same phenomenon as the factorially divergent solutions of Theorem 18.13, and it carries the same disclaimer (Theorem 18.14): a strictly descending support is a summability certificate, not convergence in any topology on $\mathbf{No}[i]$.

Put $b = \partial a + ix^{1/2}a$ and $G = \begin{pmatrix} a & \bar{a} \\ b & \bar{b} \end{pmatrix}$. The two columns satisfy $\partial v = A_x v \mp i(\partial P)v$ with $A_x = \begin{pmatrix} 0 & 1 \\ -x & 0 \end{pmatrix}$, so $\partial G = A_x G - G \text{diag}(i\partial P, -i\partial P)$. The determinant $\text{Wr} = a\bar{b} - \bar{a}b$ has zero derivative, hence lies in $\mathbb{C}$, and its leading term is $-2i$ because $a \sim x^{-1/4}$ and $b \sim ix^{1/4}$; so $\text{Wr} = -2i$ exactly, G is invertible, and

$$\text{Ph}(A_x) = \left\{ \frac{2}{3}\omega^{3/2}, -\frac{2}{3}\omega^{3/2} \right\}. \quad (17.17)$$

By Theorem 15.11 equation (17.13) has no nonzero solution in $\mathbf{No}[i]$, and $\partial^2 y + \omega y = f$ has a unique solution in $\mathbf{No}[i]$ for every f . As always in this report that is a statement about surcomplex *number* solutions under ∂ ; it does not deny the ordinary Airy functions of a complex coordinate (Theorem 8.9).

The unique positive solution of $\partial^2 \rho + \omega \rho = \rho^{-3}$ is

$$\rho = \sqrt{a\bar{a}}. \quad (17.18)$$

Indeed, with $v = \begin{pmatrix} a \\ b \end{pmatrix}$ the real matrix $C = \operatorname{Re}(vv^*)$ satisfies $\partial C = A_x C + C A_x^\top$ — the scalar imaginary phase cancels — and $\det C = (\operatorname{Im}(a\bar{b}))^2 = 1$ since $\operatorname{Wr} = -2i$; so $\mathcal{M} = C^{-1}$ is a positive determinant-one horizontal metric and the bijection of Theorem 17.6 gives equation (17.18), with uniqueness because the phases equation (17.17) are nonzero. Its expansion begins

$$\rho = \omega^{-1/4} \left(1 - \frac{5}{64} \omega^{-3} + \frac{2285}{8192} \omega^{-6} - \frac{1690275}{524288} \omega^{-9} + \frac{10371973875}{134217728} \omega^{-12} + \dots \right), \quad (17.19)$$

these being coefficients of an exact Hahn series and not estimates for a convergent ordinary power series. The normalized primitive of ρ^{-2} is $B = \frac{2}{3} \omega^{3/2} - \frac{5}{48} \omega^{-3/2} + \dots$, whose purely infinite part is exactly the positive phase in equation (17.17); the remaining correction is infinitesimal and is absorbed by the rotation $R(\eta)$ of Theorem 17.7.

For a truncation $a_N = \sum_{n \leq N} c_n x^{\alpha_n}$ the recurrence cancels everything but the last second derivative, giving the exact residual

$$\partial^2 a_N + 2ix^{1/2} \partial a_N + \frac{i}{2x^{1/2}} a_N = \alpha_N(\alpha_N - 1) c_N x^{\alpha_N - 2}, \quad (17.20)$$

which the delivered suite checks at $N = 16$ (Section D). As Theorem 29.3 requires, the residual records the *type* of the omitted information: it is a formal remainder at a stated order, and the infinite-support statement is proved by the recurrence together with strict descent of exponents, not by the finite check.

17.5 First integrals from phase resonance

Let $y_1, \dots, y_n$ be algebraically independent formal variables and extend ∂ to $\mathbf{No}[i][y_1, \dots, y_n]$ by

$$\mathcal{D}_A f = \partial_{\text{coeff}} f + \sum_j (A y)_j \frac{\partial f}{\partial y_j},$$

where ∂_{coeff} differentiates coefficients only. A polynomial or rational *first integral* is an element killed by $\mathcal{D}_A$. This definition presupposes no nonzero solution in $\mathbf{No}[i]^n$. In phase coordinates $z = G^{-1}y$ one has $\mathcal{D}_A z_j = i(\partial P_j) z_j$.

Put

$$\Pi = \sum_{j=1}^n \mathbb{Z} P_j \subseteq \mathcal{P}, \quad \Lambda = \left\{ m \in \mathbb{Z}^n : \sum_j m_j P_j = 0 \right\}, \quad \Lambda^+ = \Lambda \cap \mathbb{N}^n,$$

and $r = \dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}} \{P_1, \dots, P_n\}$. This Λ is the relation lattice of equation (21.6) for the diagonal system $\operatorname{Ph}(A)$; it is saturated by Theorem 21.11, so $\operatorname{rank} \Lambda = n - r$ and Π is free abelian of rank r .

Theorem 17.8 (The resonance algebra). *In phase coordinates,*

$$\ker \mathcal{D}_A|_{\mathbf{No}[i][z]} = \mathbb{C}[z^m : m \in \Lambda^+], \quad \ker \mathcal{D}_A|_{\mathbf{No}[i](z)} = \mathbb{C}(z^m : m \in \Lambda). \quad (17.21)$$

The polynomial algebra is finitely generated, and the rational field is purely transcendental over $\mathbb{C}$ of transcendence degree $n - r$.

Proof. For $f = \sum_{\alpha} c_{\alpha} z^{\alpha}$ the derivative vanishes exactly when $\partial c_{\alpha} + i(\partial \sum_j \alpha_j P_j) c_{\alpha} = 0$ for each α , and Theorem 15.3 makes a nonzero coefficient require $\alpha \in \Lambda^+$ and $c_{\alpha} \in \mathbb{C}$.

For the rational statement work in the Laurent ring and write f/g with coprime nonzero Laurent polynomials. A zero derivative gives $f \mid \mathcal{D}_A f$ and $g \mid \mathcal{D}_A g$; since $\mathcal{D}_A$ enlarges no Laurent support and the highest and lowest degrees in each variable are additive under multiplication, one gets $\mathcal{D}_A f = hf$ and $\mathcal{D}_A g = hg$ for a common $h \in \mathbf{No}[i]$, the case $h = 0$ included. Comparing two coefficients of f shows their ratio has logarithmic derivative $i\partial(\beta \cdot P - \alpha \cdot P)$, so by Theorem 15.3 their phases agree and the ratio is constant; thus all monomials of f share one phase, and after factoring out one coefficient and one monomial the rest has constant coefficients and exponents in Λ . The same holds for g , and equality of the two values of h forces the factored phases to agree and the factored coefficient ratio to be constant. Hence f/g is a rational function of monomials with exponents in Λ ; the reverse inclusion is immediate.

For finite generation, call $0 \neq u \in \Lambda^+$ indecomposable when it is not a sum of two nonzero members of Λ^+ . Two distinct indecomposables are incomparable in the coordinatewise order on $\mathbb{N}^n$, since $u < v$ would give $v - u \in \Lambda^+ \setminus \{0\}$; Dickson's finite-antichain lemma for $\mathbb{N}^n$ — the special case of Section A in which the alphabet is $\{1, \dots, n\}$ and words are unordered — therefore makes the indecomposables finite, and induction on total degree writes every member of Λ^+ as a sum of them. Finally a $\mathbb{Z}$ -basis of Λ gives $n - r$ Laurent monomials generating the rational field, algebraically independent because distinct integer combinations of basis exponents give distinct monomials. No analytic finiteness principle is used. $\square$

Example 17.9 (Complete reducibility does not make the invariant algebra a polynomial ring). Let $x = \omega$ and

$$G = \begin{pmatrix} 1 & x^{-1} & 0 \\ 0 & 1 & x^{-2} \\ 0 & 0 & 1 \end{pmatrix}, \quad D = \text{diag}(i, i, -2i), \quad A = (\partial G)G^{-1} + GDG^{-1},$$

so that

$$A = \begin{pmatrix} i & -x^{-2} & x^{-4} \\ 0 & i & (-3ix - 2)x^{-3} \\ 0 & 0 & -2i \end{pmatrix} \quad (17.22)$$

and $\text{Ph}(A) = \{\omega, \omega, -2\omega\}$, so by Theorem 15.11 the system has no nonzero solution in $\mathbf{No}[i]^3$. Its formal first integrals are nevertheless nontrivial: with $z = G^{-1}y$ the resonance condition is $\alpha_1 + \alpha_2 = 2\alpha_3$ and

$$\ker \mathcal{D}_A|_{\mathbf{No}[i][z]} \cong \mathbb{C}[\mathcal{U}, \mathcal{V}, \mathcal{W}]/(\mathcal{U}\mathcal{W} - \mathcal{V}^2), \quad \mathcal{U} = z_1^2 z_3, \quad \mathcal{V} = z_1 z_2 z_3, \quad \mathcal{W} = z_2^2 z_3, \quad (17.23)$$

the rational invariant field being $\mathbb{C}(z_1/z_2, z_2^2 z_3)$ of transcendence degree two. Here $r = 1$, matching Theorem 21.13(i) in kind: ω and -2ω are $\mathbb{Q}$ -dependent. The three first-integral identities are checked in the *original* coupled coordinates by the delivered suite.

17.6 The phase lattice as a differential Galois torus

Theorem 21.12 classifies the Picard–Vessiot extension of a *diagonal* system by its integer relation lattice. Theorem 15.9 reduces every finite system to a diagonal one, so that theorem now applies to all of them. The point of this subsection is only to record what the reduction gives, and to add the two ingredients the diagonal treatment did not need: an explicit group-algebra presentation in rank r , and the real form. Nothing here is new Galois theory, and the eighth manuscript says so of itself.

Corollary 17.10 (Every finite system has a phase torus). *Let $A \in \text{Mat}_n(\mathbf{No}[i])$ with $\text{Ph}(A) = \{P_1, \dots, P_n\}$, phase lattice $\Pi = \sum_j \mathbb{Z}P_j$ of rank r , and relation lattice Λ as in equation (21.6). Then the number $d = n - \text{rank } \Lambda$ of independent phase generators of Theorem 5.1(d) equals r , and Theorem 21.12 applies to the normal form of A : a Picard–Vessiot extension for $\partial y = Ay$ is obtained by adjoining r algebraically independent generators, and its differential Galois group is $\mathbf{G}_m(\mathbb{C})^r \cong \text{Hom}(\Pi, \mathbb{C}^\times)$.*

Proof. Λ is the kernel of the surjection $\mathbb{Z}^n \rightarrow \Pi$, $m \mapsto \sum m_j P_j$, so $\mathbb{Z}^n/\Lambda \cong \Pi$ is free of rank r by Theorem 21.11, giving $d = r$; and $\text{Hom}(\Pi, \mathbb{C}^\times) \cong (\mathbb{C}^\times)^r$ is the description of equation (21.7) in coordinates dual to a $\mathbb{Z}$ -basis of Π . The Picard–Vessiot statement is Theorem 21.12 applied to $i \text{diag}(\partial P_1, \dots, \partial P_n)$, transported back by the gauge of Theorem 15.9, which lies in $\text{GL}_n(\mathbf{No}[i])$ and therefore generates no new constants. $\square$

Proposition 17.11 (The group algebra of the phase lattice, and its real form). *Choose a $\mathbb{Z}$ -basis $Q_1, \dots, Q_r$ of Π and form the Laurent differential algebra*

$$\mathcal{L}_\Pi = \mathbf{No}[i][T_1^{\pm 1}, \dots, T_r^{\pm 1}], \quad \partial T_k = i(\partial Q_k)T_k, \quad (17.24)$$

writing $T_Q = \prod_k T_k^{m_k}$ for $Q = \sum_k m_k Q_k \in \Pi$, so that $T_Q T_{Q'} = T_{Q+Q'}$ and $\partial T_Q = i(\partial Q)T_Q$. Then:

- (i) $\mathcal{L}_\Pi$ has no nonzero proper differential ideal;
- (ii) if G normalizes A then $Y = G \text{diag}(T_{P_1}, \dots, T_{P_n})$ is a fundamental matrix for $\partial y = Ay$, and its entries together with $(\det Y)^{-1}$ generate $\mathcal{L}_\Pi$ over $\mathbf{No}[i]$;
- (iii) for a real system Π is stable under negation, the involution $\overline{cT_Q} = \bar{c}T_{-Q}$ commutes with ∂ , the fixed combinations $(T_Q + T_{-Q})/2$ and $(T_Q - T_{-Q})/2i$ supply the entries of a real fundamental matrix in the normal form equation (17.4), and the real form of the torus is $(\text{Res}_{\mathbb{C}/\mathbb{R}}^1 \mathbf{G}_m)^r$, whose real points are $(S^1)^r$.

Every element of $\mathcal{L}_\Pi$ is a finite sum; no Hahn summability question arises, and no exponential e^{iP} is defined in $\mathbf{No}[i]$.

Proof. (i) Let $\mathfrak{a} \neq 0$ be a differential ideal and choose $f \in \mathfrak{a}$ with the fewest nonzero monomials; multiplying by a Laurent monomial and a nonzero scalar, arrange $f = 1 + \sum_{Q \neq 0} c_Q T_Q$. Differentiation kills the constant term and introduces no new monomials, so ∂f has fewer terms and must vanish; then $c_Q^\dagger = -i\partial Q$ for each remaining Q , impossible for $Q \neq 0$ by Theorem 8.4. Hence $f = 1$. This is the rank- r form of the rank-one statement proved in Theorem 21.2.

(ii) $\partial G = AG - iG \text{diag}(\partial P_j)$ gives $\partial Y = AY$ at once. Conversely the entries of $G^{-1}Y$ supply every T_{P_j} , and $(\det Y)^{-1}$ supplies the inverse of their product, so multiplying by all but one supplies each $T_{P_j}^{-1}$; these characters generate the group algebra of Π .

(iii) Stability under negation is the conjugation symmetry used in Theorem 17.4, and $\overline{\partial(cT_Q)} = (\partial \bar{c} + i\bar{c}\partial Q)T_Q = (\partial \bar{c} - i\bar{c}\partial Q)T_{-Q} = \partial(\bar{c}T_{-Q})$. The fixed combinations satisfy ∂ -relations of the rotation block $(\partial Q)\mathcal{R}$, by the computation of Theorem 21.9 applied with $a = i\partial Q$. An automorphism commuting with the involution has $\chi(Q)\overline{\chi(Q)} = 1$, which is the asserted real form. $\square$

Warning 17.12 (What the torus statement is about). Theorems 17.10 and 17.11 concern the very large base field $\mathbf{No}[i]$, with the factorization and surjectivity properties of Theorem 15.6. They say nothing comparable for the same printed equation over a small rational-function base: in particular equation (17.17) is *not* a claim that the ordinary Airy differential Galois group over $\mathbb{C}(x)$ is a

torus. Enlarging the base to $\mathbf{No}[i]$ has already supplied the nonoscillatory gauge entries. Nor is the real form of Theorem 17.11(iii) a construction of trigonometric functions on surreals; it is an algebraic-group statement, and Theorem 21.10 records what such an extension costs. Finally, when $\Pi \neq 0$ no differential embedding $\mathcal{L}_\Pi \rightarrow \mathbf{No}[i]$ over $\mathbf{No}[i]$ exists, by Theorem 21.8: the extension represents the obstruction, it does not remove it.

17.7 What this part does not settle

Warning 17.13 (The boundaries of the matrix theory). Part III classifies systems $\partial y = Ay$ for A an $n \times n$ matrix over $\mathbf{No}[i]$ with n an ordinary finite integer, and computes the invariant from algebraic eigenvalues only when $A = iH$ with H Hermitian. It does not classify non-Hermitian coefficients by their spectra — Theorem 16.10 refutes that — does not treat infinite matrices, does not supply an algorithm for the phase map or the normalizing gauge, and does not settle which nontriangular differential modules admit a filtration whose rank-one factors carry computable obstructions, which needs genuine module-theoretic information rather than the eigenvalues of an instantaneous coefficient matrix. Higher-rank differential modules remain outside the classification. All statements are for the one fixed ∂ of Theorem 3.1 and for no other derivation, no class-function derivative and none of the coordinate operators of Table 1. The consolidated list is Section 31.9.

Remark 17.14 (Proper classes, and a set-sized reading). $\mathbf{No}$ is a proper class, and Section 2.1 fixes the framing this report uses. Nothing in Part III makes a proper class an element of a set or needs a class-sized basis of $\mathcal{P}$: every phase lattice Π is finitely generated and every element of $\mathcal{L}_\Pi$ is a finite sum. There is also a set-sized reading of each finite-data argument, in the spirit of Theorem 20.8: starting from a set of given coefficients together with $\mathbb{R} \cup \{\omega\}$, form a countable chain of set-sized subfields of $\mathbf{No}$, adjoining at each stage the real algebraic roots, derivatives, real exponentials, purely infinite parts, primitives and finite angles needed by the previous stage, together with witnesses for the finitely many factorizations and first-order inhomogeneous equations of Theorem 15.6 whose coefficients appear there. The union is a real closed differential subfield $\mathbf{No}_0$ inheriting the H -field rule and the designated witnesses, whose complexification has constant field exactly $\mathbb{C}$; every finite tuple appears at some stage and its witnesses at a later one, so the elementary lemmas of Section 7 and all the finite matrix arguments hold internally. Theorem 17.11 may therefore be read as an ordinary set-sized Picard–Vessiot statement over such an $\mathbf{No}_0[i]$.

This is a size justification for the proofs and nothing more. It does *not* say that any fixed Hahn workspace already contains the required primitives or solutions — Theorem 18.2 shows one that does not — and it gives no effective presentation, no birthday bound and no minimal such subfield.

Part IV

Workspaces: Laurent, rational, Hahn, and set-sized

18 The Laurent parent and its resolvents

18.1 A differential subfield with a simple valuation

Throughout this section $t = \omega^{-1}$ and, unless stated otherwise, $\Gamma \subseteq \mathbb{R}$ is an additive subgroup with $1 \in \Gamma$, so that $\Gamma + 1 = \Gamma$. Put

$$K_\Gamma = \mathbb{C}((t^\Gamma)) \subseteq \mathbf{No}[i], \quad H_\Gamma = \mathbb{R}((t^\Gamma)) \subseteq \mathbf{No},$$

embedded by Hahn summation of Conway monomials, and write $K_0 = \mathbb{C}((t))$ for the case $\Gamma = \mathbb{Z}$, the ordinary formal Laurent series field with ordinary complex coefficients. For nonzero f let $v(f)$ be the least exponent with nonzero coefficient, with $v(0) = +\infty$.

Proposition 18.1 (The derivation on a real-exponent workspace). *K_Γ is ∂ -stable, its constants are exactly $\mathbb{C}$, and*

$$\partial \left(\sum_{r \in S} a_r t^r \right) = - \sum_{r \in S} r a_r t^{r+1}, \quad \text{so} \quad v(\partial f) \geq v(f) + 1, \quad (18.1)$$

and hence $v(\partial^n f) \geq v(f) + n$ for every $n \geq 0$. On K_0 this is exactly the operator

$$D := \partial|_{K_0} = -t^2 \frac{d}{dt}. \quad (18.2)$$

Furthermore

$$\partial K_\Gamma = \{f \in K_\Gamma : [t^1]f = 0\}, \quad (18.3)$$

and for $f = \sum a_r t^r$ with $a_1 = 0$ a primitive inside K_Γ is $\sum_{r \neq 1} \frac{a_r}{1-r} t^{r-1}$; for general $f \in K_\Gamma$ the normalized primitive in the ambient $\mathbf{No}[i]$ is

$$\mathcal{I}_0(f) = a_1 \log \omega + \sum_{r \neq 1} \frac{a_r}{1-r} t^{r-1}, \quad (18.4)$$

which is normalized because $\log \omega$ is purely infinite and so has zero constant coefficient.

Proof. The real-exponent rule $\partial t^r = -r t^{r+1}$ of equation (3.1) and strong additivity (BM3) give equation (18.1); translation of a well-ordered support by 1 preserves well ordering and distinct exponents stay distinct, and $\Gamma + 1 = \Gamma$ keeps the result in the field. A nonconstant term has a nonzero derivative at a unique shifted exponent, so the constants are the exponent-0 terms, that is $\mathbb{C}$. The only term that could contribute at exponent 1 has exponent 0, whose multiplier is zero, which proves necessity in equation (18.3); conversely the displayed primitive has a support that is a translate of a subset of $\text{supp } f$, hence well ordered and inside Γ , and termwise differentiation returns f . For equation (18.4) use $\partial \log \omega = t$. $\square$

Example 18.2 (A missing logarithmic primitive). iff:placeholderX $K_0 = \mathbb{C}((t))$ is a differential field, yet t has no primitive in it: by equation (18.1) the coefficient of t^1 in ∂f is always zero, since it could only arise from exponent 0, whose derivative coefficient vanishes. In the full surreal field $\partial \log \omega = t$. The same holds over $H_\mathbb{Q} = \mathbb{R}((t^\mathbb{Q}))$, which is ∂ -stable while t has no primitive in it, its ambient primitives being $\log \omega + c$ with $c \in \mathbb{R}$.

Warning 18.3 (Differentiation closure is not integration closure). So ∂ is *not* surjective on K_Γ , even though ∂ is surjective on $\mathbf{No}[i]$. This is a failure of closure of a chosen representation parent, not a failure of ambient surjectivity, and the whole-field surjectivity clause (BM6) must not be silently transported to a localized workspace. Algebraic localization, algebraic closure, derivative stability, and closure under primitives are four separate properties; see Section 20 for the counterexamples separating them. There is a formal differential-form reading of the obstruction: since $d\omega = -t^{-2} dt$, the coefficient of $t^{-1}dt$ in $f d\omega$ is $-a_1$. That is a formal Laurent/Hahn residue statement only; it is not an identification with a cluster residue or an actual-point contour residue from the repository's analytic reports.

Over the real form, the ideal $\mathcal{A}$ intersects the Laurent field exactly.

Corollary 18.4 (A valuation test, on real exponents only). *For real $f \in H_\Gamma$ with $\Gamma \subseteq \mathbb{R}$,*

$$f \in \mathcal{A} \iff f = 0 \text{ or } v(f) > 1,$$

where $\mathcal{A}$ is computed in the full surreal field. In particular $t^2\mathbb{R}[[t]] = \mathcal{A} \cap \mathbb{R}((t))$ on the real Laurent field.

Proof. This is Theorem 10.1 restated through equation (18.4). If $v(f) > 1$ every exponent of the primitive is positive, so it is infinitesimal. If the least exponent r is below 1, the primitive has the nonzero leading infinite term $a_r t^{r-1}/(1-r)$, which dominates any logarithmic term and cannot be removed by a real constant. If the least exponent is exactly 1 the logarithmic coefficient is nonzero and all other primitive terms are infinitesimal, so the primitive is again infinite. A logarithmic term cannot cancel a Laurent principal part, and primitives differ only by real constants. $\square$

Warning 18.5 (Not a valuation test for arbitrary coefficients). Theorem 18.4 uses *real exponents relative to $t = \omega^{-1}$* . It is not a valuation-only classification of arbitrary surreal coefficients: Theorem 10.4 is exactly the information that this real-exponent workspace cannot see.

18.2 Resolvents as exact Hahn sums

Theorem 18.6 (First-order resolvent on a real-exponent workspace). *Let $\Gamma \subseteq \mathbb{R}$ be an additive subgroup with $1 \in \Gamma$, and let $\lambda \in \mathbb{C}^\times$. Then $\partial - \lambda$ is a bijection of K_Γ , with inverse the strongly summable operator series*

$$(\partial - \lambda)^{-1} f = - \sum_{n \geq 0} \lambda^{-n-1} \partial^n f, \tag{18.5}$$

and for the truncation $y_N = - \sum_{n=0}^N \lambda^{-n-1} \partial^n f$ the residual is exactly

$$(\partial - \lambda)y_N - f = -\lambda^{-(N+1)} \partial^{N+1} f. \tag{18.6}$$

Proof. For $f \neq 0$ put $S = \text{supp } f$ and $s_0 = \min S$. By equation (18.1), $\text{supp } \partial^n f \subseteq S + n$, and $S + \mathbb{N}$ is well ordered: a decreasing sequence in it would have integer parts bounded above by its first term minus s_0 , and an infinite subsequence with the same integer part would be a decreasing sequence in S . At a fixed exponent γ , a contribution from $\partial^n f$ requires $\gamma - n \in S$, hence $n \leq \gamma - s_0$, and $\gamma - s_0$ is an *ordinary real number*, so only finitely many n occur. This proves strong summability. Strong additivity (BM3) allows ∂ to be applied termwise, and the terms cancel pairwise after reindexing, leaving f ; the finite identity equation (18.6) is the same cancellation truncated. For injectivity, if $y \neq 0$ then either $\partial y = 0$ or $v(\partial y) \geq v(y) + 1 > v(y)$ by equation (18.1), so the leading term of λy cannot be cancelled; hence $(\partial - \lambda)y \neq 0$. $\square$

Warning 18.7 (The real-exponent hypothesis is substantive). The step “ $\gamma - s_0$ is an ordinary real number, so only finitely many n occur” is where $\Gamma \subseteq \mathbb{R}$ is used. In a higher-rank value group a fixed increment 1 need not dominate all relevant scales, and no claim is made here that successive valuation improvements always converge. A support argument has not been replaced by a convergence claim, and Theorem 18.6 is not asserted for an arbitrary set-sized ordered Γ .

Proposition 18.8 (Polynomial Laurent inverse). *Let $P \in \mathbb{C}[T]$ with $P(0) \neq 0$ and formal reciprocal $1/P(T) = \sum_{n \geq 0} q_n T^n \in \mathbb{C}[[T]]$. For every $f \in K_0$ the family $(q_n \partial^n f)_{n \geq 0}$ is strongly summable, $P(\partial) : K_0 \rightarrow K_0$ is bijective, and*

$$P(\partial)^{-1} f = \sum_{n \geq 0} q_n \partial^n f. \quad (18.7)$$

If $Q_N(T) = \sum_{n=0}^N q_n T^n$ and $y_N = Q_N(\partial)f$, then $P(T)Q_N(T) - 1$ has order at least $N + 1$, so

$$v(P(\partial)y_N - f) \geq v(f) + N + 1. \quad (18.8)$$

Proof. Summability is the valuation bound of equation (18.1) exactly as in Theorem 18.6. Strong additivity permits applying each of the finitely many derivatives in $P(\partial)$ termwise; at every exponent the calculation is finite, so the formal identity $P(T) \sum q_n T^n = 1$ gives $P(\partial)y = f$. For injectivity, the leading term of $P(\partial)y$ is $P(0)$ times the leading term of y , every positive-order derivative having strictly larger valuation. Alternatively one may factor P over $\mathbb{C}$ and compose the first-order resolvents of Theorem 18.6 with multiplicities, which gives the same conclusion; an effective implementation then needs exact representations of the roots, or can derive a coefficient recurrence from P directly. $\square$

The hypothesis $P(0) \neq 0$ is essential, and its failure is exactly Theorem 18.2: when $P(0) = 0$ the factor ∂ carries the obstruction equation (18.3), which must not be handled by a division by zero inside a resolvent routine. Theorem 18.8 covers both $P(T) = T - i$ and $P(T) = T^2 + 1$. So the absence of a nonzero homogeneous oscillator coexists with unique solvability of *every* inhomogeneous oscillator equation whose forcing lies in K_0 — an explicit instance of Theorem 12.2.

Remark 18.9 (Parent selection changes homogeneous solution spaces). For nonreal λ , $\partial - \lambda$ has trivial kernel in all of $\mathbf{No}[i]$ and Theorem 18.6’s uniqueness is uniqueness in $\mathbf{No}[i]$. For real nonzero λ the Laurent solution is unique *inside* K_Γ , while the full field also contains $ce^{\lambda\omega}$, which lies outside K_Γ . The same inverse formula holds in both cases. A theorem whose uniqueness statement changes when the ambient field is enlarged must carry its parent as part of its statement, and an algorithm must return that parent.

18.3 Factorially divergent exact solutions

Since $\partial^n t = (-1)^n n! t^{n+1}$, the resolvents produce exact Hahn series whose coefficients grow factorially. Three instances occur in the source manuscripts and all three are recorded here.

Example 18.10 ($(\partial - i)y = t$). Applying Theorem 18.6 with $\lambda = i$ and $f = t$, or equivalently $1/(s - i) = i \sum_{n \geq 0} (-i)^n s^n$ in equation (18.7), gives the unique solution in K_0

$$y = \sum_{n \geq 0} i^{n+1} n! t^{n+1} = it - t^2 - 2it^3 + 6t^4 + 24it^5 - 120t^6 - \dots, \quad (18.9)$$

which by Theorem 8.8 is the unique solution in all of $\mathbf{No}[i]$. Its truncation residual is equation (18.6); explicitly, writing y_N for the sum through $n = N$, $(\partial - i)y_N - t = -i^{-(N+1)}(-1)^{N+1}(N+1)!t^{N+2}$.

Example 18.11 $((\partial + 1)Y = t$, and a beyond-all-orders ambiguity). With $\lambda = -1$ and $f = t$,

$$Y_H = \sum_{n \geq 0} n! t^{n+1}, \quad (\partial + 1)Y_H = t, \quad (18.10)$$

with support $\{1, 2, 3, \dots\}$, each exponent occurring once; and for $Y_N = \sum_{n=0}^{N-1} n! t^{n+1}$ the exact residual is

$$(\partial + 1)Y_N - t = -N! t^{N+1}. \quad (18.11)$$

Proposition 18.12 (Uniqueness depends on the ambient field). *Equation (18.10) has exactly one solution in K_Γ , but all its solutions in $\mathbf{No}[i]$ are*

$$Y_H + C e^{-\omega}, \quad C \in \mathbb{C}. \quad (18.12)$$

Every nonzero correction $C e^{-\omega}$ is smaller in modulus than t^r for every ordinary real r , and lies outside K_Γ .

Proof. Workspace uniqueness is Theorem 18.6. In the full field the difference of two solutions satisfies $\partial Z = -Z$, so Theorem 8.8 gives $Z = C e^{-\omega}$. From $\log \omega \prec \omega$ one gets $-\omega < -r \log \omega$ for every real r after allowing any fixed real multiplicative coefficient; exponentiating gives $|C| e^{-\omega} \prec t^r$. A nonzero real-exponent Hahn series has a least real exponent with a nonzero ordinary leading coefficient, so it cannot be smaller than every real power of t . $\square$

The whole sequence of algebraic-order asymptotic coefficients therefore fails to distinguish the full-field solutions in equation (18.12). The exact Hahn workspace selects one solution by its allowed support, not by any limiting process. No Borel summation, no Stokes multiplier and no analytic continuation rule is asserted by this example; those would require additional data and separate theorems.

Theorem 18.13 (An exact factorially divergent forced oscillator). *The equation*

$$\partial^2 y + y = \omega^{-1} \quad (18.13)$$

has the unique solution in $\mathbf{No}[i]$

$$y = \sum_{n \geq 0} (-1)^n (2n)! \omega^{-(2n+1)}, \quad (18.14)$$

and for $N \geq 1$ its N -term truncation y_N satisfies exactly

$$y_N'' + y_N - \omega^{-1} = (-1)^{N-1} (2N)! \omega^{-(2N+1)}. \quad (18.15)$$

Proof. The formal inverse of $1 + T^2$ is $\sum_{n \geq 0} (-1)^n T^{2n}$; apply Theorem 18.8 to $f = t$, using $\partial^k t = (-1)^k k! t^{k+1}$. Existence can also be checked directly: $\partial^2 t^{2n+1} = (2n+1)(2n+2)t^{2n+3}$, so every positive odd exponent beyond the first cancels between $\partial^2 y$ and y ; the same cancellation for a finite truncation leaves exactly equation (18.15). The difference of two surcomplex solutions satisfies $h'' + h = 0$, hence is zero by Theorem 13.2, so uniqueness holds in the whole class field and not merely in K_0 . $\square$

Warning 18.14 (An exact value that is not a limit of its truncations). The ordinary scalar series $\sum (-1)^n (2n)! t^{2n+1}$ has radius of convergence zero: for nonzero ordinary complex t the ratio of successive term magnitudes is $(2n+2)(2n+1)|t|^2$, which grows without bound. That causes no difficulty for equation (18.14): its coefficients are ordinary complex numbers and its support is the well-ordered set $\{1, 3, 5, \dots\}$, which is all Hahn summation requires. Conversely the existence of the surreal value does not make the scalar series an ordinary holomorphic germ, nor is the value a fine-topology limit of its truncations: the tail $y - y_N$ has leading term $(-1)^N (2N)! \omega^{-(2N+1)}$, so

$$|y - y_N| > \omega^{-\omega} \quad \text{for every } N \geq 1,$$

and the errors never enter that fixed infinitesimal neighbourhood of zero. The residual identities are exact algebraic identities and scale certificates; they certify a formal remainder at a stated order, not convergence of the sequence (y_N) in any surreal topology. A numerical substitution such as $t = 10^{-6}$ would not verify a Hahn identity and is not used anywhere in this report. A closed formula for a coefficient stream is not a finite list of every coefficient.

Remark 18.15 (The coefficient recursion, and its exact guarantee). Writing $(\partial - \lambda)y = f$ coefficientwise in K_0 gives

$$-\lambda y_n - (n-1)y_{n-1} = f_n, \quad \text{that is} \quad y_n = -\lambda^{-1}(f_n + (n-1)y_{n-1}), \quad (18.16)$$

and below a known lower exponent bound all coefficients vanish. Each requested finite prefix costs one inversion of λ and a linear number of coefficient operations, and the recursion works for negative lower bounds. It does *not* make equality of arbitrary program-generated coefficient streams decidable.

19 The rational-exponent workspace

A particularly informative set-sized workspace is

$$\mathcal{H} := \mathbb{C}((t^{\mathbb{Q}})), \quad t = \omega^{-1}, \quad \mathcal{H}_{\mathbb{R}} := \mathbb{R}((t^{\mathbb{Q}})), \quad (19.1)$$

embedded in $\mathbf{No}[i]$ through normal forms. Every element has a well-ordered *rational* support: this is a *full* Hahn field, not merely the union of the finite-ramification Laurent-series fields, and in particular well-ordered supports need not have a common denominator. By Theorem 18.1 it is ∂ -stable with $\partial(\sum c_q t^q) = \sum (-q) c_q t^{q+1}$, so the restriction of the abstract derivation is completely explicit here. That explicitness does not mean the full Berarducci–Mantova construction has been implemented: the special rational monomial subgroup is doing essential work (Theorem 2.2).

Remark 19.1 (A companion report that uses this workspace). The companion report *Tail-Spans and Differential Transcendence in Surreal and Surcomplex Hahn Fields* ([docs/surreal/tail-spans-and-differential-transcendence/](#), [23]) takes equation (19.1) and Theorem 18.1 as its starting point rather than reproving them, and works inside the real-algebraic subfield $\mathbb{A}_{\mathbb{R}}((t^{\mathbb{Q}})) \subseteq \mathcal{H}_{\mathbb{R}}$, where the displayed formula still applies and keeps that subfield ∂ -stable, since it multiplies each coefficient by a rational number and shifts its exponent by one. Over the base $\mathbb{Q}((t^{\mathbb{Q}}))$ it produces a family of actual positive infinitesimal surreal numbers, of cardinality $2^{\aleph_0}$, whose ∂ -jets are all algebraically independent; the numbers are $\sum_{n \in A} \sqrt{p_n} \omega^{-n}$ for an explicit almost disjoint family of infinite sets $A \subseteq \mathbb{N}_{>0}$. That is a statement about differential transcendence over a *set-sized* Hahn base and does not bear on the image and primitive questions settled here, nor does it supply any derivative formula for a monomial outside the rational exponents (Theorem 2.2 still applies).

Theorem 19.2 (Exact derivative image: one resonant coefficient). *For $f = \sum_q c_q t^q \in \mathcal{H}$,*

$$f \in \partial(\mathcal{H}) \iff [t^1]f = 0. \quad (19.2)$$

If $c_1 = 0$ the normalized primitive inside $\mathcal{H}$ is $-\sum_{q \neq 1} \frac{c_q}{q-1} t^{q-1}$; for arbitrary $f \in \mathcal{H}$ the normalized primitive in the ambient $\mathbf{No}[i]$ is $c_1 \log \omega - \sum_{q \neq 1} \frac{c_q}{q-1} t^{q-1}$. Hence $\mathcal{H}$ is closed under differentiation but not under antidifferentiation, and $\log \omega \notin \mathcal{H}$.

Proof. This is Theorem 18.1 with $\Gamma = \mathbb{Q}$. The formula remains valid for infinite rational supports: translation of an already well-ordered support is the precise support certificate, and no denominator bound or discrete increasing sequence of exponents is needed. No appeal to topology, cardinality or computational limitation is involved; the obstruction is the single resonant coefficient $[t]f$. $\square$

Theorem 19.3 (Exact logarithmic-derivative image in $\mathcal{H}$).

$$\{y^\dagger : y \in \mathcal{H}^\times\} = \{-rt + g : r \in \mathbb{Q}, g = 0 \text{ or } v(g) > 1\}. \quad (19.3)$$

For $q = -rt + g$ in this class, put

$$k = -\sum_{\alpha > 1} \frac{[t^\alpha]g}{\alpha - 1} t^{\alpha-1}, \quad (19.4)$$

which has positive valuation (or is 0) and satisfies $\partial k = g$. Then all solutions in $\mathcal{H}$ of $\partial y = qy$ are

$$y = ct^r E(k), \quad c \in \mathbb{C}. \quad (19.5)$$

Proof. A nonzero $y \in \mathcal{H}$ has a unique leading-term factorization $y = ct^r(1+h)$ with $c \in \mathbb{C}^\times$, $r \in \mathbb{Q}$ and $v(h) > 0$ (possibly $h = 0$), and then

$$y^\dagger = -rt + \frac{\partial h}{1+h} = -rt + \partial L(1+h)$$

by Theorem 6.1. Since $L(1+h)$ has positive valuation or is zero, every nonzero term of its derivative has exponent strictly greater than 1 by equation (18.1). This gives containment in the right-hand class. Conversely, for $v(g) > 1$ the series equation (19.4) has positive valuation and $\partial k = g$, so $E(k) \in \mathcal{H}$ is defined by a positive-support Hahn sum and $(t^r E(k))^\dagger = -rt + g$. The ratio of two nonzero solutions lies in the constant field $\mathbb{C}$. $\square$

Proposition 19.4 (Workspace versus ambient). *Let $q = a + ib \in \mathcal{H}$ with $a, b \in \mathcal{H}_\mathbb{R}$. A nonzero solution of $\partial y = qy$ exists in the ambient $\mathbf{No}[i]$ if and only if*

$$b = 0 \quad \text{or} \quad v(b) > 1. \quad (19.6)$$

Proof. This is Theorem 18.4 with $\Gamma = \mathbb{Q}$, equivalently Theorem 10.1; the growth comparison $\log \omega \ll \omega^\delta$ for positive rational δ is what makes a nonzero $\log \omega$ term infinite. $\square$

So there are *three* distinct outcomes, not a binary solvable/unsolvable flag: a solution in the declared workspace, a solution only in the ambient field, and no nonzero solution at all.

The first two rows mean that no *nonzero* solution belongs to $\mathcal{H}$; multiplication by an ordinary complex constant cannot remove either obstruction, since it does not change a logarithmic derivative. Note that the second row is the workspace failure announced in Theorem 8.6: the real coefficient never obstructs *ambiently*, but A or e^A may leave a declared workspace.

Table 5: Where a solution lives. $\mathcal{H} = \mathbb{C}((t^{\mathbb{Q}}))$, $t = \omega^{-1}$.

Coefficient q in $\partial y = qy$	Location	Explanation, or one solution
1	$\mathbf{No}[i]$, not $\mathcal{H}$	e^ω ; the exponent-0 coefficient violates equation (19.3)
$\sqrt{2}t$	$\mathbf{No}[i]$, not $\mathcal{H}$	$\exp(\sqrt{2}\log \omega)$; the exponent-1 coefficient is irrational
it	Neither	The accumulated phase is $\log \omega$
it^2	$\mathcal{H}$	$E(-it)$
$-\frac{3}{2}t + (1 + 2i)t^2$	$\mathcal{H}$	$t^{3/2}E(-(1 + 2i)t)$

Remark 19.5 (A resonant normal form). Theorem 19.3 has a normal-form reading. A scalar equation over $\mathcal{H}$ has a fundamental solution in the same field only when its logarithmic coefficient splits into a rational monomial contribution $-rt$ and a higher-valuation perturbation: the monomial supplies the exponent-1 residue, and the infinitesimal exponential absorbs the perturbation. Lower exponents require larger-scale real exponentials, and lower *imaginary* exponents face the stronger ambient phase obstruction. The condition at exponent 1 is much more restrictive for logarithmic than for additive integration: a logarithmic derivative in $\mathcal{H}$ has no exponents below 1, its exponent-1 coefficient is a real *rational* number, and its higher exponents may carry arbitrary ordinary complex coefficients. This resembles a one-dimensional normal-form calculation, but every assertion is algebraic and exact: there is no asymptotic error term in equation (19.5), and the rational exponent r is not an approximation to a more general exponent.

20 Differentially stable workspaces for set-sized data

The repository’s localization principle is algebraic: a set of surcomplex values fits inside a set-sized Hahn workspace [20]. That is a *size* statement. It does not say that the workspace contains the derivatives, primitives, exponentials or finite phases of its members. Two genuinely different refinements are available, and both survive the merge: a *full Hahn field* indexed by a closed exponent group, and a set-sized *subfield* closed under a specified list of finitary operations. They prove different things and neither implies the other.

20.1 Full Hahn fields: the monomial criterion and the hull

Definition 20.1. A set-sized additive subgroup $\Gamma \subseteq \mathbf{No}$ is *differentially closed on monomials* if $\text{supp } \partial(t^\gamma) \subseteq \Gamma$ for every $\gamma \in \Gamma$, the support being taken in the t -exponent convention of equation (2.2).

Lemma 20.2 (Monomial criterion). *For a set-sized additive group $\Gamma \subseteq \mathbf{No}$,*

$$\partial K_\Gamma \subseteq K_\Gamma \iff \Gamma \text{ is differentially closed on monomials,}$$

and the same equivalence holds for $H_\Gamma = \mathbb{R}((t^\Gamma))$.

Proof. Necessity: apply the closure condition to each t^γ . Sufficiency: for $x = \sum_{\gamma \in S} c_\gamma t^\gamma$ with $S \subseteq \Gamma$, strong additivity (BM3) gives the summable family $(c_\gamma \partial t^\gamma)_{\gamma \in S}$ and $\partial x = \sum_{\gamma \in S} c_\gamma \partial t^\gamma$. Every term has support in Γ , so the sum does too; ordinary real and complex coefficients are constants of ∂ . $\square$

Theorem 20.3 (Differential Hahn hull). *Let $\Gamma_0 \subseteq \mathbf{No}$ be a set-sized additive subgroup. Define*

$$\Gamma_{n+1} = \text{span}_{\mathbb{Q}} \left(\Gamma_n \cup \bigcup_{\gamma \in \Gamma_n} \text{supp } \partial(t^\gamma) \right), \quad \Gamma_* = \bigcup_{n < \omega} \Gamma_n. \quad (20.1)$$

Then Γ_ is a set-sized divisible additive group, K_{Γ_*} is algebraically closed, and $\partial K_{\Gamma_*} \subseteq K_{\Gamma_*}$. It is the least such divisible monomial-closed exponent group containing Γ_0 . Consequently every set $A \subseteq \mathbf{No}[i]$ lies in a set-sized algebraically closed Hahn field stable under ∂ : take Γ_0 to be the group generated by the union of the normal-form supports of the members of A .*

Proof. Each $\partial(t^\gamma)$ is an individual surreal and has a set support, so by replacement and union the displayed union is a set, as is its rational span. The stages increase, so their countable union is a set-sized additive group; every member lies in a stage whose successor contains all its rational multiples, so the union is divisible. If $\gamma \in \Gamma_*$ it lies in some Γ_n , and then $\text{supp } \partial(t^\gamma) \subseteq \Gamma_{n+1}$. Now take any $f = \sum_{\gamma \in S} c_\gamma t^\gamma \in K_{\Gamma_*}$: strong additivity gives the summable family $(c_\gamma \partial t^\gamma)_{\gamma \in S}$, each member has support in Γ_* , so their sum has support in Γ_* and equals ∂f . Divisibility makes H_{Γ_*} real closed and K_{Γ_*} algebraically closed; this is the standard Hahn field fact, an *algebraic* statement, not a differential one. Minimality: any divisible monomial-closed group containing Γ_0 contains each stage by induction, hence their union. $\square$

Warning 20.4 (The proof must not use a union of fields). The last step of the proof appeals to strong additivity applied to every series over Γ_* *simultaneously*. It does not use

$$K_{\bigcup_n \Gamma_n} = \bigcup_n K_{\Gamma_n},$$

which is generally *false*: one Hahn series can use exponents appearing at arbitrarily late stages. The correct route is the one displayed, through the monomial supports.

Warning 20.5 (Three scope limits on Theorem 20.3). (i) Minimality is only among *full Hahn fields indexed by divisible exponent groups*, not among all differential subfields. (ii) The hull group Γ_* need not be countable: it is the number of closure *stages* that is countable, and the countably many stages do not constitute a terminating algorithm for every input representation. (iii) The theorem is a semantic existence construction, not an effective algorithm for computing $\partial(t^\gamma)$ at arbitrary surreal exponents, and it supplies no optimal birthday or cardinal bound. It is not primitive closure, and it asserts no closure under real exponentiation, arbitrary logarithms, or solutions of all first-order linear equations; Theorem 18.2 is the witness.

Proposition 20.6 (The real-exponent case, exactly). *Let $0 \neq \Gamma \subseteq \mathbb{R}$ be an additive subgroup. Then K_Γ is ∂ -stable if and only if $1 \in \Gamma$; and if Γ is divisible its differential Hahn hull is $\Gamma + \mathbb{Q}$. The group $\{0\}$ gives the constant field and is already stable.*

Proof. For nonzero $\gamma \in \Gamma$, equation (3.1) gives $\partial t^\gamma = -\gamma t^{\gamma+1}$, so closure requires $\gamma + 1 \in \Gamma$, hence $1 \in \Gamma$; conversely containing 1 provides every exponent $\gamma + 1$. The hull assertion is Theorem 20.3. $\square$

Example 20.7 (Divisible, algebraically closed, and not derivative-stable). $\Gamma = \mathbb{Q}\sqrt{2}$ is divisible, so K_Γ is algebraically closed and contains $t^{\sqrt{2}}$; but

$$\partial t^{\sqrt{2}} = -\sqrt{2} t^{\sqrt{2}+1} \notin K_{\mathbb{Q}\sqrt{2}}, \quad \text{since } \sqrt{2} + 1 \notin \mathbb{Q}\sqrt{2}.$$

Its hull is $\mathbb{Q} + \mathbb{Q}\sqrt{2}$. So a divisible Hahn value group need not define a derivative-stable workspace, and an arbitrary algebraically closed Hahn workspace does not automatically support the ambient

derivation. Together with Theorem 18.2 this separates *four* requirements: input containment, algebraic closure, derivative stability, and closure under primitives. This example does not dispute the repository's theorem that any set of surcomplex inputs fits inside *some* set-sized Hahn workspace [20]; it shows that the workspace must be chosen with the derivation in view. Closure properties of surreal subfields under exponential, logarithmic, derivative and antiderivative operations are studied much more finely by Bournez and Guilmant [9]; that literature addresses constraints not imposed by the elementary constructions here, and no theorem of it is used to justify Theorem 20.3.

20.2 Set-sized subfields closed under a named operation list

Theorem 20.8 (Closure localization). *Let $S \subseteq \mathbf{No}[i]$ be a set. There is a set-sized real closed subfield $F \subseteq \mathbf{No}$ with $\mathbb{R} \cup \{\omega\} \subseteq F$ and $S \subseteq F[i]$ such that F is closed under*

$$\partial, \quad \mathcal{I}_0, \quad \text{pp}, \text{ct}, \text{fin}, \quad \exp, \quad \log \text{ on positive inputs},$$

and under the real and imaginary components of E and L on their stated domains — equivalently, under finite-angle sine and cosine on $F \cap \mathcal{O}$. One may arrange

$$|F| \leq \kappa := \max\{|S|, |\mathbb{R}|, \aleph_0\}.$$

Then $L = F[i]$ is algebraically closed, is a differential subfield of $\mathbf{No}[i]$ with $\partial L = L$ and constants exactly $\mathbb{C}$, and is closed under componentwise normalized primitives; $\ker(\partial|_F) = \mathbb{R}$ and $\partial F = F$. Moreover the criterion of Theorem 8.4 holds internally: for $a, b \in F$ a nonzero solution of $\partial y = (a + ib)y$ exists in L if and only if b has a finite accumulated phase in $\mathbf{No}$, and then $\mathfrak{o}$ takes values in $F \cap \mathcal{P}$ and is onto it, so the gauge classification and the exact sequence of Section 9 read as ordinary set-level results for this workspace.

One may in addition require that $f(h) \in L$ for every $f \in \mathbb{C}[[X]]$ with ordinary complex coefficients and every $h \in L \cap \mathfrak{m}_{\mathbb{C}}$, at the same cardinality bound, using $|\mathbb{C}|^{\aleph_0} = |\mathbb{R}|$ to count the formal series.

Proof. Let F_0 be the real closure inside $\mathbf{No}$ of the field generated by $\mathbb{R}$, ω and the real and imaginary parts of the members of S . Given F_n , adjoin the images of all of its elements under the listed unary class functions — with their declared domains: positive arguments for $\log$, finite arguments for the phase components, infinitesimal arguments for E, L and for the optional formal-series values — take the field generated by those images, and let F_{n+1} be its real closure inside $\mathbf{No}$, that is, adjoin all real roots in $\mathbf{No}$ of nonzero polynomials over the current field.

Each stage is a set: images under fixed definable class functions are sets by replacement, field generation uses finite expressions, and a set of nonzero polynomials of finite degree has a set of roots, finitely many each. Each stage has size at most κ , since the operations have finite arity; for the optional clause there are $|\mathbb{C}|^{\aleph_0} = |\mathbb{R}|$ formal series, so at most κ elements are adjoined per stage and a real algebraic closure of a field of that cardinality has the same cardinality.

Put $F = \bigcup_{n < \omega} F_n$, a set of size at most κ . Any input to one of the operations lies in a stage and its image in the next; any finite tuple of polynomial coefficients lies in a single stage and all its real algebraic roots are adjoined at the next. So F has the advertised closures and is real closed, and $L = F[i]$ is algebraically closed. Differential closure holds componentwise, $\partial F = F$ because $\mathcal{I}_0 F \subseteq F$, and the constant fields are $F \cap \mathbb{R} = \mathbb{R}$ and $L \cap \mathbb{C} = \mathbb{C}$ because $\mathbb{R} \subseteq F$.

For the internal criterion, necessity is inherited from the ambient statement. For sufficiency, if $b \in F$ has a finite accumulated phase then $\mathcal{I}_0(b) \in F \cap \mathfrak{m}$ by Theorem 7.5(ii); for $a \in F$ we have $\exp(\mathcal{I}_0 a) \in F$, and $E(i\mathcal{I}_0 b) \in L$ because its real and imaginary coordinates are finite-angle cosine

and sine of an element of $F \cap \mathcal{O}$. So the witness equation (8.2) lies in L . Whenever a nonzero solution exists, *every* solution lies in L as well, since the multipliers are in $\mathbb{C} \subseteq L$; the same applies to the solution spaces of Theorems 13.2 and 13.5. Finally pp and $\mathcal{I}_0$ preserve F , so $\mathfrak{o}$ maps F into $F \cap \mathcal{P}$, and $\mathfrak{o}(\partial P) = P$ for $P \in F \cap \mathcal{P}$ gives surjectivity. $\square$

Warning 20.9 (What Theorem 20.8 does not give). It adjoins values of *specified finitary operations*, some of which are defined in the ambient field by Hahn sums. It does *not* assert closure under all set-indexed summable families: an arbitrary-support closure is a different and strictly stronger demand, and F need not contain every Hahn series over its own monomials. The optional clause adjoins formal-series values with *constant* coefficients at a *single* infinitesimal argument; that is a specified set of operations, not Hahn completeness. A particular coherent family and its requested evaluations can be included as set-sized data, but arbitrary future families are not automatically covered. The theorem is not proved to produce a full Hahn field of the form $\mathbb{C}((t^\Gamma))$ — Theorem 20.3 is the construction that does that, and the two are not interchangeable — and it provides no finite data representation, no algorithm for the adjoined maps, no birthday bound, no computable enumeration, and no termination claim for symbolic integration. Normalization of primitives need not be computable from a supplied symbolic representation. Structured fields with simultaneous exponential, logarithmic, differential and integration closure are studied much more finely in [9].

Remark 20.10 (A weaker but sometimes sufficient variant). If a full Hahn field is not needed, a lighter construction suffices: start from the field generated by the data and the ordinary constants; at each finite stage adjoin the values of a specified finitary operation set — for instance ∂ , $\mathcal{I}_0$, conjugation, and real $\exp$ and $\log$ where defined — and take the generated field; the union over $\mathbb{N}$ is a set-sized field stable under each listed operation, because any finite tuple from the union occurs at a common stage and its operation values at the next. This variant is *not* proved to be a full Hahn field and need not contain every summable family of its own members. The distinction is useful for formalization: a small operational workspace can support a particular proof even when it is not closed under all possible Hahn sums.

Remark 20.11 (Localization cannot supply an oscillator). Adjoining a solution of $T' = iT$ cannot be accomplished by passing to a larger workspace of either kind *inside* $\mathbf{No}[i]$: the proof of Theorem 8.8 already applies to the full class. The next section changes the differential field itself rather than its set-sized container.

21 Where an oscillatory solution can live

We use Picard–Vessiot terminology in the standard sense of [13, 14]: a solution-generated differential extension with *no new constants*. Five explicit constructions occur across the source manuscripts, over five different base fields and by four different proofs. They are genuinely different theorems and all are recorded; none of them appeals to an unqualified existence theorem for differential extensions of a proper-class field.

21.1 The missing-solution extension over $\mathbf{No}[i]$

Theorem 21.1 (Adjoining one missing rank-one solution). *Let $a \in \mathbf{No}[i]$ with $\Phi(a) \neq 0$. Introduce a transcendental T and extend ∂ to the rational function field $\mathbf{No}[i](\mathsf{T})$ by*

$$\partial \mathsf{T} = a\mathsf{T}, \tag{21.1}$$

using the product rule on polynomials and then the quotient rule. Then the constant field of $\mathbf{No}[i](\mathbb{T})$ is exactly $\mathbb{C}$; $\mathbf{No}[i](\mathbb{T})$ is a Picard–Vessiot extension for $\partial y = ay$; its differential automorphisms over $\mathbf{No}[i]$ are exactly $\mathbb{T} \mapsto c\mathbb{T}$ with $c \in \mathbb{C}^\times$; and its differential Galois group is $\mathbf{G}_m(\mathbb{C})$. If $\Phi(a) = 0$ the corresponding extension is trivial.

Proof. The product and quotient rules extend the derivation uniquely from $\mathbf{No}[i][\mathbb{T}]$ to $\mathbf{No}[i](\mathbb{T})$. Let $R = P/Q \neq 0$ have derivative zero, with nonzero coprime $P, Q \in \mathbf{No}[i][\mathbb{T}]$. Then $Q\partial P = P\partial Q$, and coprimality gives $P \mid \partial P$ and $Q \mid \partial Q$; since $\partial \mathbb{T} = a\mathbb{T}$, differentiation does not increase the degree in $\mathbb{T}$, so $\partial P = hP$ and $\partial Q = hQ$ for the same $h \in \mathbf{No}[i]$. If $p_j \mathbb{T}^j$ and $p_k \mathbb{T}^k$ are two nonzero terms of P with $j \neq k$, comparing coefficients gives $\partial p_j/p_j + ja = h = \partial p_k/p_k + ka$, that is

$$(p_j/p_k)^\dagger = (k - j)a.$$

Applying Φ and using Theorem 9.4 yields $(k - j)\Phi(a) = 0$, impossible since $\mathcal{P}$ is torsion-free. So P is a single term, and likewise Q , whence $R = c\mathbb{T}^m$ with $c \in \mathbf{No}[i]^\times$ and $m \in \mathbb{Z}$. The constancy equation is $c^\dagger + ma = 0$; applying Φ again forces $m = 0$, and then $\partial c = 0$ gives $c \in \mathbb{C}^\times$.

An automorphism over $\mathbf{No}[i]$ sends $\mathbb{T}$ to another nonzero solution of equation (21.1); the quotient by $\mathbb{T}$ has derivative zero, hence lies in $\mathbb{C}^\times$. Conversely every such scaling is a differential automorphism. When $\Phi(a) = 0$ a nonzero solution already lies in $\mathbf{No}[i]$ by Theorem 8.4. $\square$

Transcendence is necessary: algebraic closedness of $\mathbf{No}[i]$ leaves no nontrivial algebraic extension in which a missing solution could appear, and an algebraic root cannot cancel a nonzero obstruction (Theorem 9.4). $\mathbb{T}$ is a genuinely new differential element, not an alternative notation for a pre-existing surcomplex number.

Theorem 21.2 (The same extension, with differential simplicity). *For $a = i$ the conclusions of Theorem 21.1 hold, and in addition the Laurent polynomial ring $\mathbf{No}[i][\mathbb{T}, \mathbb{T}^{-1}]$ has no nonzero proper differential ideal; so $\mathbf{No}[i](\mathbb{T})$ has the explicit Picard–Vessiot properties for $\partial y = iy$ with differential Galois group $\mathbb{C}^\times$. A second, independent proof of the constant-field statement runs as follows.*

Proof. Let $R \in \mathbf{No}[i](\mathbb{T})^\times$ be constant. Since $\mathbf{No}[i]$ is algebraically closed, factor numerator and denominator:

$$R = c \mathbb{T}^m \prod_{j=1}^k (\mathbb{T} - a_j)^{n_j},$$

with $c \in \mathbf{No}[i]^\times$, $m \in \mathbb{Z}$, the $a_j \in \mathbf{No}[i]^\times$ distinct and the n_j nonzero integers. Logarithmic differentiation gives

$$0 = c^\dagger + i \left(m + \sum_j n_j \right) + \sum_j \frac{n_j (ia_j - \partial a_j)}{\mathbb{T} - a_j}. \quad (21.2)$$

Each $ia_j - \partial a_j$ is nonzero by Theorem 8.8, and distinct simple poles cannot cancel, so $k = 0$. What remains is $c^\dagger = -im$, which Theorem 8.4 excludes unless $m = 0$; then $c \in \mathbb{C}^\times$.

For differential simplicity, suppose $\mathfrak{q}$ is a nonzero proper differential ideal of $\mathbf{No}[i][\mathbb{T}, \mathbb{T}^{-1}]$. Choose a nonzero element of $\mathfrak{q}$ with the fewest nonzero monomials and multiply by a unit to make its constant coefficient 1. If it has another term $a_j \mathbb{T}^j$ then $j \neq 0$ and $\partial(a_j \mathbb{T}^j) = (\partial a_j + ija_j)\mathbb{T}^j \neq 0$, again by Theorem 8.4. Its derivative is therefore a nonzero element of $\mathfrak{q}$ with one fewer nonzero monomial, the constant term differentiating to zero, contradicting minimality. So the chosen element is 1 and $\mathfrak{q}$ is the whole ring. $\square$

21.2 Four set-sized bases, and a non-embedding

The proper-class base can be avoided entirely. Each of the following is a complete construction over a set-sized base, with its own no-new-constants proof.

Theorem 21.3 (Over a localized workspace). *Let F be as in Theorem 20.8, put $L = F[i]$, and let $\mathbb{T}$ be transcendental over L with $\partial\mathbb{T} = i\mathbb{T}$, the derivation extended by the product and quotient rules. Then the constant field of $L(\mathbb{T})$ is exactly $\mathbb{C}$, $L(\mathbb{T})$ is a scalar Picard–Vessiot extension for $\partial y = iy$, and its differential automorphisms over L are exactly $\mathbb{T} \mapsto c\mathbb{T}$ with $c \in \mathbb{C}^\times$, so the group is $\mathbf{G}_m(\mathbb{C})$.*

Proof. Every rational $R(T) \in L(\mathbb{T})$ has an injective formal Laurent expansion at $\mathbb{T} = 0$, $R = \sum_{n \geq n_0} a_n T^n \in L((T))$, with supports bounded below in $\mathbb{Z}$; the induced derivation is

$$\partial R = \sum_{n \geq n_0} (\partial a_n + ni a_n) T^n,$$

which agrees with the rational-function derivation by the product and inverse rules. If $\partial R = 0$ then $\partial a_n = -ni a_n$ for every n . For $n \neq 0$ the constant $-ni$ is nonreal, so $a_n = 0$ by Theorem 3.11 applied in the ambient field. For $n = 0$, $\partial a_0 = 0$ gives $a_0 \in \mathbb{C}$. Hence $R = a_0$ and there are no new constants. The automorphism statement is as in Theorem 21.1. No black-box existence theorem is used: the field and its constants have been constructed and checked. $\square$

Theorem 21.4 (Over the rational Hahn field). *Let $K_0^{\mathbb{Q}} = \mathbb{C}((t^{\mathbb{Q}}))$ with its explicit coefficientwise derivation $\partial(\sum_q c_q t^q) = \sum_q (-q)c_q t^{q+1}$. This field is set-sized and algebraically closed, its differential constants are $\mathbb{C}$, it contains $\omega = t^{-1}$ with $\partial\omega = 1$, and — being a differential subfield of $\mathbf{No}[i]$ — it contains no nonzero solution of $\partial y = niy$ for $0 \neq n \in \mathbb{Z}$. Adjoin a transcendental $\mathbb{T}$ with $\partial\mathbb{T} = i\mathbb{T}$. Then the differential constants of $K_0^{\mathbb{Q}}(\mathbb{T})$ are exactly $\mathbb{C}$, it is a Picard–Vessiot field for $\partial y = iy$, and its differential Galois group is $\mathbf{G}_m$ over $\mathbb{C}$.*

Proof. Let $f \in K_0^{\mathbb{Q}}(\mathbb{T})^\times$ have $\partial f = 0$. Write $f = cP/Q$ with $c \in (K_0^{\mathbb{Q}})^\times$ and monic coprime $P, Q \in K_0^{\mathbb{Q}}[\mathbb{T}]$ of degrees m, n . Clearing denominators gives

$$\frac{\partial c}{c} PQ + (\partial P)Q - P(\partial Q) = 0,$$

and reduction modulo P and modulo Q gives $P \mid \partial P$ and $Q \mid \partial Q$. The derivation does not raise the degree in $\mathbb{T}$, and on a monic polynomial of degree m its leading coefficient is im . Therefore

$$\partial P = imP, \quad \partial Q = inQ, \quad \frac{\partial c}{c} = i(n - m),$$

and the last equation is impossible in $K_0^{\mathbb{Q}}$ unless $m = n$, whence $c \in \mathbb{C}^\times$ by the constant-field statement. Comparing the coefficient of $\mathbb{T}^j$ in P for $j < m$ gives $\partial a_j = i(m - j)a_j$, again impossible for a nonzero a_j ; so $P = \mathbb{T}^m$, similarly $Q = \mathbb{T}^n$, and coprimality with $m = n$ forces $m = n = 0$. Hence $f \in \mathbb{C}^\times$. The automorphism computation is as before. $\square$

Remark 21.5 (Field-transcendental is not differentially transcendental). Since $K_0^{\mathbb{Q}}$ is already algebraically closed, $\mathbb{T}$ is necessarily transcendental as a *field* element. It is nevertheless *differentially algebraic*, satisfying the nonzero differential polynomial $\partial\mathbb{T} - i\mathbb{T} = 0$. The two notions must not be confused. One might try to identify $\mathbb{T}$ with $\sum_{n \geq 0} (i\omega)^n / n!$; that fails for the support reason of Theorem 11.9, and choosing a larger exponent group does not repair the defect. The differential extension is therefore not obtained by appending a missing value to a list of permitted exponents.

Theorem 21.6 (Over the rational function field $\mathbb{C}(X)$). *Let $F = \mathbb{C}(X)$ with $\partial X = 1$, let T be transcendental over F , and define a derivation on $E = F(\mathsf{T})$ by $\partial X = 1$, $\partial \mathsf{T} = i\mathsf{T}$, extended by the polynomial and quotient rules; this is well defined because X and T are algebraically independent. Then the differential constant field of E is exactly $\mathbb{C}$, E is a rank-one Picard–Vessiot field for $\partial Y = iY$ over F , its differential automorphisms over F are exactly $\mathsf{T} \mapsto c\mathsf{T}$ with $c \in \mathbb{C}^\times$, and its differential Galois group is $\mathbf{G}_m(\mathbb{C})$.*

Lemma 21.7. *For $r \in \mathbb{C}(X)^\times$, the identity $r'/r = c$ with $c \in \mathbb{C}^\times$ is impossible.*

Proof. Write $r = p/q$ with nonzero complex polynomials of degrees d, e . Expansion at infinity gives

$$\frac{r'}{r} = \frac{p'}{p} - \frac{q'}{q} = \frac{d-e}{X} + O(X^{-2}),$$

an identity of formal Laurent expansions in $1/X$ whose constant term is zero, so it cannot equal a nonzero constant. Equivalently, compare degrees after clearing denominators in $p'q - pq' = cpq$. $\square$

Proof of Theorem 21.6. Expand $R \in E$ at $\mathsf{T} = 0$ as an injective formal Laurent series $R = \sum_{n \geq n_0} r_n(X) \mathsf{T}^n \in \mathbb{C}(X)((\mathsf{T}))$ — a formal expansion, with no analytic convergence asserted — on which the derivation acts by

$$\partial \left(\sum_n r_n \mathsf{T}^n \right) = \sum_n (r'_n + in r_n) \mathsf{T}^n,$$

compatible with the rational subfield. If $\partial R = 0$ then $r'_n + in r_n = 0$ for all n ; for $n \neq 0$ Theorem 21.7 forces $r_n = 0$, and for $n = 0$ the equation $r'_0 = 0$ gives $r_0 \in \mathbb{C}$. Injectivity of the expansion gives $R \in \mathbb{C}$. The automorphism statement is as before. $\square$

Corollary 21.8 (No differential embedding). *$F = \mathbb{C}(X)$ embeds differentially into $\mathbf{No}[i]$ by $X \mapsto \omega$, since $\partial \omega = 1$. That embedding does not extend to $E = \mathbb{C}(X)(\mathsf{T})$: the image of T would be a nonzero solution of $\partial Y = iY$, contradicting Theorem 8.8. The same argument applies to the extension $\mathbb{C}(\omega)(\mathcal{E})$ with $\partial \omega = 1$ and $\partial \mathcal{E} = i\mathcal{E}$, defined by the polynomial product rule and then the quotient rule: there is no differential embedding of it into $\mathbf{No}[i]$ fixing $\mathbb{C}(\omega)$, and this conclusion requires no assertion whatever about that extension’s full constant field. Likewise no differential embedding of $\mathbf{No}[i](\mathsf{T})$ into $\mathbf{No}[i]$ over $\mathbf{No}[i]$ exists, and T cannot be algebraic over $\mathbf{No}[i]$.*

So a field can be algebraically closed, contain the required base coefficients, and possess additive primitives, and still fail to contain this simplest oscillatory differential extension. The role of T or T is not that of an algebraic root missing from $\mathbf{No}[i]$; it is a new differential transcendental, and no change of normal-form encoding inside the same field can make it exist. A symbolic constructor for such an extension is mathematically reasonable, but it must announce a change of ambient differential field: calling its new element a surcomplex number would be a mathematical error, not a harmless choice of representation.

21.3 The recovered oscillator, its real form, and what it costs

Inside any of the extensions above, with $\partial \mathsf{T} = i\mathsf{T}$, put

$$\mathsf{C} = \frac{\mathsf{T} + \mathsf{T}^{-1}}{2}, \quad \mathsf{S} = \frac{\mathsf{T} - \mathsf{T}^{-1}}{2i}. \quad (21.3)$$

Table 6: Five Picard–Vessiot constructions for $\partial y = iy$ (or $\partial y = ay$).

Base field	Size	Route to “no new constants”	Reference
$\mathbf{No}[i] = \mathbf{No}[i]$	proper class	Coprimalty, then Φ on coefficient ratios, using torsion-freeness of $\mathcal{P}$	Theorem 21.1
$\mathbf{No}[i] = \mathbf{No}[i]$	proper class	Partial fractions: distinct simple poles cannot cancel; plus differential simplicity of $\mathbf{No}[i][\mathbf{T}, \mathbf{T}^{-1}]$	Theorem 21.2
$L = F[i]$, F localized	set	Formal Laurent expansion at $\mathbf{T} = 0$, coefficientwise, plus Theorem 3.11	Theorem 21.3
$\mathbb{C}((t^{\mathbb{Q}}))$	set	Degree and divisibility in $K_0^{\mathbb{Q}}[\mathbf{T}]$, leading coefficient im	Theorem 21.4
$\mathbb{C}(X)$, $\partial X = 1$	set	Formal Laurent expansion at $\mathbf{T} = 0$ plus Theorem 21.7	Theorem 21.6

Proposition 21.9 (A genuine oscillator). *The elements equation (21.3) satisfy*

$$\partial \mathbf{C} = -\mathbf{S}, \quad \partial \mathbf{S} = \mathbf{C}, \quad \mathbf{C}^2 + \mathbf{S}^2 = 1.$$

They are linearly independent over $\mathbb{C}$, and every solution of $\partial^2 y + y = 0$ in the extension is a $\mathbb{C}$ -linear combination of them.

Proof. $\partial(\mathbf{T}^{-1}) = -i\mathbf{T}^{-1}$ gives the differential identities, and direct expansion the quadratic one. $\mathbf{T}$ and $\mathbf{T}^{-1}$ are $\mathbb{C}$ -independent, since a nontrivial relation would make the transcendental $\mathbf{T}$ algebraic, and the linear change equation (21.3) is invertible. For the spanning statement, coprimeness of $\partial - i$ and $\partial + i$ decomposes the oscillator kernel as in Theorem 13.2; any solution of $\partial y = iy$ is a constant multiple of $\mathbf{T}$ and any solution of $\partial y = -iy$ is a constant multiple of $\mathbf{T}^{-1}$, because the quotient has derivative zero and the constants are exactly $\mathbb{C}$. $\square$

These are genuine oscillator elements in a field that actually contains them. They are *not* values of surcomplex sine and cosine at ω : identifying $\mathbf{T}$ with an element of $\mathbf{No}[i]$ would contradict Theorem 3.11. Calling $\mathbf{T}$ an exponential is harmless only when its status as a new differential element is retained.

For the real form, extend conjugation by $\bar{\mathbf{T}} = \mathbf{T}^{-1}$. This is an involution commuting with the derivation, since $\partial \bar{\mathbf{T}} = \bar{i}\bar{\mathbf{T}} = -i\mathbf{T}^{-1} = \partial(\mathbf{T}^{-1})$, and it fixes $\mathbf{C}$ and $\mathbf{S}$. Two equivalent real presentations occur in the sources. First, over a real base F one may take a transcendental $\mathbf{T}_{\mathbb{R}}$ with

$$\partial \mathbf{T}_{\mathbb{R}} = \frac{1 + \mathbf{T}_{\mathbb{R}}^2}{2}, \quad \mathbf{C} = \frac{1 - \mathbf{T}_{\mathbb{R}}^2}{1 + \mathbf{T}_{\mathbb{R}}^2}, \quad \mathbf{S} = \frac{2\mathbf{T}_{\mathbb{R}}}{1 + \mathbf{T}_{\mathbb{R}}^2}, \quad (21.4)$$

the three identities following by the quotient rule, with $\mathbf{T} = (1 + i\mathbf{T}_{\mathbb{R}})/(1 - i\mathbf{T}_{\mathbb{R}})$; the real rational function field then has constant field $\mathbb{R}$. Second, and equivalently, the conjugation-fixed Cayley coordinate

$$\mathbf{v} = \frac{\mathbf{T} - 1}{i(\mathbf{T} + 1)}, \quad \mathbf{T} = \frac{1 + i\mathbf{v}}{1 - i\mathbf{v}}, \quad \mathbf{C} = \frac{1 - \mathbf{v}^2}{1 + \mathbf{v}^2}, \quad \mathbf{S} = \frac{2\mathbf{v}}{1 + \mathbf{v}^2}, \quad \partial \mathbf{v} = \frac{1 + \mathbf{v}^2}{2}, \quad (21.5)$$

is transcendental over F , so $L(\mathbf{T}) = F(\mathbf{v})[i]$ and the fixed field of the involution is $F(\mathbf{v})$: an ordinary rational function field over an ordered field, which admits orderings extending the order of F .

Theorem 21.10 (Oscillation cannot retain the H -field ordering rule). *No ordering of $F(\mathbf{v})$ extending the given order of F can satisfy*

$$h > \mathbb{R} \implies \partial h > 0$$

with the inherited derivation and $\partial\omega = 1$.

Proof. In any such ordering the identity $C^2 + S^2 = 1$ bounds C and S in absolute value by 1. The proof of Theorem 3.8 uses the displayed positivity rule, $\partial\omega = 1$, the inverse rule (3.1) and $\ker \partial \supseteq \mathbb{R}$, and nothing else; all four hold in this ordered extension, so the lemma applies there and forces ∂C and ∂S to be infinitesimal relative to $\mathbb{R}$. Since $\partial C = -S$ and $\partial S = C$, both C and S are then infinitesimal, so $C^2 + S^2$ is infinitesimal and cannot equal 1. $\square$

Equivalently, in the presentation equation (21.4): $F(\mathbb{T}_{\mathbb{R}})$ admits orders extending that of F , but no such realization can preserve $\partial\mathcal{O} \subseteq \mathfrak{m}$ with constant field $\mathbb{R}$, both oscillator coordinates being bounded by 1 and so having infinitesimal derivatives. So it is not the bare possibility of ordering a real field that blocks oscillation. The obstruction is the *compatibility* of that ordering with the Hardy-type differential positivity rule. One may enlarge the differential field to obtain the oscillator, but one cannot keep all the structure that made the base a surreal H -field. This says precisely what must be surrendered.

21.4 Several phases: the differential Galois torus

Let $a_1, \dots, a_m \in \mathbf{No}[i]$ and consider the diagonal system $\partial y_j = a_j y_j$. Write $\Phi_j = \Phi(a_j) \in \mathcal{P}$ and define the *phase-relation lattice*

$$\Lambda = \left\{ n = (n_1, \dots, n_m) \in \mathbb{Z}^m : \sum_j n_j \Phi_j = 0 \right\}. \quad (21.6)$$

If $n \in \Lambda$ then $\sum_j n_j a_j$ is a logarithmic derivative in $\mathbf{No}[i]$, so a product of integer powers of solutions can be normalized to an element of the base field. These are *integer* relations because rational functions in solutions involve integral monomial exponents; real linear dependence of the obstructions is not enough.

Lemma 21.11 (The lattice is saturated). *Λ is saturated in $\mathbb{Z}^m$: if $kn \in \Lambda$ for a nonzero integer k then $n \in \Lambda$. Hence $\mathbb{Z}^m / \Lambda$ is free abelian of rank $d = m - \text{rank } \Lambda$.*

Proof. If $k \sum_j n_j \Phi_j = 0$, torsion-freeness of the real vector class $\mathcal{P}$ gives $\sum_j n_j \Phi_j = 0$. So the quotient is finitely generated and torsion-free, hence free; equivalently the elementary integer normal-form theorem extends a basis of the saturated subgroup to a basis of $\mathbb{Z}^m$. $\square$

Theorem 21.12 (The diagonal phase torus). *The diagonal system has a Picard–Vessiot extension obtained by adjoining d algebraically independent solution symbols after base-field gauges and an invertible integer monomial change of coordinates. Its differential Galois group is isomorphic to $\mathbf{G}_m(\mathbb{C})^d$, and in the original coordinates its action on the solutions is described by*

$$G = \left\{ (c_1, \dots, c_m) \in (\mathbb{C}^\times)^m : \prod_j c_j^{n_j} = 1 \text{ for every } n \in \Lambda \right\}. \quad (21.7)$$

Proof. Choose an integer basis of $\mathbb{Z}^m$ whose first $\ell = \text{rank } \Lambda$ vectors form a basis of Λ (Theorem 21.11). The associated unimodular matrix changes the diagonal coefficients to integer linear combinations $a'_1, \dots, a'_m$. The first ℓ have zero obstruction, hence nonzero solutions $g_j \in \mathbf{No}[i]$ by Theorem 8.4.

The remaining obstructions $\Phi'_{\ell+1}, \dots, \Phi'_m$ are $\mathbb{Z}$ -independent, since any relation among them would be an element of Λ with nonzero complementary coordinates.

For those d coefficients introduce algebraically independent symbols $\mathsf{T}_1, \dots, \mathsf{T}_d$ with $\partial \mathsf{T}_k = a'_{\ell+k} \mathsf{T}_k$. If a nonzero constant of $\mathbf{No}[i](\mathsf{T}_1, \dots, \mathsf{T}_d)$ is written P/Q with coprime multivariate polynomials, the argument of Theorem 21.1 gives $\partial P = hP$ and $\partial Q = hQ$ with $h \in \mathbf{No}[i]$, total degree not increasing under this derivation. For distinct nonzero terms $p_\alpha \mathsf{T}^\alpha$ and $p_\beta \mathsf{T}^\beta$ of P , coefficient comparison gives

$$(p_\alpha/p_\beta)^\dagger = \sum_{k=1}^d (\beta_k - \alpha_k) a'_{\ell+k},$$

and applying Φ contradicts integer independence unless $\alpha = \beta$. So P and Q are monomials, and the same argument on their quotient shows that a rational constant has zero exponent vector and a coefficient in $\mathbb{C}$.

Combining the g_j with the T_k and inverting the unimodular monomial change supplies nonzero solutions of all the original diagonal equations, generating the same differential field. Each independent symbol may be scaled by an arbitrary element of $\mathbb{C}^\times$, giving $\mathbf{G}_m(\mathbb{C})^d$; in the original coordinates the trivialized integer combinations must stay fixed, which is exactly equation (21.7), and those equations leave precisely d independent constant multipliers. $\square$

Example 21.13 (Three separations). (i) For $a_1 = i$, $a_2 = 2i$ the obstructions are ω and 2ω , the lattice is generated by $(2, -1)$, one can choose solutions with $y_2 = y_1^2$, and $G = \{(c, c^2) : c \in \mathbb{C}^\times\}$ is one-dimensional. (ii) For $a_1 = i$, $a_2 = \sqrt{2}i$ the obstructions ω and $\sqrt{2}\omega$ are dependent over $\mathbb{R}$ but have no nonzero integer relation, so there are two algebraically independent generators and the group is $\mathbf{G}_m(\mathbb{C})^2$; the field exponential on real arguments does not supply a canonical irrational power of a new oscillatory symbol inside a rational Picard–Vessiot extension. (iii) For $a_1 = i/\omega$, $a_2 = i/(\omega \log \omega)$ the obstructions are $\log \omega$ and $\log \log \omega$; the first dominates the second by an infinite factor, so no nonzero integer relation exists and the torus again has dimension two — smallness of both coefficients does not reduce the number of independent missing phases.

The theorem treats diagonal systems completely and supplies useful information for triangular extensions. Taken by itself it is *not* a classification of Picard–Vessiot groups for arbitrary nontriangular matrices over $\mathbf{No}[i]$. It becomes one only in combination with the matrix normal form proved later: Theorem 15.9 reduces an arbitrary finite system to a diagonal one over $\mathbf{No}[i]$, and Theorem 17.10 then reads off its torus, identifying the count d of Theorem 21.11 with the rational rank r of the phase lattice $\Pi \subseteq \mathcal{P}$ generated by $\text{Ph}(A)$. Theorem 17.11 adds the group-algebra presentation in rank r and the real form; the differential simplicity proved there in rank r is the general form of the rank-one statement of Theorem 21.2. For a computational backend the lattice is additional required data: treating $i\omega$ and $2i\omega$ as independent phases would produce the wrong Galois group, and treating $i\omega$ and $i\sqrt{2}\omega$ as algebraically dependent merely because their obstructions are real multiples of one another would also be wrong.

22 The one genuine bridge: ω -series and composition

There is an established setting in which intrinsic differentiation really is recoverable as differentiation of a represented function. It is not all of $\mathbf{No}$ with an arbitrary choice of functions.

Let $\mathscr{W} = \mathbb{R}\langle\langle\omega\rangle\rangle$ denote the ω -series: the smallest subfield of $\mathbf{No}$ containing $\mathbb{R}$ and ω and closed under exponential, positive logarithm, and set-indexed summable families. It is a *proper class*,

explicitly distinct from the set-sized workspaces of Sections 19 and 20. Berarducci and Mantova construct a composition $f \circ x$ for $f \in \mathscr{W}$ and positive infinite x ; their results identify $(\partial f) \circ x$ with the *fine* derivative of $x \mapsto f \circ x$ and give a compatible chain rule inside that setting, and their Taylor theorem states that for sufficiently small h

$$f \circ (x + h) = \sum_{n \geq 0} \frac{(\partial^n f) \circ x}{n!} h^n, \quad (22.1)$$

the sum being support-controlled and the permitted size of h depending on f and x [2, Cor. 7.6–7.7, Thms 7.14, 8.2].

Remark 22.1 (How the two derivatives agree, and where). For $f = \log \omega$ the represented function is $x \mapsto \log x$ and $\partial f = 1/\omega$; composition sends the latter to $1/x$, exactly its fine function derivative. For $f = \omega^3$ both descriptions give $3x^2$. Equation (22.1) explains this agreement in a much larger class than finite elementary expressions. This is a different construction from a formal polynomial: in $P(Z) = \omega Z$ the symbol ω is a coefficient held fixed by ∂_Z , whereas in the ω -series $f = \omega^2$ both occurrences belong to the distinguished compositional scale and $f \circ x = x^2$. A representation must record which interpretation is intended before any chain rule can be chosen. The local agreement of derivative operations does not say that every surreal scalar is an ω -series or that every primitive belongs to that class: the full-field surjectivity clause (BM6) and the restricted composition theorem have different domains and must be applied separately.

Warning 22.2 (Imported, with its hypotheses intact). The composition, Taylor and incompatibility results are *separate imported theorems*; they do not follow from Theorem 3.1, and they are not hypotheses hidden inside the phase criterion. In particular $f \circ x$ is an additional operation, not the result of substituting into an arbitrary normal form. Berarducci and Mantova also prove that the simplest derivation on *all* of $\mathbf{No}$ admits no composition compatible with the composition and compatibility axioms of that paper [2, Thm 8.4]. Their hypotheses matter — the compatibility includes a constant rule, an order condition for positive derivative, and a chain rule at positive infinite arguments — and the theorem is quoted here for this particular derivation. It is *not* strengthened into an assertion about all conceivable notions of composition, or about every other choice of derivation, and nothing here settles alternative global composition structures. The present finite-phase obstruction is a direct differential-algebraic argument, not a restatement or a strengthening of that composition theorem.

The practical lesson is to retain at least three input kinds: a formal expression with named coefficient constants, an ω -series with a distinguished compositional scale, and a general scalar with an intrinsic derivation. They can overlap as *values* while supporting different operations, and identifying their carriers in memory does not justify identifying their semantic interfaces.

Remark 22.3 (Four tempting inferences, each refuted by a witness). The elementary embedding of the ordered differential field of logarithmic-exponential transseries into $(\mathbf{No}, \partial)$ [3] places this derivation in a powerful nonoscillatory asymptotic setting; it is a theorem about a specified ordered differential language and must not be silently upgraded to a statement about every simultaneously added analytic, compositional or complex exponential operation, and it is not a theorem that every algebraic differential equation has a solution. Four specific inferences fail, each with an explicit witness:

- (1) algebraic closedness does not imply surjectivity of the logarithmic derivative — i is missing from the image (Theorem 8.1);
- (2) surjectivity of ∂ does not imply closure under exponential integrals — $i\omega$ is a primitive of i , and no compatible exponential of it exists in $\mathbf{No}[i]$ (Theorem 11.4);

- (3) differential stability of a Hahn workspace does not imply primitive closure — $\log \omega \notin H_{\mathbb{Q}}$ (Theorem 18.2);
- (4) an infinitesimal imaginary part is necessary but not sufficient for a nonzero rank-one solution — i/ω is excluded while i/ω^2 is included (Theorem 8.5).

Part V

Coordinate differential equations in common-domain Hahn rings

We now change operators. Throughout Part V the derivative is D_z of Table 1: it differentiates holomorphic coefficient *functions* in an ordinary complex coordinate z and holds every Hahn monomial and every scalar parameter constant. No scalar derivation on the coefficient field is involved in Sections 23 to 27; Section 28 then puts the two operators side by side and proves the compatibility statements that do hold.

The two halves of this report have different constant fields. The general solution of a coherent linear equation is parametrized by *Hahn* constants; the general solution of a scalar equation for ∂ is parametrized, when it exists at all, by *ordinary complex* constants. A number can be a coefficient constant without being a differential-field constant — $D_z \omega = 0$ while $\partial \omega = 1$ — and that difference is the whole reason the two theories coexist.

23 The fixed-common-domain ring

Let Γ be a set-sized ordered abelian subgroup of the additive surreals, interpret $t^\gamma = \omega^{-\gamma}$ as in equation (2.2), and put

$$K_\Gamma = \mathbb{C}((t^\Gamma)) = \left\{ \sum_{\gamma \in S} c_\gamma t^\gamma : S \subseteq \Gamma \text{ well ordered, } c_\gamma \in \mathbb{C} \right\},$$

with valuation v the least exponent and $v(0) = +\infty$; so $v(t^\gamma) > 0$ means the monomial is infinitesimal. Here Γ may have higher rank than $\mathbb{R}$, and *no divisibility assumption on Γ is needed* for any construction in this part. Let $U \subseteq \mathbb{C}$ be an ordinary *connected* open set, let $\mathcal{O}(U)$ denote the ordinary holomorphic functions on U , and define

$$\mathcal{H}_\Gamma(U) = \mathcal{O}(U)((t^\Gamma)) = \left\{ F(z) = \sum_{\gamma \in S} f_\gamma(z) t^\gamma : S \text{ well ordered, } f_\gamma \in \mathcal{O}(U) \right\}. \quad (23.1)$$

Every coefficient is holomorphic on the *same* U . Differentiation is coefficientwise,

$$D_z F = \sum_{\gamma \in S} f'_\gamma(z) t^\gamma,$$

a derivation of the ring that creates no new exponents; its support is a subset of the original support. For vectors and matrices, support means the union of the supports of the finitely many entries. A “nonzero coefficient function” means one not identically zero, so the global support is not redefined each time a coefficient happens to vanish at a particular z .

Warning 23.1 (Two neighbours this ring is not, and one notation clash). $\mathcal{H}_\Gamma(U)$ is *not* the larger ring with an unrelated convergent germ of shrinking radius for each exponent and no common domain, and it is *not* the formal power-series ring $\mathbb{C}[[z]]((t^\Gamma))$ or $K_\Gamma[[z]]$. These distinctions are already emphasized in the repository's analytic and foundational reports [18, 20, 16], and Theorem 24.7 and Section 28.1 exhibit the separating witnesses. There is no requirement that the family (f_γ) be uniformly bounded in γ , and no numerical series over exponents need converge: holomorphicity and summability impose different conditions and are used at different steps. Notation: in equation (23.1) the symbol $\mathcal{O}(U)$ denotes ordinary holomorphic functions; it is *not* the finite-element valuation class $\mathcal{O}$, which never carries an argument.

Evaluation at an ordinary $z_0 \in U$ is a ring homomorphism to K_Γ , coefficientwise, and

$$\ker(D_z : \mathcal{H}_\Gamma(U) \rightarrow \mathcal{H}_\Gamma(U)) = K_\Gamma, \quad (23.2)$$

because an ordinary holomorphic function with zero derivative on a connected U is constant. If U is simply connected, define the normalized primitive coefficientwise,

$$I_{z_0}F = \sum_\gamma \left(\int_{z_0}^z f_\gamma(\zeta) d\zeta \right) t^\gamma,$$

the ordinary integrals being path independent and holomorphic; they do not enlarge the support.

A section also has a value at a finite-halo point, after enlarging the exponent workspace if necessary: for $z_0 \in U$ and infinitesimal h ,

$$F^\sharp(z_0 + h) = \sum_{\gamma \in S} \sum_{n \geq 0} \frac{f_\gamma^{(n)}(z_0)}{n!} t^\gamma h^n. \quad (23.3)$$

The union of supports lies in $S + (\text{supp } h)^*$, and the two-support lemma together with the ordered-word count of Theorem 2.6 and Section A gives strong summability of the double family. No uniform bound on the ordinary Taylor coefficients is required. This is *infinitesimal Taylor evaluation only*, not evaluation at an arbitrary infinite coordinate.

24 Linear coherent systems with positive perturbation

Theorem 24.1 (Positive-perturbation fundamental matrix). *Let U be simply connected, $z_0 \in U$, and*

$$A = A_0 + E \in \text{Mat}_d \mathcal{H}_\Gamma(U), \quad A_0 \in \text{Mat}_d \mathcal{O}(U), \quad S = \text{supp } E \subseteq \Gamma_{>0}.$$

Then there is a unique Hahn-coherent matrix X with

$$D_z X = AX, \quad X(z_0) = I_d. \quad (24.1)$$

It is invertible in $\text{Mat}_d \mathcal{H}_\Gamma(U)$, its support is contained in S^ , and its coefficient at exponent 0 is the ordinary normalized fundamental matrix Υ_0 of $D_z \Upsilon_0 = A_0 \Upsilon_0$, $\Upsilon_0(z_0) = I_d$. Uniqueness holds among all Hahn-coherent matrices, not only those whose support was prescribed in advance.*

Proof. Ordinary holomorphic linear theory gives Υ_0 on U , normalized at z_0 , with everywhere nonzero determinant: local existence and uniqueness, analytic continuation along ordinary paths, and simple connectedness [12, Ch. 4]. Put $G = \Upsilon_0^{-1} E \Upsilon_0$, whose support lies in S , and $TH = I_{z_0}(GH)$. For

each n the support of $T^n I_d$ lies in the set of sums of n members of S , and at a fixed exponent only finitely many ordered words in S contribute, by Theorem 2.6. Hence

$$V = \sum_{n \geq 0} T^n I_d \quad (24.2)$$

is strongly summable with support in S^* and satisfies $V = I_d + TV$, so $D_z V = GV$ and $V(z_0) = I_d$; then $X = \Upsilon_0 V$ solves equation (24.1). Writing $V = I_d + W$ with W of positive support, the geometric matrix series $\sum_{n \geq 0} (-W)^n$ is strongly summable by the same lemma and is a two-sided inverse, so X is invertible with the claimed exponent-0 coefficient.

For uniqueness let Z be the difference of two coherent solutions and put $H = \Upsilon_0^{-1} Z$. If $H \neq 0$, take the least exponent γ of its global support. The equation $D_z H = GH$ forces $H'_\gamma = 0$: any contribution on the right would need a coefficient of H at $\gamma - s < \gamma$ for some $s \in S$, $s > 0$. The initial condition gives $H_\gamma(z_0) = 0$, so $H_\gamma = 0$ — a contradiction. Note that this argument never used a prescribed support monoid. $\square$

The proof yields an explicit recursion, which is a *support certificate* as well as an existence proof: with $V_0 = I_d$ and all unlisted coefficients zero,

$$V'_\gamma = \sum_{s+\delta=\gamma} G_s V_\delta, \quad V_\gamma(z_0) = 0 \quad (\gamma > 0), \quad (24.3)$$

where every dependency satisfies $\delta < \gamma$ and every displayed sum is finite.

Warning 24.2 (Not an imported convergence proof). The familiar iterated-integral form equation (24.2) does *not* import an ordinary convergence proof into the surreal fine topology. The primitive is an ordinary coefficientwise primitive and the infinite sum is a strongly summable Hahn family. At higher rank the successive support levels need not become cofinal in the value group, and no such cofinality is claimed or used.

Corollary 24.3 (Inhomogeneous coherent systems). *Under the hypotheses of Theorem 24.1, for $f \in \mathcal{H}_\Gamma(U)^d$ and $c \in K_\Gamma^d$ the unique solution of $D_z y = Ay + f$, $y(z_0) = c$, is*

$$y = X(c + I_{z_0}(X^{-1}f)),$$

with support contained in $(\text{supp } c + S^) \cup (\text{supp } f + S^*)$.*

Proof. The ordinary product rule holds coefficientwise because each Hahn coefficient is a finite sum, which verifies the formula; the support follows from the supports of X and X^{-1} and from $S^* + S^* = S^*$; uniqueness follows from Theorem 24.1 applied to differences. $\square$

Corollary 24.4 (Liouville determinant identity). *With the same normalization,*

$$\det X = \det \Upsilon_0 \cdot E(I_{z_0}(\text{tr } E)).$$

Here E is the formal exponential of a positive-support coherent section, not an unrestricted scalar complex exponential.

Proof. Finite matrix algebra gives Jacobi's formula $D_z \det X = (\text{tr } A) \det X$. The displayed expression satisfies the same scalar equation and equals 1 at z_0 , so the $d = 1$ case of Theorem 24.1 identifies them. The exponent has positive support, so Theorem 2.6 admits its exponential. $\square$

Proposition 24.5 (A negative leading exponent forbids a coherent scalar exponential). *If $a \in \mathcal{H}_\Gamma(U)$ has $v(a) < 0$, then $D_z Y = aY$ has no nonzero solution in $\mathcal{H}_\Gamma(U)$.*

Proof. Differentiation can delete coefficient functions but cannot create smaller exponents, so $v(D_z Y) \geq v(Y)$. Since U is connected, $\mathcal{O}(U)$ is an integral domain and $v(aY) = v(a) + v(Y) < v(Y)$ for $Y \neq 0$. The leading exponents cannot agree. $\square$

In particular $D_z Y = i\omega Y$ has no nonzero common-domain Hahn-coherent solution on *any* connected ordinary domain. The argument excludes not merely one series representation but every nonzero coherent solution in the stated ring.

Example 24.6 (Positive support is not necessary for matrix systems). If $N^2 = 0$ then

$$D_z X = t^{-1}NX, \quad X(0) = I_d, \quad X = I_d + z t^{-1}N$$

is a coherent *polynomial* matrix despite its negative exponent. Nilpotence permits a finite algebraic cancellation that the scalar argument of Theorem 24.5 excludes. So the positive-support hypothesis is sufficient, not necessary, and negative-exponent matrix systems need finer analysis; this example bounds Theorem 24.5 deliberately.

Example 24.7 (A scale obstruction, and how rescaling resolves it). Consider

$$D_z Y = t^{-1}Y, \quad Y(0) = 1, \tag{24.4}$$

with $1 \in \Gamma$. It has no nonzero solution in $\mathcal{H}_\Gamma(U)$ on any ordinary nonempty domain, by Theorem 24.5: $v(D_z Y) \geq v(Y)$ while $v(t^{-1}Y) = v(Y) - 1$. The formal z -series $\sum_{n \geq 0} t^{-n} z^n / n!$ *does* exist as an element of $K_\Gamma[[z]]$, but its coefficient-family exponents are unbounded below, so it is not an element of the fixed-domain ring equation (23.1): formal existence and coherent existence have separated. Now rescale $z = tw$; since t is D_z -constant the equation becomes $D_w Y = Y$, $Y(0) = 1$, whose solution on an ordinary w -domain is the ordinary holomorphic e^w . That describes a *microscopic* z -domain, not a common ordinary one: at $v(z) > 1$ the original Taylor family is also strongly summable, because z/t is infinitesimal, whereas at $z = tw$ with nonzero ordinary w one uses the ordinary coefficient value e^w and *not* a Hahn sum of infinitely many exponent-zero terms $w^n/n!$. This is a domain-and-scale obstruction, not a failure of the formal differential equation, and it is why the positive-support hypothesis of Theorem 25.2 cannot simply be dropped.

25 A polynomially nonlinear initial-value theorem

A nonlinear theorem needs both a domain on which an ordinary background solution exists and a reason why every new Hahn coefficient is determined by earlier ones. Positivity of the support supplies the latter. The source manuscripts prove this theorem twice, in two notations, with two different admissibility hypotheses on the nonlinearity: polynomial dependence on the dependent variables, or holomorphic dependence on a neighbourhood of the graph of the background solution with Taylor evaluation in the vector variable. Both versions are recorded; the second needs the following admissibility lemma.

Lemma 25.1 (Admissibility of nonlinear evaluation). *Let $d \geq 1$, let $y_0 : U \rightarrow \mathbb{C}^d$ be holomorphic, let $W \subseteq \mathbb{C} \times \mathbb{C}^d$ be an ordinary open neighbourhood of its graph, let $S \subseteq \Gamma_{>0}$ be well ordered, and let F_0 and each F_s , $s \in S$, be holomorphic maps $W \rightarrow \mathbb{C}^d$. Write formally*

$$F(z, Y) = F_0(z, Y) + \sum_{s \in S} t^s F_s(z, Y). \tag{25.1}$$

If $u \in \mathcal{H}_\Gamma(U)^d$ has positive support, the Taylor evaluation of equation (25.1) at $y_0 + u$ is well defined and every coefficient is holomorphic on U . If $\text{supp } u \subseteq M$ and $M = E^*$ for a positive well-ordered $E \supseteq S$, then the result has support in M .

Proof. Let T be the finite union of the supports of the components of u ; then $S \cup T$ is well ordered and positive. A term in a Taylor coefficient has exponent $s + \beta_1 + \cdots + \beta_k$ with $s \in S \cup \{0\}$ and each $\beta_j \in T$. Theorem 2.6 gives a well-ordered union and only finitely many ordered decompositions at each exponent, and there are finitely many vector-component choices at each fixed length k . So each coefficient is a finite sum of products of holomorphic derivatives of the F_s along y_0 with coefficients of u , hence holomorphic on U . The support containment follows from closure of M under addition. $\square$

No ordinary analytic convergence theorem is applied to a sum indexed by surreal exponents: the ordinary Taylor coefficients supply holomorphic functions, and Theorem 2.6 supplies the summation permission.

Theorem 25.2 (Support-certified positive-support initial-value problem). *In the setting of Theorem 25.1, assume U is simply connected, $z_0 \in U$, and $y'_0(z) = F_0(z, y_0(z))$. Let $\mathbf{v} \in K_\Gamma^d$ have positive support, and put*

$$E = S \cup \text{supp } \mathbf{v}, \quad M = E^*.$$

Then there exists a solution of the form

$$Y = y_0 + u, \quad u \in \mathcal{H}_\Gamma(U)^d \text{ of positive support,} \quad (25.2)$$

to

$$D_z Y = F(z, Y), \quad Y(z_0) = y_0(z_0) + \mathbf{v}, \quad (25.3)$$

and it is unique. Its support satisfies the explicit certificate

$$\text{supp } u \subseteq M \setminus \{0\}. \quad (25.4)$$

Convention 25.3 (The uniqueness class, stated exactly). Uniqueness in Theorem 25.2 is uniqueness among coherent solutions of the form $y_0 + (\text{positive-support vector})$ on U — that is, among all positive-support coherent solutions with the specified ordinary reduction y_0 . The support containment equation (25.4) is a *conclusion*, not a hypothesis: no support was prescribed, and uniqueness therefore holds among all positive-support coherent solutions and not merely among those whose support was assumed to lie in M . Nothing wider is claimed. In the *linear* case the least-exponent argument delivers more, and Theorem 24.1 states that stronger conclusion separately: uniqueness there is among all Hahn-coherent matrices.

Proof. Put $A(z) = D_Y F_0(z, y_0(z))$, the Jacobian along the background solution. Ordinary complex linear theory gives an invertible holomorphic fundamental matrix Υ on U with $\Phi' = A\Phi$, $\Phi(z_0) = I_d$; simple connectedness removes monodromy [12, Ch. 3–4].

M is well ordered by Theorem 2.6. Perform recursion along $M_{>0}$. For a proposed positive-support u , expansion gives $F_0(z, y_0 + u) = F_0(z, y_0) + Au + R_0(z, u)$ where every monomial of R_0 has degree at least two in u , and each perturbation term $t^s F_s(z, y_0 + u)$ carries the additional positive exponent s . Hence at a fixed exponent γ the equation reads

$$u'_\gamma = A(z)u_\gamma + R_\gamma(z), \quad u_\gamma(z_0) = \mathbf{v}_\gamma, \quad (25.5)$$

where R_γ depends only on coefficients u_β with $0 < \beta < \gamma$: a contribution from R_0 has at least two positive exponents summing to γ , so each is smaller, and a contribution from a perturbation has $s > 0$ before any factor of u , so every remaining exponent is smaller. The only term containing u_γ is the isolated linear term Au_γ .

At a fixed γ only finitely many contributions occur: expand each previously admitted exponent as a finite word in E ; all contributions then determine ordered words in E with sum γ , of which Theorem 2.6 permits only finitely many, and there are finitely many vector indices at each fixed length. Each F_s has finitely many polynomial monomials, or finitely many relevant Taylor terms at each order by Theorem 25.1, and only finitely many s can participate. So unbounded degrees across different s do not introduce an infinite coefficient sum, and *no common bound on the polynomial degrees of the F_s is required*.

By induction R_γ is an ordinary holomorphic vector on U , and the ordinary linear inhomogeneous problem equation (25.5) has the unique holomorphic solution

$$u_\gamma(z) = \Upsilon(z) \left(\mathbf{v}_\gamma + \int_{z_0}^z \Upsilon(\zeta)^{-1} R_\gamma(\zeta) d\zeta \right) \quad (25.6)$$

on U : the integral is an *ordinary* complex integral of a holomorphic integrand on a simply connected set, so it is path independent and holomorphic there. This completes the recursion; the recursion is set-sized and no limit value at an accumulation exponent is needed, each coefficient being assigned by its own equation. The resulting support lies in $M_{>0}$, hence is well ordered, and Theorem 25.1 makes the substitution summable, so equation (25.3) holds coefficientwise.

For uniqueness in the sense of Theorem 25.3, let u and $\tilde{u}$ be two positive-support coherent solutions with nonzero difference. The finite union of their supports is well ordered, so the difference has a least exponent γ , below which the coefficients agree. At γ all nonlinear and perturbation difference contributions therefore cancel, leaving $(u_\gamma - \tilde{u}_\gamma)' = A(z)(u_\gamma - \tilde{u}_\gamma)$ with zero initial value; ordinary linear uniqueness makes that coefficient identically zero, a contradiction. $\square$

Warning 25.4 (What the proof does not assume, and what the theorem does not promise). There is no Banach-space norm on the collection of coefficients, no contraction in the surreal fine topology, no convergence of Picard iterates, and no shrinking domain per exponent: every R_γ and every u_γ in equation (25.6) is holomorphic on the *original* U . The recursion uses well-founded support dependence and ordinary linear uniqueness, and nothing else.

The ordinary background solution is a *hypothesis*. The theorem does not promise to continue it past its ordinary singularities; its existence on an arbitrary prescribed domain is not asserted, and it may encounter poles or other obstructions beyond U . Simple connectedness is required. Negative-valued perturbations and arbitrary radius-free coefficient germs are not covered (Theorems 24.6 and 24.7), and unrestricted substitution into an arbitrary holomorphic function of Y without specifying its domain and support conditions is out of scope. This is a differential-equation specialization of the support-recursion viewpoint already present in the repository, not a claim to have invented positive-support recursion; its additional content is the explicit initial-value contract, the common-domain argument through a single fundamental matrix, and the uniqueness clause of Theorem 25.3.

Remark 25.5 (The linear case, and one practical rule). For $F_0(z, Y) = A_0(z)Y + b_0(z)$ with perturbations affine in Y , the nonlinear terms vanish and every higher coefficient needs one linear inhomogeneous solve with the *same* fundamental matrix Υ : the expensive ordinary homogeneous problem is solved once, not once per Hahn exponent. An effective implementation still needs a way to enumerate the finitely many decompositions of a requested exponent. A well-ordered support is

a *semantic* certificate, not automatically an effective presentation; for a finite positive grid, finite truncations often reduce the requirement to integer-composition enumeration, but for an arbitrary program-described high-rank support no general terminating enumeration algorithm is claimed.

25.1 Two Riccati witnesses

Example 25.6 (A coherent entire solution with a pole at an infinite coordinate). Take $d = 1$, $F_0(Y) = Y^2$, $y_0 = 0$, no coefficient perturbation, and initial value $\mathbf{v} = t$. Then

$$y = \frac{t}{1 - tz} = \sum_{n \geq 0} t^{n+1} z^n \quad (25.7)$$

satisfies $D_z y = y^2$, $y(0) = t$. Every coefficient is a *polynomial* in z , so this is a coherent section on the whole ordinary complex plane. Yet the rational expression has a pole at the infinite point $z = t^{-1} = \omega$, which is not in the finite halo of $\mathbb{C}$: evaluating the Hahn series there would produce infinitely many copies of t and is not a strongly summable evaluation. So being coherent on the whole ordinary plane does not mean being regular at every surcomplex coordinate. The location of the pole explains, rather than contradicts, the exact geometric-series identity.

Example 25.7 (An entire coefficient family with an exact recursion). Take $F_0(z, Y) = Y^2$, $y_0 = 0$, $U = \mathbb{C}$, $S = \{1\}$, so

$$D_z Y = Y^2 + t, \quad Y(0) = 0. \quad (25.8)$$

Theorem 25.2 produces a unique positive-support coherent solution in $\mathcal{O}(\mathbb{C})((t^{\mathbb{Z}}))$; writing $Y = \sum_{n \geq 1} a_n z^{2n-1} t^n$, the coefficient equations give

$$a_1 = 1, \quad (2n - 1)a_n = \sum_{j=1}^{n-1} a_j a_{n-j} \quad (n \geq 2), \quad (25.9)$$

so

$$Y = tz + \frac{t^2 z^3}{3} + \frac{2t^3 z^5}{15} + \frac{17t^4 z^7}{315} + \frac{62t^5 z^9}{2835} + \cdots, \quad (25.10)$$

every coefficient an entire polynomial. Formally this is $\sqrt{t} \tan(\sqrt{t} z)$; that notation denotes the displayed Taylor identity, and the final series has integer exponents even when the chosen workspace does not contain $\sqrt{t}$. For finite surcomplex z the argument $\sqrt{t} z$ is infinitesimal and a small-argument tangent interpretation is legitimate in an enlarged workspace; but substituting $z = \omega = t^{-1}$ gives exponents $1 - n$, unbounded below, and the family is not summable. Entire coefficient functions on the ordinary plane do not justify evaluation at every infinite surcomplex input. This is an exact coherent solution on a common ordinary domain, not an all-scale surcomplex entire function, and it therefore does not contradict an all-scale rigidity theorem stated in a different function class.

26 Monodromy stays an ordinary-domain construction

On a non-simply-connected ordinary domain U , lift the coefficient system to its ordinary universal covering surface and choose a lift of the base point. The proof of Theorem 24.1 applies there: ordinary holomorphic primitives of coefficient one-forms are path independent on the cover, and the support calculation is unchanged.

Theorem 26.1 (Monodromy valued in $\mathrm{GL}_d(K_\Gamma)$). *Analytic continuation of the normalized fundamental matrix around an ordinary based loop ℓ yields another fundamental matrix X_ℓ , and their ratio has zero coordinate derivative, so*

$$X_\ell = X M_\ell, \quad M_\ell \in \mathrm{GL}_d(K_\Gamma). \quad (26.1)$$

The coefficient of M_ℓ at exponent 0 is the ordinary monodromy matrix of A_0 , and all its other exponents lie in $S^ \setminus \{0\}$. Under a fixed convention for composition of based loops these matrices form the usual monodromy representation; reversing the loop-composition convention reverses the multiplication order.*

Proof. Immediate from equation (23.2) applied on the cover, together with $X(z_0) = I_d$, which lets M_ℓ be read off at the lifted endpoint; the support statement is Theorem 24.1. $\square$

No fine-continuous surcomplex path is asserted: the loop is an ordinary complex path, continuation is coefficientwise, and the result is a matrix of Hahn constants. This is the same domain separation that makes coefficientwise contour theory meaningful.

Example 26.2 (A residue invisible to ordinary monodromy). On a branch domain for the ordinary logarithm containing 1, consider

$$D_z Y = \frac{\lambda + \varepsilon}{z} Y, \quad Y(1) = 1, \quad \lambda \in \mathbb{C}, \quad v(\varepsilon) > 0. \quad (26.2)$$

The coherent solution is $Y = z^\lambda E(\varepsilon \log z)$, with $z^\lambda = \exp(\lambda \log z)$ in ordinary complex analysis. A positive loop about 0 changes $\log z$ by $2\pi i$ and gives

$$M = e^{2\pi i \lambda} E(2\pi i \varepsilon). \quad (26.3)$$

If $\lambda \in \mathbb{Z}$ the *ordinary* monodromy is 1, while the full monodromy is not 1 when $\varepsilon \neq 0$; by equation (6.3) its leading change is exactly $2\pi i \varepsilon$. So reduction to the ordinary coefficient can erase a genuine global obstruction to a single-valued coherent solution.

Warning 26.3 (The cross-category pair). In equation (26.2) the element ε is constant for D_z but need not be constant for the scalar ∂ . It is *not* an infinitesimal phase residue in the sense of Theorem 5.1(b); it is a Hahn coefficient of a coordinate problem. The finite-primitive criterion of Theorem 8.4 is therefore not the criterion for solving equation (26.2). Concretely:

$$\varepsilon = \omega^{-1} \quad \text{is permitted in the coordinate problem } D_z Y = \frac{\lambda + \varepsilon}{z} Y,$$

$$\varepsilon = \omega^{-1} \quad \text{is prohibited as the imaginary coefficient of } \partial y = i\omega^{-1}y.$$

This is the sharpest available illustration that the two halves of this report are not competing answers to one problem. They are answers to precisely typed problems that happen to share differential notation.

27 A genuinely two-scale, noncommuting example

Let $\Gamma = \mathbb{Z}^2$ with the lexicographic order and use monomials with

$$v(t) = (0, 1), \quad v(u) = (1, 0), \quad \text{so } u \prec t^n \text{ for every ordinary } n \geq 1.$$

This is a Hahn workspace for *coordinate* differentiation: no formula for ∂u is assumed, and none is needed. Set

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad M = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad A(z) = tN + uzM.$$

Both matrices are nilpotent and $NM \neq MN$. The normalized fundamental matrix has the expansion $X(z) = \sum_{p,q \geq 0} t^p u^q X_{pq}(z)$ with $X_{00} = I_2$, and the recursion

$$X'_{pq} = NX_{p-1,q} + zMX_{p,q-1}, \quad X_{pq}(0) = 0 \quad (p+q > 0), \quad (27.1)$$

a negative index meaning zero. Ordinary polynomial integration gives

$$X_{10} = zN, \quad X_{01} = \frac{z^2}{2}M, \quad X_{11} = z^3 \left(\frac{NM}{6} + \frac{MN}{3} \right) = \begin{pmatrix} z^3/6 & 0 \\ 0 & z^3/3 \end{pmatrix},$$

and since $\text{tr } A = 0$, Theorem 24.4 gives $\det X = 1$ exactly.

Warning 27.1 (The naive exponential is wrong, and total degree is not a valuation cutoff). It would be incorrect to replace X by $E(tzN + uz^2M/2)$: that expression's mixed coefficient is $z^3(NM + MN)/4$, not X_{11} , and the difference at tu is

$$\frac{z^3}{12}(MN - NM) \neq 0.$$

The order of multiplication inside iterated integrals matters even though every individual coefficient calculation is finite. This is exactly why Section A counts *ordered* words: an unordered or multiset version of the support count would not carry Theorems 24.1 and 25.2.

Separately: the delivered verification program computes all X_{pq} with $p+q \leq 6$, checks the coefficient equations and initial conditions, and verifies $\det X = 1$ to the same total degree. *Total degree is not a valuation cutoff here*: t^{100} has higher total degree but lower valuation than u . The support theorem justifies the complete series; the finite jet is a different, explicitly declared representation of part of it.

28 Formal, microscopic, and coherent: three different promises

28.1 Formal solvability is the weakest of the three

Proposition 28.1 (Formal existence over any characteristic-zero field). *Let K be any field of characteristic zero and $A(Z) = \sum_{n \geq 0} A_n Z^n \in \text{Mat}_d(K[[Z]])$. The formal problem $\partial_Z X = A(Z)X$, $X(0) = I_d$ has a unique solution $X = \sum_{n \geq 0} X_n Z^n$, given by*

$$X_0 = I_d, \quad (n+1)X_{n+1} = \sum_{k=0}^n A_k X_{n-k}. \quad (28.1)$$

Division by $n+1$ is permitted and the right side uses only known coefficients; no topology and no analytic estimate is involved. In particular the formal problem is always solvable for $K = \mathbf{No}[i]$.

Warning 28.2 (Three promises, worked through $A = i\omega$). Theorem 28.1 does not assert that the solution can be *evaluated* at a given surcomplex number. For $A = i\omega$ it gives

$$X(Z) = \sum_{n \geq 0} \frac{(i\omega Z)^n}{n!}.$$

At any nonzero *ordinary* z the valuations of the terms are unbounded below, so this family is not strongly summable (Theorem 11.9). At an infinitesimal h with ωh infinitesimal it *is* summable, and equals $E(i\omega h)$. And common-domain coherence is excluded outright by Theorem 24.5, which rules out not just this series representation but every nonzero coherent solution of $D_z Y = i\omega Y$ in the stated ring. So

$$\text{formal solvability} \neq \text{microscopic evaluability} \neq \text{common-domain coherence},$$

and the three must never be substituted for one another.

28.2 The coefficient derivation on a common-domain family

The one place where the two operators of this report legitimately meet is the following. Fix a common-domain family $F(z) = \sum_{\gamma \in S} f_\gamma(z) t^\gamma$ as in equation (23.1) and write, for each monomial,

$$\partial t^\gamma = \sum_{\delta} d_{\delta\gamma} t^\delta, \quad d_{\delta\gamma} \in \mathbb{R}. \quad (28.2)$$

These scalars form a support-controlled matrix; only set-sized columns are ever used at a time, and it is not a matrix indexed by all surreals as a set.

Lemma 28.3 (Finite incoming contributions). *For every well-ordered exponent set S , the union $T_S = \bigcup_{\gamma \in S} \text{supp } \partial t^\gamma$ is well ordered, and for each δ only finitely many $\gamma \in S$ have $d_{\delta\gamma} \neq 0$.*

Proof. The family $(t^\gamma)_{\gamma \in S}$ is summable. Apply (BM3), including its clause that the *derived* family is summable; the two conclusions are exactly the two clauses of Theorem 2.4. $\square$

Theorem 28.4 (Differentiation preserves common-domain coherence). *The rule*

$$\partial_H F = \sum_{\delta \in T_S} \left(\sum_{\gamma \in S} d_{\delta\gamma} f_\gamma \right) t^\delta \quad \left(\text{equivalently } \partial_H F = \sum_{\gamma \in S} f_\gamma(z) \partial t^\gamma \right) \quad (28.3)$$

defines a $\mathbb{C}$ -linear derivation of the common-domain Hahn algebra. Every inner sum is finite; each new coefficient is holomorphic on the same ordinary domain U , so differentiating the scalar coefficients neither shrinks nor destroys the common domain; and

$$\partial_H D_z F = D_z \partial_H F. \quad (28.4)$$

If the exponent group Γ is differentially stable (Theorem 20.1) the values stay in $\mathcal{H}_\Gamma(U)$; for an arbitrary initial Γ the same assertions hold after passing to its differential Hahn hull (Theorem 20.3).

Proof. Theorem 28.3 gives the well-ordered output support and the finiteness of each inner sum, so each new coefficient is a finite real-linear combination of functions holomorphic on the same U , hence holomorphic there. Linearity is immediate. For the product rule, expand the Hahn product of two families and use $\partial(t^{\gamma+\eta}) = t^\gamma \partial t^\eta + t^\eta \partial t^\gamma$; standard product summability and (BM3) make the expanded families summable, and coefficient comparison then uses only finite sums. Finally ordinary differentiation is $\mathbb{C}$ -linear and commutes with each finite inner sum in equation (28.3), which gives equation (28.4). Stability of Γ , or passage to the hull, keeps the supports inside the workspace. $\square$

Warning 28.5 (The derivative may need a larger monomial workspace). ∂_H need *not* preserve a previously chosen smaller monomial group: the support of ∂t^γ can contain monomials outside a preselected subgroup (Theorem 20.7). Its representing matrix $(d_{\delta\gamma})$ is only row-finite on each specified summable input support; no claim is made that an individual derivative has finite support or that the matrix has a finite global representation. A derivative routine should therefore return both a value *and* the parent in which that value is represented.

Theorem 28.6 (Total intrinsic derivative of an evaluation). *Let F be a common-domain family, let $c \in U$ and $h \in \mathfrak{m}_{\mathbb{C}}$, and let $F^\sharp(c+h)$ be the finite-halo evaluation equation (23.3). Then*

$$\partial(F^\sharp(c+h)) = (\partial_H F)^\sharp(c+h) + \partial h \cdot (D_z F)^\sharp(c+h). \quad (28.5)$$

Equivalently, for $z \in \mathcal{O}_{\mathbb{C}}$ with $c = \text{st}(z) \in U$ and $h = z - c$, $\partial(\text{ev}_z F) = \text{ev}_z(\partial_H F) + \text{ev}_z(D_z F) \partial z$. All terms may be placed in one set-sized differentially stable Hahn workspace after adjoining $\text{supp } h$ and taking the hull (Theorem 20.3).

Proof. Apply (BM3) to the summable double family equation (23.3). Since c is an ordinary complex point, all the numbers $f_\gamma^{(n)}(c)/n!$ are constants of ∂ , and $\partial h = \partial z$. The product rule splits the derivative of each term into

$$(\partial t^\gamma) \frac{f_\gamma^{(n)}(c)}{n!} h^n \quad \text{and} \quad t^\gamma \frac{f_\gamma^{(n)}(c)}{(n-1)!} h^{n-1} \partial h \quad (n \geq 1),$$

the second omitted for $n = 0$. The first family is summable by Theorem 28.3 followed by the same positive-support Taylor argument; the second is the corresponding Taylor family for $D_z F$ multiplied by the fixed scalar ∂h . So the split and the rearrangement are legitimate (Theorem 2.7 is satisfied), and reindexing gives the two terms of equation (28.5). $\square$

An ordinary holomorphic lift $f^\sharp$ has ∂ -constant coefficients, so equation (28.5) reduces to $\partial(f^\sharp(z)) = (f')^\sharp(z) \partial z$. A general coherent family carries the additional coefficient term, and this distinction is independent of whether the evaluated value has a short expression.

Example 28.7 (Two worked total derivatives). (i) $F(z) = t^2 e^z$ at $z = t$: here $\partial_H F = -2t^3 e^z$, $D_z F = t^2 e^z$, $\partial t = -t^2$, and

$$\partial(t^2 E(t)) = (-2t^3 - t^4) E(t),$$

the extra $-t^4 E(t)$ being the moving-argument term. Differentiating only the scalar prefactor, or only the analytic argument, would miss part of the answer. (ii) $F(Z) = \omega Z^2 + tZ$ at $z = 1+t$: here $\partial_H F = Z^2 - t^2 Z$, $D_z F = 2\omega Z + t$, $\partial z = -t^2$, and

$$F(z) = \omega + 2 + 2t + t^2, \quad \partial(F(z)) = 1 - 2t^2 - 2t^3,$$

which equation (28.5) reproduces exactly; keeping only the argument derivative would lose the leading constant 1. The family $F(Z) = \omega$ already disproves that omission.

28.3 Fixed contours, fixed centres, and moving poles

Corollary 28.8 (Coefficientwise linear naturality). *Any fixed $\mathbb{C}$ -linear operation on ordinary coefficient functions commutes with ∂_H wherever that operation is defined coefficientwise. In particular, for a fixed ordinary piecewise smooth contour $\gamma_c \subseteq U$,*

$$\partial \left(\int_{\gamma_c} F(z) dz \right) = \int_{\gamma_c} \partial_H F(z) dz, \quad \int_{\gamma_c} F dz := \sum_{\gamma} \left(\int_{\gamma_c} f_{\gamma} dz \right) t^{\gamma}, \quad (28.6)$$

and the analogue holds for a coefficientwise Laurent residue at a fixed ordinary centre, whenever all coefficient functions belong to the stated punctured-domain class.

Proof. Apply the fixed linear operation to the finite inner sums of equation (28.3): both orders of application give the coefficient $\sum_{\gamma} d_{\delta\gamma} L(f_{\gamma})$ at δ . $\square$

No surcomplex-valued Riemann integral is introduced: each side of equation (28.6) is a coefficientwise functional. For example, on a punctured ordinary disk about $a \in \mathbb{C}$ the family $F(z) = t/(z - a)$ has contour integral $2\pi it$ around a positive simple loop, and both sides of equation (28.6) equal $-2\pi it^2$; this is an exact Hahn identity.

Warning 28.9 (A moving contour or a moving pole is a different statement). The contour in Theorem 28.8 is fixed and ordinary. A moving contour, moving endpoints, or a residue with a different centre convention requires additional terms or a different statement; none is hidden in the corollary. At an *actual moving pole* the coefficient term cannot be dropped: for $a \in \mathbf{No}[i]$ and $F(Z) = 1/(Z - a)$,

$$\partial_H F = \frac{\partial a}{(Z - a)^2}, \quad D_Z F = -\frac{1}{(Z - a)^2}, \quad \partial \frac{1}{z - a} = -\frac{\partial z - \partial a}{(z - a)^2}, \quad (28.7)$$

these rational identities holding whenever the denominators are nonzero and needing no coherent expansion. A fixed-centre residue must not be silently replaced by an actual residue at a . The repository already distinguishes those residues; intrinsic differentiation supplies another reason to keep them apart. Likewise, the formal Hahn residue reading of Theorem 18.3 is not an identification with a cluster residue or an actual-point contour residue.

Remark 28.10 (A pointwise-only evaluation identity). For a *fixed* ordinary $z \in U$, strong additivity does give

$$\partial(F(z)) = \sum_{\gamma} f_{\gamma}(z) \partial(t^{\gamma}),$$

a pointwise scalar identity. Turning it into a statement about a common-domain coherent family, differentiating again in z , or integrating it along a path requires separate control of the coefficient-family supports and of the domain — which is precisely what Theorems 28.4, 28.6 and 28.8 supply, under their hypotheses. The distinct coefficient rings documented in the repository remain distinct after ∂ is added.

28.4 Formal power series need their own admissibility condition

On $K[[Z]]$ the formal coefficient and variable derivatives commute:

$$\tilde{\partial} F = \sum_n (\partial c_n) Z^n, \quad \partial_Z F = \sum_{n \geq 1} n c_n Z^{n-1}, \quad \tilde{\partial} \partial_Z F = \partial_Z \tilde{\partial} F.$$

Evaluation at a scalar a requires summability. If the families defining $F(a)$, $(\tilde{\partial} F)(a)$ and $(\partial_Z F)(a)$ are all summable, then the same computation gives

$$\partial(F(a)) = (\tilde{\partial} F)(a) + \partial a \cdot (\partial_Z F)(a). \quad (28.8)$$

Theorem 4.5 is the sharper version, which assumes only the summability of the evaluated family and *derives* the other two.

Warning 28.11 (Do not apply this to an unevaluable germ). Equation (28.8) must not be applied to an arbitrary unevaluable formal germ. The finite-halo theorem Theorem 28.6 supplies its own summability proof; an unrestricted formal coefficient sequence does not. The limiting case is Theorem 6.2(iii): the evaluation rule $f(h)$ of Theorem 6.1 requires *ordinary constant* coefficients, and $\sum_{n \geq 0} \omega^{n^2} X^n$ is not summable at $X = \omega^{-1}$ because its monomials ω^{n^2-n} are not reverse well ordered. The infinitesimal chain rule must not be exported to arbitrary $\mathbf{No}[i]$ -coefficient streams without a joint support certificate. On the specific ring $\mathcal{O}(U)((t))$ one may define $\partial_H = -t^2 D_t$ and D_z coefficientwise, and they commute because both sides have identical coefficient formulas; that statement need not hold as a meaningful endomorphism assertion on an arbitrarily chosen K_Γ , since the scalar derivative can require monomials outside that workspace (Theorem 28.5).

28.5 The four categories, side by side

Table 7: Four typed categories that share differential notation. They are not competing answers to one problem.

Category	Constants	Constructive positive result	Boundary
Scalar $(\mathbf{No}[i], \partial)$	$\mathbb{C}$	Finite-primitive criterion, finite-phase solution formula, gauge normal form, constant-coefficient classification	$\partial y = iy$ has no nonzero solution; no global compatible exponential
Real-exponent Hahn scalar K_Γ , $\Gamma \subseteq \mathbb{R}$, $1 \in \Gamma$	$\mathbb{C}$	$\partial - \lambda$ bijective for $\lambda \in \mathbb{C}^\times$; $P(\partial)$ bijective on $\mathbb{C}((t))$ when $P(0) \neq 0$; exact residuals	t has no primitive there; the real-exponent hypothesis is substantive
Formal $K[[Z]]$	K	Linear coefficient recursion, always solvable	Evaluation may not be summable
Common-domain $\mathcal{H}_\Gamma(U)$	K_Γ	Positive-perturbation linear and polynomially nonlinear initial-value theorems, support certificates, monodromy in $\mathrm{GL}_d(K_\Gamma)$	A negative scalar leading exponent prevents a coherent exponential; nilpotence is an exception

Part VI

Computation, formalization, and the register of non-claims

29 Certificate-oriented computational contracts

29.1 Theorem-driven outputs, not a universal solver

The exact criterion suggests a usable symbolic interface. Given a represented coefficient $q = a + ib$, a backend may try to produce primitives A, B with $\partial A = a$ and $\partial B = b$ together with a certified decomposition $B = P + r + \eta$ with $P \in \mathcal{P}$, $r \in \mathbb{R}$, $\eta \in \mathfrak{m}$. It need not use the normalization $\mathcal{I}_0$ to do

so, since $\text{pp}(B) = \mathfrak{o}(b)$ is unchanged by the real integration constant (Theorem 5.7). The logical workflow is

$$q = a + ib \mapsto B \mapsto \text{pp}(B) \mapsto \begin{cases} \text{a base-field solution,} & \text{pp}(B) = 0, \\ \text{a certified obstruction and an extension symbol,} & \text{pp}(B) \neq 0. \end{cases} \quad (29.1)$$

Warning 29.1 (This is a specification, not a terminating algorithm). Equation (29.1) is a mathematical specification. For arbitrary program-described surreal normal forms, integration, exact support access and zero testing may all be unavailable. A failed search for a primitive is not a proof that none exists — in the full field every element has one — and a failed representation of an infinitesimal exponential is not a proof that no mathematical solution exists. A backend should return a well-defined symbolic primitive and state that the purely infinite part has not been decided; it must not report “no solution” merely because its representation cannot compute that part. The repository’s capability-tier distinctions are therefore especially important at this differential boundary [19].

Three logically distinct outcomes arise for the ambient question, and a fourth once a workspace is declared.

Table 8: Result contracts. The last row is not a nonexistence theorem.

Result	Certificate required	Mathematical conclusion
Workspace solution	Rational exponent r , a finite k with positive support and $-rt + \partial k = q$	A certified nonzero solution $t^r E(k)$ lies in the declared Hahn field (Theorem 19.3)
Ambient-only solution	A verified primitive A of a , a <i>finite</i> accumulated phase B of b with verified $\text{pp}(B) = 0$, and an admissible infinitesimal-residue expression	A nonzero solution $\exp(A)E(i\eta)$ exists in $\mathbf{No}[i]$, but the workspace obstruction is nonzero (Theorem 8.4)
Ambient phase obstruction	A verified accumulated phase B of b with certified nonzero $\text{pp}(B)$	No nonzero surcomplex solution exists; the gauge normal form returns P and the removable factor g (Theorem 9.1)
Undetermined representation	A required primitive, decomposition, equality or support certificate is unavailable	<i>No</i> existence or nonexistence conclusion

For the finite exact core — finite rational Hahn polynomials with Gaussian-rational coefficients, represented as a finite map $q \in \mathbb{Q} \mapsto c_q \in \mathbb{Q}(i)$ with zero coefficients deleted — the first three outcomes are decidable. Theorem 18.1 gives a terminating derivative; additive integration inside $\mathcal{H}$ succeeds exactly when the exponent-1 coefficient vanishes, and otherwise equation (18.4) gives an ambient primitive using one explicit $\log \omega$ constructor. Two tests must be offered *separately*: the *workspace* test rejects every nonzero term below exponent 1 and rejects a nonreal exponent-1 coefficient (with Gaussian-rational coefficients a real exponent-1 coefficient is automatically rational), and the *ambient* test examines the imaginary part b and succeeds precisely when $b = 0$ or every exponent of $\text{supp } b$ exceeds 1. For arbitrary computably described Hahn supports none of this is advertised as a total decision procedure, and the semantic formula $\Phi(q) = \text{pp}(\mathcal{I}_0 \text{Im } q)$ remains exact without promising an algorithm for $\mathcal{I}_0$ or for equality of arbitrary represented scalars.

Warning 29.2 (What a certificate must actually establish). A solution certificate consists of exact derivative identities plus support and size evidence. For a finite expression the identities may reduce

to field algebra plus the real exponential and logarithm rules. For an infinitesimal Taylor expression the certificate must identify positive support and justify the summations used in the derivative rule. For a *nonsolvability* certificate it is not enough to report a large leading term numerically: one must prove that an accumulated phase has nonzero purely infinite part. The output $t^r E(k)$ is a finite exact expression denoting an infinite Hahn series, not a finite list of its coefficients.

A finite-phase certificate need not compute the whole normalized primitive: it suffices to exhibit a finite B and prove $\partial B = \text{Im } q$. This is useful when semantic existence is known but a general symbolic integration procedure is unavailable.

The two inputs $q_1 = i\omega^{-2}$ and $q_2 = i\omega^{-1}$ are distinguished by the exact accumulated phases $-\omega^{-1}$ and $\log \omega$. A numerical specialization $\omega := M$ for a large real M would instead give ordinary complex equations with nonzero oscillatory solutions in *both* cases. Such a specialization is not an embedding of differential structures and cannot certify the surreal answer.

29.2 A derivation-aware representation interface

Every exact parent should record both its ambient embedding and its closure capabilities. A Laurent parent may advertise: closed under ∂ ; closed under $(\partial - \lambda)^{-1}$ for $\lambda \neq 0$; *not* closed under primitives. A log-exponential extension can support more primitives without thereby supporting arbitrary global phases, and adding a constructor named **Exp** is not enough: its domain and its derivative contract must be stated.

Warning 29.3 (A truncation must record the *type* of omitted information). A truncation of Y_H in equation (18.10) can carry the exact residual equation (18.11) and a real-power support tail. That certificate does *not* rule out an exponentially small homogeneous correction in the full field (Theorem 18.12). A computation that returns “unique solution” must say whether uniqueness is in K_Γ , in the full $\mathbf{No}[i]$, in an oscillatory extension, or among coherent sections of a named ring. Likewise, returning $\exp(\int A)$ is not a certificate for a complex integrating factor: the software must either verify the finite-phase criterion, select a differential extension, or report that the required operation is not implemented. An unevaluated expression and a proven element of the requested field are different outputs. A matrix jet specified by total degree, a Laurent jet specified by valuation, and an exact recursively defined series must remain three different output types (Theorem 27.1). Branch continuation and monodromy should use ordinary-domain path data rather than an implied fine-continuous path in the surreal class (Theorem 26.1). For several extension symbols, the integer relation lattice of equation (21.6) is additional required data.

A realistic first implementation can stay inside an explicit differential algebra: rational expressions in a finite collection of real-power and logarithmic scales; finite constant-coefficient operators; formal Laurent series represented by a recurrence or by coefficient access. At each extension a derivative routine should return both a value and its parent, and an integration routine should distinguish a proved primitive, a proved obstruction *in the selected parent*, and a failure of the available algorithm to decide. The missing t -coefficient of equation (18.3) is a proved parent obstruction; failure to recognize a complicated primitive in the full surreal field is not.

29.3 Formalization boundaries

A proof-assistant development can begin from a set-sized ordered field with a derivation, a chosen infinite scale w with $\partial w = 1$, real constants, and specified local exponential and logarithm operations with their support theorems. The bounded-derivative argument (Theorem 3.8), the complexification

Table 9: Operations, required certificates, and result boundaries.

Object or request	Required certificate	Result boundary
Intrinsic derivative	A selected derivation and a parent field, with a supported image in a known parent	May require a larger parent: a monomial derivative need not lie in the old one (Theorem 28.5)
Coordinate derivative	One common coefficient domain	Every Hahn monomial and scalar parameter stays constant
Hahn evaluation	A joint reverse-well-ordered support with finite multiplicity	A formal expression may not denote a sum at the input (Theorem 11.9)
Coherent total derivative	One common coefficient domain and <i>both</i> derivatives	Dropping ∂_H changes the answer (Theorem 28.7)
Finite-phase integrating factor	A real accumulated phase B with verified $\text{pp}(B) = 0$	A nonzero solution in $\mathbf{No}[i]$
Phase obstruction	A verified nonzero purely infinite part of an accumulated phase	Nonexistence of a nonzero homogeneous solution
Hahn resolvent	A real-exponent group with $1 \in \Gamma$, and a support-shift certificate; a lower exponent bound for a prefix	An exact Hahn solution <i>in that workspace</i> ; prefix computation is not stream equality
Coherent initial-value solver	Base solution, common domain, positive support, finite decomposition access	A certified coefficient recursion
Global phase	An explicit phase convention with its supported domain	A group law does not imply compatibility with ∂ (Theorem 11.7)
Oscillatory extension	A distinct parent and the rule $\partial T = iT$	Not a surcomplex scalar (Theorem 21.8)

(Theorem 3.5), the phase decomposition (Theorem 6.3) and the first-order criterion (Theorem 8.4) then become lemmas over explicit hypotheses. Surjectivity of the derivation is a *genuine additional hypothesis*, not something inferred from real closedness.

Four concrete proof obligations are worth naming, in increasing order of difficulty.

- (1) The explicit rational monomial derivative formula equation (18.1) on $\mathcal{H} = \mathbb{C}((t^{\mathbb{Q}}))$. The primitive-image and logarithmic-image theorems (Theorems 19.2 and 19.3) then reduce to support translation, leading-term factorization and the infinitesimal exponential theorem. This is a tractable independent target.
- (2) The implication “monomial support closure $\Rightarrow$ closure of the full small-support Hahn type under ∂ ”, which is the content of Theorems 20.2 and 20.3. It complements, rather than duplicates, a generic Hahn convolution development.
- (3) Theorem 28.3: for each admissible input support, each output coefficient receives only finitely many monomial-derivative contributions. Once available, the common-domain coefficient interface becomes finite linear algebra at every output exponent.
- (4) The represented quotient $\mathcal{P}$ of Theorem 2.1, so that the rank-one classification does not accidentally require a type whose elements are proper classes.

The interfaces should keep ∂ , ∂_Z , $\tilde{\partial}$, ∂_H , D_z and fine differentiability separate. Evaluation is a map with a support certificate, and Theorem 4.5 is a compatibility theorem for that map; it is not a coercion identifying operators. An exponential operation should carry the domain predicate $\text{Im } z \in \mathcal{O}$ whenever it promises the intrinsic differential law. The difficult Berarducci–Mantova existence construction can remain an explicit imported interface until separately formalized; it is a much larger dependency and is not needed to check the logical consequences of a stated interface.

Warning 29.4 (No formalization is delivered). No Lean source and no machine-checked formalization is included or asserted to exist, here or in the repository or in a library. The preceding paragraphs specify proof obligations and a possible order of development, nothing more. This report also does not claim that the concrete surreal carrier has already been connected to every required Hahn, real-closed-field, exponential or derivation interface.

30 Relation to the literature and to the companion reports

The proof has three layers. At the foundation are the established surreal field and exponential constructions and the chosen strongly additive, surjective derivation [1, 7]. The normal-form splitting and the infinitesimal formal calculus connect that structure to the repository’s finite-angle geometry [17]. Above them are the explicitly proved differential consequences: the finite-primitive criterion, the gauge normal form, the maximal strip, the scalar and constant-coefficient classifications, the workspace images, and the coordinate initial-value theorems.

The universal H -field results [3] explain why the differential viewpoint is a major part of surreal theory rather than an ad hoc extra operation; through Part IV they are cited for context and are not a substitute for an unstated linear-solvability theorem. In Part III that changes: the elementary embedding of [3] is used as a transfer principle, and the splitting and surjectivity facts of [4, Lemma 2.4.19] are used as premises (Theorem 15.6, item N53). The amplitude–phase results of [5] over Hardy fields, and the classical Pinney parametrization [6], are the antecedents of Theorems 17.6 and 17.7 and are kept visible there; no solution of the Boshernitzan conjecture named in [5] is claimed. The ω -series composition results [2] show that an external function interpretation is valid

on a substantial specified class while warning against unrestricted composition (Theorem 22.2). The global exponential construction [8] supplies the crucial contrasting structure: algebraic global exponentiation and number-derivation compatibility are different requirements (Theorem 11.6). Bournez and Guilmant [9] study closure properties of selected surreal fields far beyond the elementary constructions of Section 20, and no theorem of theirs is used to justify them. Ordinary complex initial-value theory [12] supplies the analytic ingredient of Part V, and [13, 14] supply the Picard–Vessiot and differential-module terminology in their standard sense. Neumann [10] and Higman [11] are the published sources for Section A.

The repository documents are treated as unrefereed project manuscripts that motivate the gap, not as replacements for the published differential foundations; none of their analytic-geometry or contour theorems is needed for the main proofs here. Three cross-links matter.

- **Trigonometry.** That report supplies the finite-angle phase group and explicitly declines to choose a derivation [17]. Theorem 6.5 reproves the part needed here and adds the derivative identity; Theorem 11.6 states precisely what the present no-go theorems do and do not say about its global phase extensions. Its existing constructions need not be removed or renamed as false; their contracts need only be distinguished from the intrinsic differential one, and the derivation should be named wherever it is used.
- **Analysis.** That report’s derivation-obstruction theorem, for one chosen canonical global surcomplex exponential, is *prior coverage* [18]. Theorem 11.4 strengthens it in phase-independence and does not contradict it; the two are compatible. Part V reuses its common-domain coherent-section language and refines the germ-versus-common-domain-versus-formal distinction into the explicit separations of Table 7.
- **Computer algebra and foundations.** The capability taxonomy and the subsection “Differential equations must name their category” [19], and the four-interface discipline of [20], are answered directly by Tables 1, 8 and 9 and by the tiering contract of Section 29.1.
- **Spectral theory.** The companion report *Surcomplex Spectral Theory* [21] supplies the finite algebraic spectral vocabulary that Section 16.1 uses, and its “spectrum” is the algebraic eigenvalue and singular-value list. That is *not* the differential phase spectrum of Theorem 5.1(e), and Theorem 5.2 states the distinction in the terms both reports need. Theorem 16.5 is the bridge between them for Hermitian coefficients; Theorem 16.10 shows there is no bridge in general. **Cross-link both ways.**

31 The register of non-claims

This section is part of the result. Each item below was stated explicitly by at least one of the seven source manuscripts; all of them survive the merge. Several are also stated in place above, and the cross-references say where.

31.1 Recovered on audit

An audit of this merge against its seven sources found two scope disclaimers stated in a source and missing here. They are restored, since each limits a claim the report otherwise appears to make.

- R1. **The interface table describes nothing that exists.** One source introduces its certificate and interface contract with the explicit disclaimer that these are *proposed* interfaces, not

claims about functions already implemented in this repository’s symbolic packages. That applies verbatim to Table 9: nothing in it is implemented anywhere in the collection, and its prescriptive voice is a proposal throughout. The companion computer-algebra report holds the existing capability taxonomy, and it implements none of this either.

- R2. **A recorded run is not a reproduction promise.** One source notes that a local environment’s Python version and elapsed time need not match its bundled run. Section D reproduces the delivered records and states the versions they were recorded under; it does not assert that a rerun elsewhere will reproduce those versions or timings, and a mismatch in either is not evidence of a failed check.

31.2 Provenance of the mathematics

- N1. The Berarducci–Mantova derivation is *imported*, not constructed and not reproved (Theorem 3.1). Its construction via log-atomic numbers, a pre-derivation there, path derivatives and nested truncation ranks is not reproduced. The summability and surjectivity proofs are substantial imported inputs, not consequences of the product rule alone. Assigning a derivative to a few familiar monomials is not a construction of a strongly additive derivation on every surreal, and this report uses the published conclusions rather than a pretended implementation.
- N2. No uniqueness characterization of ∂ is offered (Theorem 3.3). Other surreal derivations exist; none of them is classified here; the listed structural properties do not characterize every possible surreal derivation; and the word “canonical”, where it survives, is a naming convention for this distinguished construction.
- N3. Every intrinsic statement is relative to this one fixed ∂ with the normalization $\partial\omega = 1$, which is indispensable for discussing a named equation such as $\partial y = iy$. The results are *not* derivation-agnostic.
- N4. Smallness is *derived* here from the H -field positivity clause (Theorem 3.8), and is therefore not advertised as part of the imported theorem. Two of the source manuscripts instead import it, one of them flagging it as specific to the chosen derivation and not a general fact about derivations of non-Archimedean fields; that route is recorded in Theorem 3.9. Either way, smallness is a structural hypothesis about this derivation.
- N5. The positive-support summability lemma is imported in its standard additive-exponent form (Theorem 2.6) and is additionally proved in Section A. One source audit records that the Neumann paper was *not retrieved in full* and that the publisher page did not provide accessible full text in that session; the article therefore imports the standard lemma rather than claiming to reprove or re-audit the original paper, and Section A exists so that nothing depends on an uninspected passage. Higman’s publication metadata was checked against the publisher. Gonshor’s book is used as a standard reference for normal forms and the surreal exponential; only its publisher catalogue was identified, and the whole book was not downloaded or audited.
- N6. The restricted composition, Taylor and global-incompatibility results for ω -series are *separately* imported (Theorem 22.2); they do not follow from Theorem 3.1 and are not hypotheses hidden inside the phase criterion. The composition obstruction is quoted with its hypotheses intact, for this particular derivation, and is *not* strengthened into an assertion about every

conceivable notion of composition or about every other choice of derivation. Nothing here settles alternative global composition structures. The finite-phase obstruction of this report is a direct differential-algebraic argument, not a restatement or strengthening of that composition theorem.

- N7. The universal H -field results are cited for context, not as a substitute for an unstated linear-solvability theorem, and the elementary embedding of transseries is a theorem about a specified ordered differential language which must not be upgraded to a statement about every simultaneously added analytic, compositional or complex exponential operation (Theorem 22.3). Bournez–Guilmant is cited as contextual work only; no theorem of it justifies Theorems 20.3 and 20.8.
- N8. The class-theoretic framing is an *assumption* (Section 2.1): definable classes, or a class-theoretic presentation with enough replacement for the stated constructions. The exact sequences are elementwise and are not the formation of a set of proper-class cosets (Theorem 9.3).

31.3 Novelty, refereeing, verification

- N9. No priority or novelty claim is made, and no claim to resolve a named published open problem. These are proved deductions from explicitly stated hypotheses; some may admit formulations already familiar in the general theory of H -fields, and the deductions are *not* asserted to be absent from every published treatment of H -fields, transseries or differential algebra. Ordinary constant-coefficient differential algebra and formal resolvents have a long history; their specialization here is exposition.
- N10. No independent refereeing and no peer review is claimed. The source manuscripts were research expositions prepared with AI assistance; this merge is of the same character.
- N11. No proof-assistant verification, no Lean source, no machine-checked proof object (Theorem 29.4). The general proofs remain human-readable mathematical arguments.
- N12. The literature consultation was targeted, not an exhaustive novelty or priority search, and the bibliography is correspondingly targeted.
- N13. The seven repository audits were *targeted coverage audits*, pinned to one revision, and read-only. None was a line-by-line verification of the fifteen reports or of their archived sources; none asserts that the word “derivation”, or Berarducci–Mantova, or transseries, is absent from every source manuscript; none assesses the correctness of the repository’s other claims. The reading scopes recorded in the sources include the documentation catalogue and manifest; the analysis, trigonometry, foundations and computer-algebra READMEs; the opening foundational section of the analysis source; selected computer-algebra source ranges; and lines 1–220 of the trigonometry article. Several of the audits record that code-search or connector calls returned no usable matches or were truncated, and that those empty or partial responses were *not* used as evidence of absence and not treated as a complete inventory. The assessments are revision-specific: no claim is made that the identified gap remains unfilled on a later commit or branch. The repository’s existing phase decomposition and chosen-exponential obstruction are acknowledged as prior coverage, not presented as missing results. No repository file was modified by any of the seven source packages, and the placement suggestions they made were suggestions.

31.4 Effectivity: what is exact but not computable

- N14. $\mathcal{I}_0$ is a normalization defined by a uniqueness property, not an algorithm (Theorem 7.2). It is not asserted to be multiplicative, not asserted to preserve arbitrary Hahn sums, and not asserted to be effective for arbitrary surreal inputs or computable from a supplied symbolic representation. No implementation of it on arbitrary program-described surreals, no universal zero test, and no full differential-algebraic solver is supplied.
- N15. $\mathfrak{o}$ and Φ are exact but not presented as computable functions on arbitrary finitely described surreal inputs. The semantic formula $\Phi(q) = \text{pp}(\mathcal{I}_0 \text{Im } q)$ is exact without promising an algorithm for $\mathcal{I}_0$ or for equality of arbitrary represented scalars. An unevaluated primitive, or an undecided finiteness test, is *not* a negative solvability proof (Theorem 29.1).
- N16. Equation (29.1) is a specification, not a terminating algorithm. Only the first three outcomes of Table 8 are claimed decidable, and only for the finite exact core; for arbitrary computably described Hahn supports they are not advertised as total decision procedures.
- N17. The rational decision procedure is restricted to $\mathbb{Q}(X)$ although Theorem 10.6 is stated over $\mathbb{R}(X)$, and it *raises* on unsupported input rather than reporting mathematical nonexistence (Theorem 10.7). No algorithm for an arbitrary real coefficient is asserted.
- N18. A well-ordered support is a *semantic* certificate, not automatically an effective presentation (Theorem 25.5). No general terminating enumeration algorithm for arbitrary high-rank program-described supports is claimed. A closed formula for a coefficient stream is not a finite list of every coefficient, and the coefficient recursion of equation (18.16) does not make equality of arbitrary program-generated streams decidable.
- N19. ∂_H is represented by a matrix that is only *row-finite on each specified summable input support*; no claim is made that an individual monomial derivative has finite support, or that the matrix has a finite global representation, or that ∂_H preserves a previously chosen smaller monomial group (Theorem 28.5).
- N20. Theorem 20.3 is a semantic existence construction, not an effective algorithm for computing $\partial(t^\gamma)$ at arbitrary surreal exponents, and its countably many stages do not constitute a terminating algorithm for every input representation (Theorem 20.5).

31.5 Scope of the scalar theory

- N21. The criterion classifies *scalar homogeneous first-order* multiplicative equations, and nothing more. It is not a claim that all nonlinear equations have solutions, that $\mathbf{No}[i]$ is differentially closed — it is not (Theorem 8.12) — that arbitrary nonlinear scalar equations or all higher-order operators on the full class are classified, or that primitives are computable from finite input descriptions.
- N22. No general inhomogeneous existence theorem is claimed *by Part II*. Outside the admissible locus $\Phi(q) = 0$ its only assertion is *uniqueness* (Theorems 12.1 and 12.2); the absence of a nonzero homogeneous solution does not imply inhomogeneous nonexistence, and the additional differential-field structure that a general inhomogeneous existence theorem would need is not invoked there. Nor is it claimed that such broader solvability questions are open in the literature. Theorem 12.1 proves surjectivity of $\partial - q$ only in the admissible case, and no claim is made about surjectivity of all linear differential operators on $\mathbf{No}[i]$: a trivial kernel

is explicitly distinguished from a failure of surjectivity, and $\partial - i$ on $\mathbb{C}((t))$ is the witness (Theorem 18.9). Part III does prove first-order inhomogeneous existence (equation (15.2)) and, from it, existence for every forcing of every finite system (Theorem 15.11); that conclusion rests *entirely* on the separately imported Theorem 15.6 and not on the rank-one criterion, as Theorem 15.12 states. It leaves the present item standing for the scalar theory as such, it does not make any homogeneous equation solvable, and it changes none of the witnesses of Theorem 12.2: what it adds is that their uniqueness is now known to be accompanied by existence. No effectivity follows (item N57).

- N23. The constant-coefficient theorem and its matrix corollary concern *ordinary complex constant* coefficients and matrices only, not arbitrary surcomplex-coefficient systems (Theorem 13.4). The kernel theorems classify *homogeneous* solutions in $\mathbf{No}[i]$ only; a full kernel does not settle an inhomogeneous equation over a smaller subfield.
- N24. The Hermitian case of the nontriangular matrix problem is settled; the non-Hermitian case is not. Theorem 15.9 gives a complete phase normal form for every finite system, and Theorem 16.5 computes its invariant from ordinary algebraic eigenvalues *when the coefficient is iH with H Hermitian*. Outside that hypothesis no spectral formula is claimed, and Theorem 16.10 refutes the obvious candidate: a nonnormal coefficient with instantaneous eigenvalues $\pm i$ whose system nevertheless has a fundamental matrix in $\mathrm{GL}_2(\mathbf{No}[i])$, so that integrating imaginary parts of eigenvalues would predict $\{\omega, -\omega\}$ instead of the correct $\{0, 0\}$. Within the scalar theory of Part II taken alone, none of this is available: there the trace condition is only the first invariant (Theorem 14.2), the torus theorem treats *diagonal* systems only (Theorem 21.12), constant matrices are settled while arbitrary matrices with surreal entries are not, and no general matrix normal form follows from the scalar result — the normal form of Part III needs the separate import of Theorem 15.6. Higher-rank differential modules remain unclassified, and so does the question of which nontriangular differential modules admit a filtration whose rank-one factors carry computable obstructions, which needs genuine module-theoretic information rather than the eigenvalues of an instantaneous coefficient matrix (Theorems 16.11 and 17.13).
- N25. No differential closedness, no universal Picard–Vessiot closure, no classification of all differential extensions, and no decision procedure for arbitrary nonlinear differential equations is claimed. The Picard–Vessiot constructions of Section 21 are explicit; the set-sized ones deliberately avoid any blanket application of Picard–Vessiot existence theory to a proper-class base, and the proper-class ones are proved by hand for exactly that reason. The extension $\mathbb{C}(\omega)(\mathcal{E})$ of Theorem 21.8 is used only to exclude a differential embedding, and no assertion whatever is made about its full constant field.
- N26. Theorem 11.2’s maximality does not claim that the differential law alone uniquely specifies every normalization on $\mathcal{S}$; ratios of admissible values are differential constants.
- N27. All nonexistence results concern surcomplex *number* solutions under ∂ (Theorem 8.9). They do not deny ordinary oscillatory functions, do not say that $\sin z$ and $\cos z$ fail their ordinary differential equations, and do not deny algebraically defined global surcomplex phase extensions or the Ehrlich–Kaplan canonical global surcomplex exponentials, which satisfy different structural contracts and should not be described as nonexistent (Theorem 11.6). The obstruction is only to *also* satisfying pointwise differential compatibility for this ∂ .

- N28. No general composition law on **No** is constructed and unrestricted substitution “replace ω by x in every surreal” is never used (Theorem 4.7).
- N29. Splitting a differentiated infinite sum into a coefficient-derivative part and a variable-derivative part requires *both* separated families to be summable independently (Theorem 2.7); summability of the sums does not imply it.
- N30. ct is not a multiplicative standard part (Theorem 2.3), and the real-power rule $\partial\omega^r = r\omega^{r-1}$ is asserted for ordinary real r only and must not be extended by silently replacing r with an arbitrary surreal exponent (Theorem 2.2).
- N31. The infinitesimal exponential’s injectivity is *local* to $\mathfrak{m}_{\mathbb{C}}$ and is not a global injectivity claim; the infinitesimal evaluation rule requires *ordinary constant* coefficients and must not be exported to arbitrary **No**[i]-coefficient streams without a joint support certificate (Theorems 6.2 and 28.11).
- N32. Strong summability is not valuation convergence and is not fine-topological convergence (Theorem 2.5). No cofinal growth of $n\sigma$ in a higher-rank value group is established, and no argument uses one. The truncation residual identities certify a formal remainder at a stated order, not convergence of the truncation sequence in any surreal topology, and the factorial Hahn series is not an ordinary analytic function germ and not a fine-topology limit of its truncations (Theorem 18.14). Numerical substitution is declined as a surrogate for a Hahn identity.

31.6 Scope of the workspaces

- N33. Neither localization theorem gives closure under *all* set-indexed summable families (Theorem 20.9): an arbitrary-support closure is a different and stronger demand, a localized field need not contain every Hahn series over its own monomials, and there is no Hahn completeness, no finite data representation, no algorithm for the adjoined maps, no birthday or cardinal bound, no computable enumeration and no termination claim for symbolic integration. Arbitrary future coherent families are not automatically covered. Theorem 20.8 is not proved to produce a full Hahn field with a fixed exponent group, and the lighter variant of Theorem 20.10 is proved to be neither.
- N34. Theorem 20.3’s minimality holds only among full Hahn fields indexed by divisible exponent groups, not among all differential subfields; the hull group itself need not be countable; and the theorem is not primitive closure, nor closure under real exponentiation, arbitrary logarithms, or solutions of all first-order linear equations (Theorems 18.2 and 20.5).
- N35. The sharp support threshold is deliberately restricted to real-exponent Hahn coefficients, and the criterion provably does *not* globalize to **No** by valuation alone (Theorem 10.5). The valuation test of Theorem 18.4 is not a valuation-only classification of arbitrary surreal coefficients. The threshold must not be merged into the general resolvent.
- N36. Theorem 18.6’s real-exponent hypothesis is substantive: in a higher-rank group a fixed increment 1 need not dominate all relevant scales, and no claim replaces the support argument by an assertion that successive valuation improvements converge (Theorem 18.7). Theorem 18.8 is confined to the rank-one Laurent field with $P(0) \neq 0$, and its residual bound is a certificate for a finite truncation: it does not assert that the truncation equals the full solution.

- N37. The factorial examples assert no Borel summation, no Stokes multiplier and no analytic continuation rule (Theorems 18.12 and 18.14); the formal Hahn residue reading of Theorem 18.3 is not an identification with a cluster residue or an actual-point contour residue from the repository's analytic reports.
- N38. Higher rank is open here. The obstruction interacts with invariant subspaces and changes of basis in ways this report does not determine.

31.7 Scope of the coordinate theory

- N39. The coherent theory requires *positive* support of the perturbation. That hypothesis is sufficient, not necessary: the nilpotent example Theorem 24.6 shows a coherent solution with a negative exponent, and negative-exponent matrix systems need finer analysis (Theorem 25.4).
- N40. Theorem 25.2 requires either polynomial dependence on the dependent variables or holomorphic dependence with the Taylor admissibility of Theorem 25.1, together with a *supplied* ordinary background solution on U and U simply connected. The background solution's existence on an arbitrary prescribed domain is not asserted, the theorem does not continue it past its ordinary singularities, negative-valued perturbations and arbitrary radius-free coefficient germs are not covered, and unrestricted substitution into an arbitrary holomorphic function of Y without specifying its domain and support conditions is out of scope. Uniqueness is exactly the class of Theorem 25.3 and nothing wider.
- N41. No general irregular-singularity classification, no Stokes classification and no Riemann–Hilbert correspondence over arbitrary Hahn fields is offered.
- N42. Monodromy (Theorem 26.1) is an *ordinary*-domain construction on the ordinary universal cover with coefficientwise continuation; no fine-continuous surcomplex path is asserted. Monodromy for the *nonlinear* coherent problem on non-simply-connected domains remains future work.
- N43. The finite-halo evaluation equation (23.3) is infinitesimal Taylor evaluation only, not evaluation at an arbitrary infinite coordinate (Theorems 25.6 and 25.7).
- N44. The analytic interface is stated for a *fixed* ordinary open domain and a *fixed* ordinary contour or residue centre. Moving contours, moving endpoints, moving poles and other residue conventions are explicitly outside it (Theorem 28.9), and statements are deliberately not transported between arbitrary formal germs and fixed-domain holomorphic families.
- N45. Formal existence, microscopic evaluability and common-domain coherence are three different promises and are never substituted for one another (Theorem 28.2). The formal recursion asserts nothing about evaluability at a given surcomplex number.
- N46. The Riccati example of Theorem 25.7 gives an exact coherent solution on a common ordinary domain, not an all-scale surcomplex entire function; it does not contradict an all-scale rigidity theorem stated in a different function class.
- N47. Theorem 24.1 assumes no norm on coefficients, no contraction, no convergence of Picard iterates and no shrinking domain per exponent (Theorem 25.4); nothing here claims to have invented positive-support recursion.

31.8 Scope of the executable checks

- N48. The eight delivered verification suites check *finite exact symbolic identities and finite jets*, and nothing else. In total they record $1,395 + 115 + 149 + 258 + 291 + 150 + 259 + 62$ passing checks (Section D); a matrix identity or a whole checked coefficient prefix counts as one check in two of the suites. They do not construct the surreal numbers, do not implement Conway normal forms or the Berarducci–Mantova path construction, do not verify arbitrary Hahn summability or strong additivity for arbitrary supports, do not verify class-size or proper-class statements, do not verify the Hahn–Neumann support lemma or Higman’s lemma, do not verify transfinite recursion, and do not verify any of the general nonexistence or infinite-support theorems. They are regression suites for examples, not proof assistants and not implementations of a general surreal differential solver.
- N49. The suites model ∂ by ordinary symbolic differentiation: as $-t^2 d/dt$ on t -expressions, as d/dx on expressions in an ordinary variable standing in for ω , and as $f \mapsto \text{cancel}(iT \partial f / \partial T)$ on the oscillatory extension. The primitive tests replace ω by an ordinary positive variable and check finite algebraic differential identities; they do not infer that an ordinary exponential is a scalar-compatible surcomplex exponential. A finite Taylor check is a formal coefficient check up to its stated order.
- N50. Declared cutoffs are part of the claim, not incidental: one suite uses a formal-series cutoff of 16 with a deterministic seed; another uses matrix *total degree* 6 — explicitly not a valuation cutoff, since t^{100} has higher total degree and lower valuation than u — a Laurent reciprocal-series order of 12, and nonlinear checks through t^{12} ; another checks resolvent residuals for λ in a listed finite set and N in a listed finite range; another checks factorial residuals to $N = 16$. A successful check of a residual at finitely many orders is not a proof of its general formula, and symbolic agreement of finite Taylor coefficients is not a proof of Hahn summability.
- N51. The restricted rational classifier accepts exact Gaussian-rational or rational coefficients and rejects floats, strings, non-Gaussian-rational coefficients, unbound parameters, and non-rational functions by *raising*; those rejections are counted as passing checks in one suite, and the script is explicit that it is not an expression parser (Theorem 10.7).
- N52. Two artifact-level discrepancies in the source packages are recorded rather than silently repaired: one package’s PowerShell build script was supplied but not executed on Windows in the preparation environment, and one package’s README lists a checksum file that is not present among the delivered files. Neither affects any mathematical statement.

31.9 Scope of the matrix theory

The eighth manuscript states twelve explicit non-claims of its own. All survive here, together with the audit findings recorded against its package.

- N53. Theorem 15.6 is an *import*, not a result of this report. The splitting of monic scalar operators into first-order factors and the surjectivity of nonzero scalar operators are taken from [4, Lemma 2.4.19] and transferred along the elementary embedding of [3]; neither is reproved. The Berarducci–Mantova derivation and that elementary embedding are likewise imported wholesale (items N1 and N7). “Linearly closed” in the cited source means the splitting property and *not* that every homogeneous equation has a full solution set in the base field; reading it the second way would contradict Theorem 8.8 (Theorem 15.7).

- N54. The rank-one phase obstruction, the bounded-primitive ideal, the diagonal Galois tori and the universal exponential algebra used in Part III are existing repository or literature theory, reproved or specialized where a proof was wanted, and are not discoveries of the eighth manuscript; its own README and its comparison table say so (Theorems 15.1 and 15.2 and Section 17.6). The classical Pinney parametrization equation (17.10) and the abstract Schwarz-closed-field existence principle behind it are likewise disclaimed as antecedents, not contributions [6, 4, 5].
- N55. No named published open problem is claimed solved by Part III, and in particular no new solution of the Boshernitzan conjecture discussed in [5] is claimed. Its own novelty assessment is provisional and is narrowed to three statements — the arbitrary-dimensional Hermitian spectral–integral formula equation (16.3), the sharp entrywise perturbation criterion Theorem 16.7 with Theorem 16.9, and the normalized non-Abelian integration theorem Theorem 17.1 — on the basis of a *targeted* literature search that did not locate an exact prior statement. That is evidence for treating them as candidate new results, not a priority determination, and this report makes no novelty claim of its own (item N9).
- N56. No proof-assistant verification, no Lean formalization and no independent refereeing of Part III is claimed; its strongest claims require independent mathematical review. This repeats items N11 and N12 for the eighth member, whose own text states it in those words.
- N57. Non-effectivity. The obstruction map $\mathfrak{o}$ and the normalizing gauges G and W are exact *semantic* operations, as $\mathcal{I}_0$ is (Theorem 7.2). Even for finite matrices, arbitrary surreal input carries no assumed finite computable encoding, and no factorization algorithm follows from the first-order transfer in Theorem 15.6. Identifying restricted, support-certified input languages for which eigenvalue primitives, their purely infinite parts and a normalizing unitary gauge admit certified effective computation is posed as an open problem, not solved; the exact recurrences of Theorem 17.3 and equation (17.16) are instances, not a solution.
- N58. Every statement of Part III is relative to the one fixed intrinsic ∂ of Theorem 3.1 with $\partial\omega = 1$. None is a claim about every surreal derivation, about derivatives of a class function of a surreal variable, or about any of the coordinate operators of Table 1. Infinite-dimensional matrices, topological initial-value problems on $\mathbf{No}$, Stokes data over smaller analytic fields and a global surcomplex exponential are all outside its assertions (Theorem 17.13).
- N59. All dimensions in Part III are ordinary finite integers, and every displayed Hahn sum there — equation (17.3), equation (17.14), equation (17.19) — is a support-certified formal sum, not a limit in any topology on $\mathbf{No}[i]$. This is item N32 applied to the matrix part.
- N60. The metric theorems make no Lyapunov and no uniform-equivalence claim. Theorem 15.17 exhibits a positive horizontal metric of scale $e^{-2\omega}$ for printed coefficients whose ordinary solutions grow, and Theorem 15.16 has no hypothesis relating a horizontal metric to any fixed norm.
- N61. Theorem 16.5 does not remove the connection term $W^*\partial W$ of equation (16.5); that term is present and is merely invisible in the ordered additive quotient by $\mathcal{A}$. It is not the assertion that diagonalization commutes with differentiation, and the classification is not the assertion that every phase grade has a horizontal vector in $\mathbf{No}[i]^n$ — by Theorem 15.11 only the zero grade does (Theorem 16.6).

- N62. The proper-class handling of Theorem 17.14 is a size justification for the proofs, not an algorithm. It does not say that any fixed Hahn workspace already contains the needed primitives or solutions (Theorem 18.2), and it gives no effective presentation, no birthday or cardinal bound, and no minimal witness-closed subfield.
- N63. The eighth suite’s own scope statement is narrow by its authors’ wording: its checks do not verify the arbitrary-support Hahn arguments, the published model-theoretic inputs, the existence of a gauge for an arbitrary surreal matrix, or the general theorems. Its deliberate counterexamples are asserted *as* counterexamples — the nonnormal coefficient is required to have eigenvalues $\pm i$ together with a base-field fundamental matrix, and the noncommuting exponential defect is required to be nonzero — so a suite that passed while those assertions failed would be reporting the opposite of what it appears to report.
- N64. The eighth package’s verification program *rewrites its own delivered evidence*. Its closing lines compute the package root as the parent of the script’s directory, create `data/` there, and write `data/verification.json`; its README documents the behaviour as regeneration. Section D therefore reproduces the record *as delivered* and does not re-run it, and Section E repeats the instruction to run any source suite on a copy. A byte-identity check made after such a run compares two equally modified copies and establishes nothing. This is an artifact-level hazard, like the two in item N52; it affects no mathematical statement, but it is the reason no count in this report is presented as independently reproduced.

31.10 Open continuations, hedged

The following are directions for further development. They are *not* assertions that the corresponding questions are unresolved in the literature.

Two items that earlier editions of this list posed have since been answered inside the report, by Part III, and are recorded here as answered rather than deleted.

- *Which variable-coefficient matrix equations keep their fundamental matrices in $\mathbf{No}[i]$.* Theorem 15.11 answers this for every finite system: a fundamental matrix exists in $\mathrm{GL}_n(\mathbf{No}[i])$ exactly when every member of $\mathrm{Ph}(A)$ is zero, and $\dim_{\mathbb{C}} \ker(\partial - A) = m_0(A)$ in general. For Hermitian coefficients Theorem 16.8 makes the test explicit in terms of ordinary algebraic eigenvalues: the dimension is the number of eigenvalues lying in $\mathcal{A}$. The distinction the item asked for — genuine phase obstruction against workspace obstruction — is exactly the distinction between $\mathrm{Ph}(A) \neq \{0, \dots, 0\}$ and the workspace phenomena of Part IV, and Theorem 16.7 shows that the former is stable under every entrywise perturbation by $\mathcal{A}_{\mathbb{C}}$.

What the item also asked — which nontriangular differential modules admit a filtration whose rank-one factors have *computable* obstructions — remains open, and for the reason the item gave: that needs genuine module-theoretic information, not the eigenvalues of an instantaneous coefficient matrix. Theorem 16.10 confirms the reason rather than removing it. Higher-rank differential modules likewise remain unclassified.

- *Connecting the phase invariant with the strongest available linear-solvability theorems for this H -field.* This is done in Theorems 15.6, 15.11 and 16.8, and it separates existence from dimension exactly as the item asked: existence for *every* forcing holds unconditionally, while the dimension of the homogeneous solution space is the multiplicity $m_0(A)$ of the phase zero. The separation is not free — existence comes from the imported Theorem 15.6 and not from the rank-one criterion (Theorem 15.12, item N53) — and no effective version of either half is claimed (item N57).

The remaining continuations stand. Classify logarithmic derivatives in multi-scale ∂ -stable Hahn workspaces where support shifts lack the uniform rational form of equation (18.1). Extend the phase spectrum to non-Hermitian coefficients, or show that no spectral formula can exist there, the obstruction being that a nonnormal similarity gauge need not be entrywise finite (Theorem 16.11). Identify restricted, support-certified input languages on which the phase map, the purely infinite parts of eigenvalue primitives and a normalizing unitary gauge become effectively computable. Build explicitly typed analytic and coefficient interfaces that prove exchange rules with contour functionals under their correct common-domain and summability hypotheses, always fixing whether centres and contours move. Extend the workspace algorithms from real-exponent Hahn fields to logarithmic and higher-rank parents with explicit support-change certificates, preserving the distinction between a representation of a primitive and a decision about its infinite part. Separate negative-exponent coherent systems into finite nilpotent cases, cases reducible by a gauge transformation, and genuinely non-coherent cases, each proposed reduction carrying its own support proof. Develop monodromy for coherent coordinate equations on non-simply-connected domains in the nonlinear case.

32 Conclusion

The internal derivative does not turn a surcomplex number into an ordinary complex-valued function at infinity. Real exponential integration is extremely rich; unit-circle phases are severely constrained, because only a constant ordinary phase and an infinitesimal residue already live in the base field. The exact consequence is

$$\{y^\dagger : y \in \mathbf{No}[i]^\times\} = \mathbf{No} + i\partial\mathbf{m} = \mathbf{No} + i\partial\mathcal{O}, \quad \partial\mathbf{m} \subsetneq \mathbf{m}.$$

The purely infinite part of an accumulated phase of the imaginary coefficient is therefore the complete rank-one obstruction. It supplies a canonical normal form, distinguishes algebraic closure from differential solution closure, explains the missing oscillator, determines the maximal domain of any derivation-compatible exponential, removes every nonreal characteristic root from a constant-coefficient solution space, and fixes how many formal phase generators a diagonal Picard–Vessiot extension needs. At the workspace level two sharper identities are available,

$$\partial(\mathcal{H}) = \{f \in \mathcal{H} : [t]f = 0\}, \quad \{y^\dagger : y \in \mathcal{H}^\times\} = \mathbb{Q}t + \{g : g = 0 \text{ or } v(g) > 1\},$$

giving exact auditable tests rather than heuristic simplification rules, and showing why derivative closure, primitive closure and exponential closure must be separate capabilities.

The same obstruction survives the passage to finite matrices, and there it becomes a multiset. Every finite system over $\mathbf{No}[i]$ has a differential phase spectrum $\text{Ph}(A) \subset \mathcal{P}$, complete for gauge equivalence, of which the trace test equation (14.3) sees only the sum; and when the coefficient is iH with H Hermitian the spectrum is computed by integrating the ordinary algebraic eigenvalues of H and keeping the purely infinite parts of their primitives. The connection term left behind by pointwise diagonalization is neither ignored nor approximated: by Theorem 16.3 a unitary gauge is entrywise finite, so that term lands in $\text{Mat}_n(\mathcal{A}_\mathbb{C})$, which is exactly the kernel the invariant annihilates. That is why the perturbation theorem needs no spectral gap. What the argument needs instead is the strictness $\mathcal{A} \subsetneq \mathbf{m}$ that Theorem 5.4 insists on, and an additional imported input; and it needs Hermitian symmetry, which Theorem 16.10 shows cannot be dropped.

A distinct theory of coordinate initial-value problems remains available and useful. On a common ordinary complex domain, positive-support Hahn perturbations of a holomorphic problem are solved

uniquely by support recursion through ordinary linear equations, with explicit support certificates, a Liouville determinant identity, and monodromy valued in $\mathrm{GL}_d(K_\Gamma)$. The two theories coexist without conflict once the derivative, the parent field, the coefficient domain and the summability certificate are part of every statement.

The resulting picture is coherent rather than paradoxical: $\mathbf{No}[z]$ is algebraically closed; ∂ is surjective; real exponential integrals exist; finite angular motion is exact; and a sustained nonzero ordinary angular velocity is nevertheless absent. These are different closure properties, and the obstruction sequence equation (9.3) records their precise boundary. The boundary worth remembering is this:

Every surcomplex direction has a finite angle, but a differential evolution prescribing a purely infinite *accumulated* phase need not have a surcomplex value. A phase convention may assign a value; it cannot force that value to satisfy the incompatible number-derivation law.

And the distinctions are not a notation policy. Their compatibility and incompatibility are theorems: an intrinsic derivative of a scalar, an external derivative of a function, and the substitution that might connect them are three pieces of structure, not one.

Part VII

Appendices

A The positive-support calculus, proved

This appendix supplies the combinatorial facts used throughout. It works in a set-sized ordered abelian group and uses no completeness assumption on the group. The finite-word principle is the well-ordered-alphabet case of Higman's lemma [11]; the support consequence is the classical Hahn–Neumann fact [10]. Both are cited as published sources, and a proof is included as well, so that nothing in this report depends on an uninspected passage. The published citations are *not* replaced by this appendix, and this appendix is not replaced by them.

A.1 Two elementary facts about well-ordered supports

First, every sequence in a well-ordered set has an infinite nondecreasing subsequence: choose an occurrence of the least value in the whole tail, then an occurrence of the least value in the tail strictly after that index, and continue; the selected indices increase and the selected values do not decrease.

Lemma A.1 (Sum of two supports). *If A, B are well-ordered subsets of an ordered abelian group, then $A + B$ is well ordered, and for each γ only finitely many pairs $(a, b) \in A \times B$ satisfy $a + b = \gamma$. A finite union of well-ordered subsets is well ordered.*

Proof. If $a_n + b_n$ were strictly decreasing, select a nondecreasing subsequence of the a_n ; the corresponding b_n would then be strictly decreasing, contradicting well ordering of B . In a linear order, the absence of an infinite strictly decreasing sequence implies well ordering in the classical

set-theoretic setting used here: a nonempty subset with no least element would produce such a sequence by recursive choice.

If infinitely many distinct pairs had a fixed sum, select a sequence of distinct pairs. Their a entries are distinct, so a nondecreasing subsequence of them is strictly increasing, and the corresponding b entries strictly decrease — impossible. Finally, an infinite decreasing sequence in a finite union would have an infinite subsequence inside one member of the union. $\square$

A.2 Finite words over a well-ordered alphabet

A word $a_1 \cdots a_m$ *embeds* in a word $b_1 \cdots b_n$ if there are indices $1 \leq j_1 < \cdots < j_m \leq n$ with $a_k \leq b_{j_k}$ for every k . The empty word embeds in every word.

Lemma A.2 (Finite-word lemma). *Every infinite sequence of finite words over a well-ordered alphabet contains two words, in their original order, such that the earlier embeds in the later.*

Proof. Call a sequence *bad* if it contains no such pair, and suppose a bad sequence exists. Choose one minimally in the following sense: after a fixed prefix $w_0, \dots, w_{n-1}$, choose w_n of smallest possible length among words that permit continuation to an infinite bad sequence. Natural-number lengths make each choice possible, and no w_n is empty, since the empty word embeds in everything.

Write $w_n = v_n a_n$ with a_n the last letter, and select indices $n_0 < n_1 < \cdots$ for which the letters a_{n_j} are nondecreasing. Consider the sequence

$$w_0, \dots, w_{n_0-1}, v_{n_0}, v_{n_1}, v_{n_2}, \dots$$

It cannot be bad, because v_{n_0} is shorter than the minimal choice w_{n_0} after the same prefix. So it has a good pair. That pair cannot lie wholly inside the prefix, since the original sequence was bad. If it runs from a prefix word to some v_{n_j} , then the prefix word also embeds in w_{n_j} , again contradicting badness. And if it runs from v_{n_i} to v_{n_j} with $i < j$, append the nondecreasing last letters to the embedding: $w_{n_i} = v_{n_i} a_{n_i}$ embeds in $v_{n_j} a_{n_j} = w_{n_j}$, once more a contradiction. Hence no bad sequence exists. $\square$

A.3 From words to Hahn sums

Let $S \subseteq \Gamma_{>0}$ be well ordered. If a word in S embeds in another, the sum of its letters is at most the sum of the other's; and equality forces the words to be identical, because any unmatched letter is strictly positive and any strict comparison of matched letters contributes a strictly positive difference, so a finite sum of such contributions cannot vanish.

An infinite strictly decreasing sequence of word sums would therefore be a bad word sequence, contradicting Theorem A.2; hence S^* is well ordered. If infinitely many distinct words had the same sum, an embedded pair among them — which Theorem A.2 supplies — would force two of them to be identical. So each exponent has only finitely many word representations. This proves Theorem 2.6.

Remark A.3 (Ordered words, and what this appendix does *not* prove). The count is over *ordered* words, not over unordered multisets. That is what makes the lemma suffice for noncommuting matrix products: each word retains the order in which its coefficient matrices were multiplied, and finite matrix dimensions introduce only finitely many intermediate indices for a word of fixed length. Theorem 27.1 is the concrete witness that the ordering matters. Combined with Theorem A.1, this

justifies multiplying an arbitrary Hahn series by a positive-support power series, the Taylor-evaluation families of equations (6.1) and (23.3), and every coefficient recursion in this report.

It does *not* establish cofinal growth of $n\sigma$ in a higher-rank value group, and no assertion of that kind is used anywhere (Theorems 2.5 and 24.2).

B Notation, and a concordance with the source manuscripts

Table 10: Notation used in this report.

Symbol	Meaning
$\mathbf{No}, \mathbf{No}[i]$	The real surreal class field and its complexification $\mathbf{No}[i]$; class-sized carriers, with all supports and summed families sets.
∂, x'	The fixed simplest Berarducci–Mantova derivation, normalized by $\partial\omega = 1$, extended to $\mathbf{No}[i]$ by $\partial i = 0$.
∂_s	The rescaled derivation $\partial f / \partial s$, with $\partial_s s = 1$.
$\partial_Z, \tilde{\partial}$	Formal differentiation of the indeterminate on $\mathbf{No}[i][[Z]]$, and coefficientwise intrinsic differentiation there.
D_z	Coordinate differentiation of holomorphic coefficient functions in an ordinary complex z ; every monomial and scalar is held constant.
∂_H	Intrinsic differentiation of the Hahn monomials of a common-domain family, holding z fixed.
D_t	Formal differentiation in a Hahn parameter treated as an indeterminate.
F'_{fine}	The pointwise fine order-field derivative of a class function; not a field derivation.
$\mathcal{O}, \mathfrak{m}$	The finite real surreals and the real infinitesimals, with $\mathcal{O} = \mathbb{R} \oplus \mathfrak{m}$.
$\mathcal{O}_{\mathbb{C}}, \mathfrak{m}_{\mathbb{C}}$	$\mathcal{O} + i\mathcal{O}$ and $\mathfrak{m} + i\mathfrak{m}$.
$\mathcal{P}$	The additive class of purely infinite normal forms, including 0; <i>not</i> the class of all infinite surreals.
pp, ct, fin, st	The purely infinite projection; the coefficient of ω^0 , defined on all of $\mathbf{No}$; ct + infinitesimal part; and the standard part, defined on $\mathcal{O}$ only. ct and st agree only on $\mathcal{O}$.
t^γ, t	The Conway monomial $\omega^{-\gamma}$, and $t = \omega^{-1}$; not a generic exponential power.
E, L	The infinitesimal Hahn exponential on $\mathfrak{m}_{\mathbb{C}}$ and logarithm on $1 + \mathfrak{m}_{\mathbb{C}}$.
cis	The finite-angle phase map on $\mathcal{O}$, with kernel $2\pi\mathbb{Z}$.
θ	A finite angle : the argument of cis, always in $\mathcal{O}$.
η	An infinitesimal phase residue : the unique element of $\mathfrak{m}$ left after factoring out the constant ordinary unit c .
B	An accumulated phase : any primitive of the imaginary coefficient. Not an angle; obstructed exactly when $\text{pp}(B) \neq 0$.
$z^\dagger$	The logarithmic derivative $\partial z / z$ for $z \neq 0$; presumes no logarithm.
$\mathcal{I}_0$	The unique primitive with $\text{ct } \mathcal{I}_0(b) = 0$; a normalization, not an algorithm. $\mathcal{I}_0 _{\mathcal{A}}$ is the unique infinitesimal primitive.
$\mathcal{A}$	The bounded-primitive ideal $\partial\mathfrak{m} = \partial\mathcal{O}$; a convex ideal of $\mathcal{O}$, strictly inside $\mathfrak{m}$.
$\mathfrak{o}, \Phi$	$\mathfrak{o}(b) = \text{pp } \mathcal{I}_0(b) \in \mathcal{P}$, and $\Phi(q) = \mathfrak{o}(\text{Im } q)$.

Symbol	Meaning
$\mathcal{S}, E_{\mathcal{S}}$	The additive strip $\mathbf{No} + i\mathcal{O}$, not a subring, and its derivation-compatible exponential.
$\mathbb{T}(\mathbf{No}), \mathbb{T}(\mathbb{C})$	$\{u \in \mathbf{No}[i] : u\bar{u} = 1\}$ and the ordinary complex unit circle.
H_{Γ}, K_{Γ}	$\mathbb{R}((t^{\Gamma}))$ and $\mathbb{C}((t^{\Gamma}))$; $K_0 = \mathbb{C}((t))$, on which $\partial = -t^2 d/dt$.
$\mathcal{H}$	The rational-exponent workspace $\mathbb{C}((t^{\mathbb{Q}}))$, a full Hahn field.
$\mathcal{H}_{\Gamma}(U)$	The fixed-common-domain ring $\mathcal{O}(U)((t^{\Gamma}))$, all coefficients holomorphic on one ordinary U .
$\mathcal{O}(U)$	Ordinary holomorphic functions on U ; <i>not</i> $\mathcal{O}$.
$\mathcal{W}$	The proper-class ω -series field with its restricted composition theory.
$\mathbf{T}$	A transcendental differential generator with $\partial \mathbf{T} = i\mathbf{T}$ (or $= a\mathbf{T}$); never a surcomplex number.
$\mathbf{C}, \mathbf{S}, \mathbf{v}$	The recovered oscillator elements and the conjugation-fixed Cayley coordinate in the extension.
$\Lambda, \mathbf{G}_m(\mathbb{C})$	The integer phase-relation lattice, and the multiplicative algebraic group.

Warning B.1 (The concordance, and three symbols that must not be reused). Across the source manuscripts the derivation is written ∂ , ∂ , D , δ , der , ∂_{BM} and with a prime; two of them write ∂ for the intrinsic derivation and D_z for the coordinate one, and one writes D for a Picard–Vessiot derivation on a wholly different field. This report fixes ∂ for the intrinsic derivation and uses D for no operator on $\mathbf{No}$ or $\mathbf{No}[i]$ — its only appearance is as ordinary polynomial differentiation on $\mathbb{C}[x]$ in the constant-coefficient discussion after Theorem 12.1, where nothing surreal is differentiated — and keeps Table 1 as the reference.

The letter J is not used in this report (Theorem 5.6). In the sources it denotes the normalized intrinsic primitive (here $\mathcal{I}_0$), the representative class of purely infinite surreals (here $\mathcal{P}$), the inverse of $\partial|_m$ (here $\mathcal{I}_0|_{\mathcal{A}}$), and — in the companion *rank-one Berkovich* report, which remains standalone — the zero-constant *coordinate* primitive of a convergent power series, whose “complete primitive obstruction” is an annular Laurent residue and is unrelated to the phase obstruction. Any cross-link between the two reports must name the operator, not the letter.

The letter U is used in this report for an ordinary complex domain (Part V), and once more, with V , as a pair of formal placeholders in the statement of the exponential and logarithm identities (Theorem 6.1), where they stand for nothing at all. One source manuscript flagged its own collision between U as a transcendental differential generator and U as an ordinary domain; the generator is $\mathbf{T}$ here, and neither use ever denotes a surreal number or a matrix over one. The eighth manuscript writes U for a unitary matrix; this report writes W , and Theorem 15.5 fixes the rest of that part’s alphabet: H for a Hermitian matrix — never the hyphenated adjective in “ H -field” and never the coefficient field H_{Γ} — G for a gauge matrix, the upper-case upgrade of the scalar gauge g of equation (9.1); $\mathcal{M}$ for a horizontal metric, where the eighth manuscript writes G ; $\mathcal{R}$ for the rotation generator, where it writes the forbidden J ; and Π for the additive phase lattice inside $\mathcal{P}$, because Λ is already the integer relation lattice of equation (21.6), which is the object that manuscript calls L . Its Ψ is this report’s $\mathfrak{o}$ (Theorem 5.1(e)) and is *not* the Ψ of Theorem 11.4, which is a placeholder for a putative derivation-compatible exponential and is the letter’s only occurrence here; its $\mathcal{I}$ is $\mathcal{A}$. In Part III the letter W denotes a unitary matrix and M a finite differential module; Part V uses the same two letters for a positive-support matrix, an ordinary neighbourhood and a well-ordered exponent set, and the two parts never share a statement. The unitary group itself is $U_n(\mathbf{No}[i])$, printed upright so that it is not read as a domain.

Finally, $\mathcal{O}$ never takes an argument. Where a source wrote $\mathcal{O}(V)$ for holomorphic functions, this report writes $\mathcal{O}(V)$.

Dependency chain

Theorem 3.1 (imported) $\implies$ Theorem 3.5, Theorem 6.1 $\implies$ Theorem 6.3, Theorem 6.7
 $\implies$ Theorem 8.1, Theorem 8.4 $\implies$ Theorem 9.1, Theorem 11.2, Theorem 13.2.

The ideal-structure theorem Theorem 7.5 needs only the constant field, surjectivity, the H -field positivity clause and $\partial\omega = 1$; the polar decomposition additionally needs real closedness, standard part and the infinitesimal exponential; the image theorem additionally needs real exponential compatibility. The bounded-derivative lemma alone gives a quick independent proof that $\partial y = iy$ forces $y = 0$ (Theorem 8.8, second proof). The constant-coefficient theorem needs only the nonreal-eigenvalue obstruction, the constant field and the real exponential. The Laurent resolvents need strong additivity and $\partial t = -t^2$. The extension theorems need the constant-coefficient obstruction and a formal Laurent expansion in a new indeterminate. Strong additivity is essential for the infinitesimal chain rule and for Theorem 28.4; the Laurent identities can be verified inside formal series algebra independently of any construction of all surreals, the embedding being what identifies their operator with ∂ . Neither the composition theorem nor the Picard–Vessiot background is a dependency of the phase criterion. So the most striking negative conclusions do not depend on delicate analytic-geometry results or on a full model-theoretic classification.

C Provenance: what each archive contributed

This report merges eight source manuscripts. The manuscripts themselves are not distributed with this report; the disposition of each source result is recorded in the provenance material below and in the package README. The identifiers below are the merge identifiers; among the first seven the original archive names collided, three pairs of them differing only by a trailing marker, so the mapping is recorded here and in the package README because it is not recoverable from the filenames. Members 01–09 are pinned to `e260237db9b71da8b74a0c13c8e6355119091100`; member 10 joined later and is pinned to `aa846271b4dcae2c055b216126a87210292ec19b`.

Table 11: Provenance by member.

Member	Distinctive contribution retained in this report
01 <i>thresholds and Galois tori</i>	The exact image as the boxed organizing identity (Theorem 8.1); the obstruction as an $\mathbb{R}$ -linear surjection with the gauge <i>uniqueness</i> theorem and the rank-one tensor corollary (Theorems 9.1 and 9.4); the sharp support threshold over a real-exponent Hahn field with strictness at exponent 1 and the valuation-insufficiency counterexample (Theorems 10.1 and 10.4); the upper-triangular criterion with its descending recursion, the Jacobi trace test and the $\text{diag}(i, -i)$ counterexample (Theorems 14.1 and 14.2); the rational and diagonal Picard–Vessiot extensions with from-scratch no-new-constants proofs, the saturated integer lattice, the torus, and the $(i, 2i)$ versus $(i, \sqrt{2}i)$ examples (Theorems 21.1, 21.12 and 21.13); the formal oscillator in complex and real rational form with the order-incompatibility argument (equation (21.4) and Theorem 21.10); the non-summability of the exponential series at ω (Theorem 11.9); the phase-independent no-global-exponential corollary (Theorem 11.4); the “at most one, not none” warning with its two witnesses (Theorem 12.2); the class-hygiene discipline (Theorem 2.1); the coefficient-domain tiering contract (Section 29.1).
02 <i>word lemma and nonlinear IVP</i>	The entire coordinate half: the common-domain ring with the explicit disowning of both the shrinking-radius germ ring and the full formal ring (equation (23.1) and Theorem 23.1); the linear fundamental-matrix theorem with uniqueness among <i>all</i> Hahn-coherent matrices, the support certificate and the explicit recursion (Theorem 24.1); the inhomogeneous corollary, the Liouville determinant identity, the negative-leading-exponent proposition and the nilpotent counterexample bounding it (Theorems 24.3 to 24.6); the nonlinear deformation theorem with no common degree bound (Theorem 25.2); monodromy valued in $\text{GL}_d(K_\Gamma)$ with the example where ordinary monodromy is trivial and full monodromy is not (Theorems 26.1 and 26.2); the two-scale noncommuting computation and the refutation of the naive exponential (Section 27); the coherent entire solution poled at an infinite coordinate (Theorem 25.6); the cross-category pair (Theorem 26.3); the three-promise separation worked through $A = i\omega$ (Theorem 28.2); the strong Laurent resolvent with its residual certificate and the exact factorial solution (Theorems 18.8 and 18.10); the four-category separation table (Table 7); and the from-scratch proof of the ordered-word lemma (Section A).
05 <i>logarithmic- derivative image</i>	The exact decision criterion for rational coefficients and its raising-rather-than-reporting contract (Theorems 10.6 and 10.7); the order-theoretic structure and convexity of the bounded-primitive ideal (Theorem 7.5(iii)); the countable-stage differential Hahn hull with its monomial criterion, its minimality caveat, the real-group criterion, the $\mathbb{Q}\sqrt{2}$ counterexample and the “differentiation closure is not integration closure” example (Theorems 18.2, 20.2, 20.3, 20.6 and 20.7); the analytic coefficient interface with coefficientwise linear naturality, the total-derivative chain rule and the separate admissibility condition for formal power series (Theorems 28.4, 28.6 and 28.8 and Section 28.4); and the algebraic unit-modulus example (Theorem 10.8).

Member	Distinctive contribution retained in this report
06 <i>finite primitive and oscillation</i>	The base skeleton, and in particular: the strip theorem with <i>pointwise</i> maximality (Theorems 11.1 and 11.2); the character-independent differential defect (Theorem 11.5); the products, duals and no-finite-order corollary (Theorem 9.4); the direct-sum description (equation (7.5)); the summary table of distinguishable situations (Table 3); the factorially divergent forced oscillator with its closed residual and anti-topological disclaimer (Theorems 18.13 and 18.14); and the theorem that oscillation cannot retain the H -field rule (Theorem 21.10).
07 <i>exact sequence and rational exponents</i>	The explicit $\mathbb{C}((t^{\mathbb{Q}}))$ workspace chapter, with the exact derivative image, the exact logarithmic-derivative image, the single resonant coefficient, the resonant normal form and the workspace-versus-ambient discussion (Theorems 19.2 to 19.5 and Table 5); the derivative taxonomy including the fine order-field derivative and the proof that a fine derivative does not determine an intrinsic one (Table 1 and Theorem 4.1); the change-of-scale observation (Theorem 4.2); the two-derivative chain rule with <i>derived</i> summability (Theorems 4.5 and 4.6); the integration-by-parts correction for normalized primitives (Theorem 7.3); the restricted ω -series composition bridge and the precise sense in which two derivatives agree (Section 22 and Theorem 22.1); the terminating finite inverse of $\partial - i$ on $\mathbb{C}[\omega]$ (equation (12.3)); the union-of-fields anti-lemma (Theorem 20.4); and the four-outcome contract that returns a reason rather than a failure flag (Table 8).
08 <i>coherent IVP and general resolvent</i>	The resolvent over a general real-exponent K_{Γ} with its substantive hypothesis (Theorems 18.6 and 18.7); the ambient-versus-workspace uniqueness of the factorial solution, with its exponentially small full-field ambiguity (Theorem 18.12); the valuation test for finite phase (Theorem 18.4); the negative-valued coefficient that destroys common-domain coherence, with the rescaling that resolves it (Theorem 24.7); “the same printed equation under different derivatives” (Theorem 4.1(7)); the Riccati example with an entire coefficient family (Theorem 25.7); the oscillatory extension proved Picard–Vessiot via the absence of proper differential ideals in the Laurent ring (Theorem 21.2); the anti-Banach disclaimer (Theorem 25.4); the certificate table (Table 9); and the rule that a truncation must record the <i>type</i> of omitted information (Theorem 29.3).
09 <i>BM bridge and derived smallness</i>	The derivation of smallness and of strict order reversal on the infinitesimals from the H -field positivity axiom — the provenance this report uses (Theorem 3.8); the theorem that intrinsic differentiation preserves common-domain coherence, with the total intrinsic derivative and the fixed-contour and moving-pole discussion (Theorems 28.4, 28.6, 28.8 and 28.9); the bounded-primitive ideal as a named object with its full structure theory, including the cut description and the induced order-preserving quotient isomorphism (Theorem 7.5); the increasingly-slow-scale examples (Theorem 10.3 and Table 3); the resolvent with its support certificate and the whole-prefix coefficient recurrence with its exact guarantee (Theorems 18.6 and 18.15); the constant-coefficient coefficient stream and the $\sum \omega^{n^2} X^n$ support counterexample (Theorem 28.11); and the derivation-aware representation interface (Table 9).

Member	Distinctive contribution retained in this report
10 <i>Hermitian phase spectral integral</i>	The whole of Part III: matrix gauge equivalence and the complete phase normal form for every finite system, with the differential phase spectrum as its complete invariant (Theorems 5.1 and 15.9(e), Theorem 5.2); the imported splitting and first-order surjectivity input with its transfer (Theorems 15.6 and 15.7); existence for every forcing and the exact kernel dimension (Theorems 15.11 and 15.12); the canonical phase projectors and the refined trace identity (Theorems 15.13 and 15.14); the complete horizontal metric cone with its anti-Lyapunov example and the unitary normal form (Theorems 15.16 to 15.18); the attained min–max, the entrywise Weyl bound and the bounded-gauge estimate (Theorems 16.1 to 16.4); the spectral–integral theorem itself with gap-free perturbation invariance, the exact kernel count and its sharpness (Theorems 16.5 and 16.7 to 16.9); the two deliberate boundary counterexamples and the coupled unitary example (Theorems 16.10 to 16.12); normalized non-Abelian integration with its cocycle law and the exact noncommuting Hahn recurrence (Theorems 17.1 to 17.3); the real rotation-block descent and the real metric cone with its parity (Theorems 17.4 and 17.5); the positive Ermakov–Pinney dichotomy and the canonical amplitude–phase reduction (Theorems 17.6 and 17.7); the exact Airy Hahn expansion with its Wronskian, amplitude jet and truncation residual (Section 17.4); the resonance algebra with its Dickson finiteness and the singular three-phase invariant algebra (Theorems 17.8 and 17.9); the phase lattice as a Galois torus, its group algebra and real form (Theorems 17.10 to 17.12); and the set-sized witness-closed reading of the proper-class arguments (Theorem 17.14).

Remark C.1 (How the merge was performed, and the count of non-claims). A result that all seven prove is printed once. Where two members proved the same statement by genuinely different routes, both routes are kept and marked: Theorem 3.8 against Theorem 3.9 for smallness; the three proofs of Theorem 8.8; the two constant-field proofs in Theorems 21.1 and 21.2; Section A against the published citation; and the two localization constructions of Section 20, which prove different things. Where two members stated the same theorem with *different* uniqueness qualifiers, the qualifier printed is the one its proof supports and no wider (Theorem 25.3), with the stronger linear statement kept separately (Theorem 24.1). Where one name covered different coefficient rings, the rings are named (Theorems 10.1, 18.6 and 18.8). Nothing is strengthened beyond what a source proves.

The first seven manuscripts state, between them, $13 + 16 + 18 + 13 + 24 + 21 + 13 = 118$ explicit non-claims, with heavy overlap, and the eighth states a further 12, giving 130. Section 31 consolidates them into 64 numbered items, organized by theme; every one of the 130 has a home there, and many items carry several sources’ phrasings at once. Two of the 118 are artifact-level discrepancies in the source packages rather than mathematical limitations, and they are recorded as such in item N52; a third, the eighth package’s self-rewriting verification script, is item N64.

Integrating the eighth member followed the same rules. Its reproofs of results already in the report — the convexity and strictness of $\mathcal{A}$, the exact logarithmic-derivative image, the saturation of the relation lattice — are recorded as agreements and cited, not printed again (Theorems 15.1 and 15.2 and Section 17.6). Its two genuinely different routes are kept and marked: the rank- r differential simplicity of Theorem 17.11(i) beside the rank-one argument of Theorem 21.2, and the metric proof of the Pinney dichotomy beside the classical parametrization equation (17.10). Where

it strengthened a statement this report had deliberately hedged, the hedge was amended rather than removed and the new hypothesis was named: item N22 now records that first-order inhomogeneous existence holds but only by the import of Theorem 15.6, and item N24 now records the Hermitian case settled and the non-Hermitian case open, keeping the counterexample that marks the line. Nothing is strengthened beyond what a source proves.

D Verification record, exactly as delivered

Each source package shipped a deterministic exact-symbolic program together with its *recorded* run. This appendix reproduces those records. It does not re-run them and does not merge their counts into a single suite: eight suites with eight scopes remain eight suites.

Table 12: The eight recorded verification runs.

Member	Recorded	Scope and declared cutoffs
01	1,395 checks passed	Standard-library Python only, exact Gaussian rationals, deterministic seed 20260921, formal-series cutoff 16. Categories: Leibniz rule 250; additivity 250; primitive identity including the logarithmic term 250; conjugation compatibility 250; power-threshold regressions 49; positive-phase primitives 100; small-phase differential equation as a jet 100; unit-circle exponential identity as a jet 100; Cayley logarithmic derivative as a jet 1; exact polynomial-forcing inverses 36; repeated real-root amplitudes 9. Not a surreal-number implementation and not a proof checker, by its own docstring.
02	115 checks passed	Python with SymPy, exact symbolic only. Groups: power primitives 7; finite-phase formula identities 3; iterated-logarithm primitives 15; Laurent resolvent residuals 10; noncommuting matrix coefficient equations and initial values 56; matrix ODE and determinant jets 5; nonlinear rational solution and jets 15; infinitesimal monodromy jets 2; total chain-rule identities 2. Matrix cutoff is <i>total degree</i> 6, not a valuation cutoff; Laurent reciprocal order 12; nonlinear checks through t^{12} .
05	149 checks passed	Python with SymPy. Exposes a restricted rational-phase classifier as an importable API that type-checks its arguments, rejects nan, complex infinity and infinities, rejects free symbols other than the declared variable, coerces to polynomials over $\mathbb{Q}$, and <i>raises</i> outside $\mathbb{Q}(\text{variable})$. Check families include the real power rule over 22 rationals; power and logarithmic primitives to depth 3; twelve hand-listed rational-phase examples; a 6×6 degree sweep; eight rejection tests; the algebraic unit $(1 + i\omega)/\sqrt{1 + \omega^2}$; truncated $E(-it)$ residuals for $N = 0.12$ against equation (10.5); the total chain rule with its moving-argument term; commutation of the two coefficient operations; a fixed-contour sample; and a constant-coefficient example.

Member	Recorded	Scope and declared cutoffs
06	258 checks passed	Python with SymPy, models ∂ as $-t^2 d/dt$ on t -expressions and the extension derivation by $f \mapsto \text{cancel}(iT \partial f / \partial T)$; uses an ordinary variable as a stand-in for ω . Categories: total chain rule 7; scale primitives 23; formal exponential/logarithm 45; phase and gauge algebra 5; oscillatory extension 8; constant-coefficient modes 18; Laurent resolvent residuals 60; factorial oscillator 35; polynomial resolvent cutoffs 21; Laurent valuation bounds 36. Covers real-power primitives at $p \in \{-2, 0, \frac{1}{2}, 1, \frac{3}{2}, 2, 3\}$, the log-scale ladder at depths 0..3 and $p \in \{0, 1, 2, 3\}$, and truncated infinitesimal identities at orders 1..9. Its own trailing scope note reads: exact finite symbolic identities only; these checks do not prove the infinite-support, class-size, or general differential-field theorems.
07	291 checks passed	Python with SymPy plus a finite Hahn core implemented as a validating map from rational exponents to Gaussian-rational coefficients, with a terminating derivative, a primitive that rejects a nonzero resonant $[t^1]$, an ambient primitive pair returning (p, c) for $p + c \log \omega$, a classifier returning <i>workspace</i> / <i>ambient-only</i> / <i>phase-obstructed</i> , and a workspace certificate (r, k) asserting that k has positive support. The normalizer <i>rejects</i> floats, strings and non-Gaussian-rational coefficients, and four such rejections are counted among the passing checks; the script states that it is not an expression parser. Thirteen groups, including rational monomials 87 and the finite product rule 81.
08	150 checks passed	Python with SymPy, deterministic, with no numerical tolerance — an assertion requires exact simplification to zero. Covers monomial derivatives and primitives at nine rational exponents $\{-\frac{5}{2}, -1, -\frac{1}{3}, 0, \frac{1}{4}, 1, \frac{4}{3}, 2, \frac{7}{2}\}$; the logarithmic primitive; sixteen product-rule checks on four sparse expressions with complex coefficients; twenty-four resolvent-truncation residuals for $\lambda \in \{1, 2, i, 1 + i\}$ and $N = 1..6$, each required to equal exactly the first omitted derivative with the alternating sign; sixteen factorial residuals for $N = 1..16$; the Riccati coefficients $a_1..a_{10}$; constant-coefficient examples; the Cayley phase identity; and the oscillator identities in the adjoined extension.
09	259 checks passed	Python with SymPy over formal symbols, with $\partial(\cdot) = \text{expand}(-t^2 \partial / \partial t)$; a matrix identity or a whole checked coefficient prefix counts as <i>one</i> check. Groups: derivation identities 41 (including $\partial(t^n)$ for $n = -12..12$ and $\partial^n t$ for $n = 0..12$ against $(-1)^n n! t^{n+1}$); primitive identities 15; formal small-series identities 24 at orders 4, 6, 8, 10; coherent total-derivative examples 8, including the moving coefficient value $1 - 2/\omega^2 - 2/\omega^3$ and a moving-pole identity; general resolvent residuals 81 for $\lambda \in \{i, 2, 1 + i\}$ and $N = 0..8$; factorial-series residuals 13; whole-prefix coefficient recurrences 9 through t^{10} with lower bounds 1, -2, -5; constant-coefficient operator examples 33; matrix identities 35.

Member	Recorded	Scope and declared cutoffs
10	62 checks passed	Python with SymPy, exact symbolic and rational arithmetic throughout, with no numerical sampling. Groups: the nonnormal counterexample 3 — unimodular gauge, $\partial G = AG$, and characteristic polynomial $\zeta^2 + 1$, asserted <i>as</i> a counterexample (Theorem 16.10); the coupled unitary example 4 (orthogonality, skew-Hermiticity, the exact phase gauge, the instantaneous determinant, Theorem 16.12); the Jordan metric 2 (Theorem 15.17); the three-phase system 6, including its invariant Hermitian metric, its gauge identity, three polynomial first integrals verified in the <i>original</i> coupled coordinates, and the quadratic relation of equation (17.23); the noncommuting recurrence 12 with 12 unitarity-coefficient checks, one naive-exponential defect identity and one assertion that the commutator defect is nonzero (Theorem 17.3); the Airy recurrence 16 with the first amplitude correction and the exact truncation residual at $N = 16$ (equations (17.16) and (17.20)); and three Pinney checks (determinant one, invariant metric, symplectic amplitude reduction) carried out on arbitrary symbolic jets under the substitution for the second derivative (equations (17.9) and (17.11)). The record also carries the degree-8 amplitude jet of equation (17.19) and the 3×3 coefficient matrix of equation (17.22) as strings, both matching the displayed forms. Its own trailing scope note reads: finite exact symbolic checks; not a proof of general or infinite-support theorems.

The first seven runs were recorded under Python 3.13.5, six of them under SymPy 1.14.0 and the seventh using the standard library only. The eighth records SymPy 1.14.0 and a status of PASS with 62 checks, but records no Python version; its package pins `sympy==1.14.0` and its README states Python 3.10 or later. All eight record zero failures. The consolidated scope limits are item N48 to item N52 and item N63 to item N64 of Section 31. Three artifact-level discrepancies are recorded there and repeated here for completeness: one package’s PowerShell build script was supplied but not executed on Windows in its preparation environment; one package’s README lists a `SHA256SUMS.txt` that is not present among its delivered files; and the eighth package’s program rewrites its own delivered `data/verification.json`, computing the package root as the parent of its own directory, so the record reproduced above is the delivered one and a rerun would overwrite it.

E Building this report

The source `article.tex` is standalone: the bibliography is internal, and no external bibliography database, graphics file, font file, shell escape or network access is required. A standard TeX installation providing the packages named in the preamble suffices. From this directory,

```
latexmk -pdf -interaction=nonstopmode article.tex
```

and, if `latexmk` is unavailable, run `pdflatex` with `-interaction=nonstopmode` three times so that the table of contents, the cross-references and the `cleveref` labels settle. The directory also contains

`README.md` and the unmodified `code/` and `data/` directories of the eight source packages. To re-run a source suite, run it *on a copy*: several of the delivered programs rewrite their own evidence files beside themselves, and the eighth writes to a `data/` directory it creates one level *above* its own, so a byte-identity check performed after a run compares two equally modified copies and tells you nothing.

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